

# Sense Amplifiers

# SENSE AMPLIFIER SELECTION GUIDE

## SENSE AMPLIFIERS

| DESCRIPTION                  | THRESHOLD SENSITIVITY | COMMON-MODE RANGE | TYPE <sup>†</sup> OF OUTPUT | t <sub>PD</sub> <sup>‡</sup> TYPICAL | DEVICE TYPE FOR TEMPERATURE RANGE |             | PACKAGE TYPE | UNITS PER PACKAGE | ADDITIONAL FEATURES                                                                                                                                                         | PAGE NO. |
|------------------------------|-----------------------|-------------------|-----------------------------|--------------------------------------|-----------------------------------|-------------|--------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|                              |                       |                   |                             |                                      | -55°C to 125°C                    | 0°C to 70°C |              |                   |                                                                                                                                                                             |          |
| CORE-MEMORY SENSE AMPLIFIERS | ±4 mV                 | ±2.5 V            | R                           | 35 ns                                | SN5520                            | SN7520      | J            | 1                 | <ul style="list-style-type: none"> <li>Provides memory data register</li> <li>Complementary outputs</li> <li>Dual input channels</li> <li>Single-ended output</li> </ul>    | 33       |
|                              |                       |                   | O-C or R                    | 30 ns                                | SN5522                            | SN7522      | J            | 1                 | <ul style="list-style-type: none"> <li>Independent strobes</li> <li>Test points for strobe timing adjustment</li> </ul>                                                     |          |
|                              |                       |                   | R                           | 25 ns                                | SN5524                            | SN7524      | J            | 2                 | <ul style="list-style-type: none"> <li>Independent strobes</li> <li>Test points for strobe timing adjustment</li> </ul>                                                     |          |
|                              |                       |                   | R                           | 25 ns                                | SN5528                            | SN7528      | J            | 2                 | <ul style="list-style-type: none"> <li>Independent strobes</li> <li>Internally compensated reference amplifier</li> </ul>                                                   |          |
|                              |                       |                   | O-C                         | 25 ns                                | SN55232                           | SN75232     | J            | 2                 | <ul style="list-style-type: none"> <li>Independent strobes</li> <li>Internally compensated reference amplifier</li> </ul>                                                   |          |
|                              |                       |                   | R                           | 25 ns                                | SN55234                           | SN75234     | J            | 2                 | <ul style="list-style-type: none"> <li>Independent strobes</li> <li>Internally compensated reference amplifier</li> </ul>                                                   |          |
| MOS-TO-TTL CONVERTER ARRAY   | ±3 mV                 | ±1.5 V            | R                           | 25 ns                                | SN55238                           | SN75238     | J            | 2                 | <ul style="list-style-type: none"> <li>Independent strobes</li> <li>Internally compensated reference amplifier</li> <li>Test points for strobe timing adjustment</li> </ul> | 89       |
|                              |                       |                   | T-P                         | 28 ns                                | SN55236                           | SN75236     | W            | 2                 | <ul style="list-style-type: none"> <li>Built in data buffer and data register</li> <li>Reference amplifier inherently stable</li> </ul>                                     |          |
| MOS-MEMORY SENSE AMPLIFIERS  | 500 µA                |                   | T-P                         | 30 ns                                |                                   | SN75270     | J,N          | 7                 | <ul style="list-style-type: none"> <li>7 single-ended noninverting drivers per package</li> <li>Single 5-V supply</li> </ul>                                                | 88       |
|                              |                       |                   | T-P                         | 17 ns                                | SN55107A                          | SN75107A    | J            | 2                 | <ul style="list-style-type: none"> <li>Independent strobes</li> </ul>                                                                                                       |          |
|                              |                       |                   | O-C                         | 19 ns                                | SN55108A                          | SN75108A    | J            | 2                 | <ul style="list-style-type: none"> <li>Independent strobes</li> </ul>                                                                                                       |          |
| TMS 4062 I/O INTERFACE       | ±10 mV                | ±3 V              | T-P                         | 17 ns                                |                                   | SN75207     | J,N          | 2                 | <ul style="list-style-type: none"> <li>Independent strobes</li> </ul>                                                                                                       | 83       |
|                              |                       |                   | O-C                         | 19 ns                                |                                   | SN75208     | J,N          | 2                 | <ul style="list-style-type: none"> <li>Independent strobes</li> </ul>                                                                                                       |          |
| TMS 4062 I/O INTERFACE       | ±50 µA                |                   | R                           | 25 ns                                |                                   | SN75370     | J,N          | 2                 | <ul style="list-style-type: none"> <li>Combined driver and sense amplifier</li> <li>Read enable and write enable controls</li> </ul>                                        | 169      |

T-P = Totem Pole, O-C = Open Collector, R = Resistor Pull-Up

†t<sub>PD</sub> = Propagation Delay Time

NOTES: 1. For additional information, contact your nearest TI field sales office.

2. For data sheet, see page 41 of "The Line Driver and Line Receiver Data Book for Design Engineers," LCC4290.

## HIGH-SPEED SENSE AMPLIFIERS FOR CONVERSION OF COINCIDENT-CURRENT MEMORY READOUT TO SATURATED DIGITAL-LOGIC LEVELS

### performance features

- High Speed and Fast Recovery Time
- Time and Amplitude Signal Discrimination
- Adjustable Input Threshold Voltage Levels
- Narrow Region of Threshold Voltage Uncertainty
- Multiple Differential-Input Preamplifiers
- High D-C Noise Margin . . . Typically One Volt
- Good Fan-Out Capability

### ease-of-design features

- Choice of Output Circuit Function
- TTL or DTL Drive Capability
- Standard Logic Supply Voltages
- Plug-in Configuration Ideal for Flow-Soldering Techniques
- Pins on 100-mil Grid Spacings for Industrial-Type Circuit Boards

### description

Series 5520/7520 monolithic sense amplifiers are designed for use with high-speed memory systems. These sense amplifiers detect bipolar differential-input signals from the memory and provide the interface circuitry between the memory and the logic section. Low-level pulses originating in the memory are transformed into logic levels compatible with standard transistor-transistor-logic (TTL) and diode-transistor-logic (DTL) circuits.

These sense amplifiers feature multiple differential-input preamplifiers and versatile gating and output circuits, permitting a significant reduction in the circuitry required to accomplish the sensing function. A unique circuit design provides inherent stability of the input threshold level over a wide range of power-supply voltage levels and temperature ranges. Independent strobing of each of the dual sense-input channels ensures maximum versatility and permits detection to occur when the signal-to-noise ratio is at a maximum. The gate and strobe inputs and the outputs are compatible with standard TTL and DTL digital logic circuits.

The SN5520 and SN7520 circuits may be used to perform the functions of a flip-flop or register that responds to the sense and strobe input conditions.

The SN5522 and SN7522 circuits feature a high-fan-out, single-ended, open-collector output stage. In addition, they may be used to expand the inputs to an SN5520 or SN7520 circuit, or to perform the wired-AND function.

The SN5524 and SN7524 circuits provide for independent, dual-channel sensing with separate outputs. SN55234 and SN75234 are similar but have inverted outputs and internal compensation. SN55232 and SN75232 are identical to the SN55234 and SN75234, respectively, except that their output gates each feature an open-collector output.

The SN5528 and SN7528 circuits are identical to the SN5524 and SN7524, respectively, except that the output of each preamplifier is available as a test point. SN55238 and SN75238 are similar to SN5528 and SN7528, respectively, but have inverted outputs and internal compensation.

Series 5520 sense amplifiers are available in the J ceramic dual-in-line package and are characterized for operation over the full military temperature range of  $-55^{\circ}\text{C}$  to  $125^{\circ}\text{C}$ . Series 7520 sense amplifiers are available in both the J (ceramic) and N (plastic) dual-in-line packages and are characterized for operation from  $0^{\circ}\text{C}$  to  $70^{\circ}\text{C}$ .

| CONTENTS                                                                           | PAGE |
|------------------------------------------------------------------------------------|------|
| Design Characteristics, Circuit Operation, and Other General Information . . . . . | 34   |
| Maximum Ratings and Recommended Operating Conditions . . . . .                     | 37   |
| Definitive Specifications:                                                         |      |
| Types SN5520, SN7520 . . . . .                                                     | 38   |
| Types SN5522, SN7522 . . . . .                                                     | 40   |
| Types SN5524, SN7524 . . . . .                                                     | 42   |
| Types SN5528, SN7528 . . . . .                                                     | 44   |
| Types SN55232, SN75232 . . . . .                                                   | 46   |
| Types SN55234, SN75234 . . . . .                                                   | 48   |
| Types SN55238, SN75238 . . . . .                                                   | 50   |
| D-C Test Circuits . . . . .                                                        | 52   |
| Switching Time Test Circuits and Voltage Waveforms . . . . .                       | 67   |
| Typical Characteristics . . . . .                                                  | 76   |
| Application Data . . . . .                                                         | 79   |

## SERIES 5520/7520 SENSE AMPLIFIERS

### design characteristics

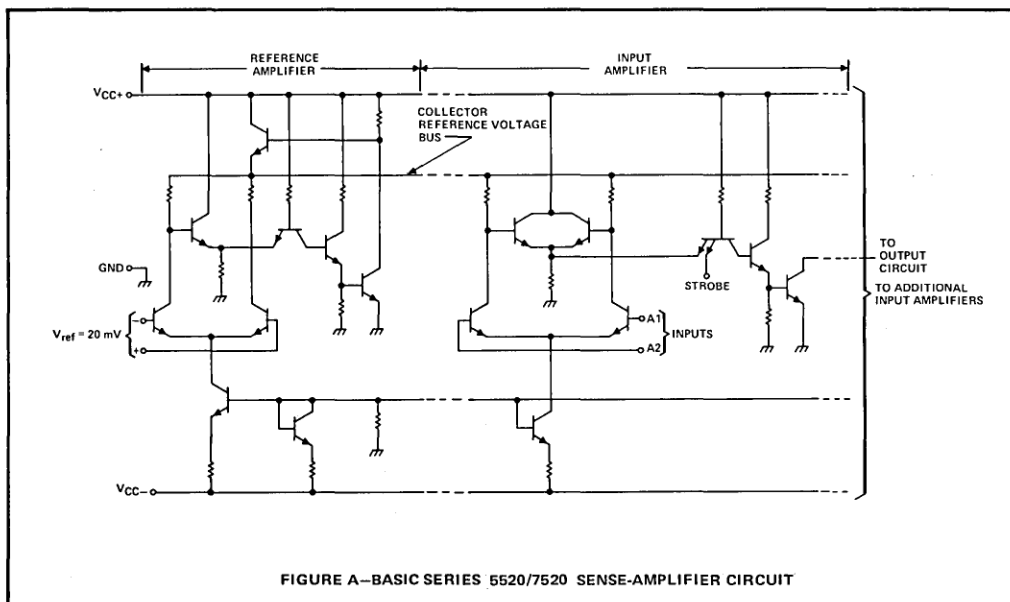
Series 5520/7520 sense amplifiers are completely d-c coupled. Previous designs have resulted in circuits in which the threshold level could not be closely controlled because they were highly sensitive to changes in the d-c levels throughout the amplifier. This was due primarily to the required tolerances on the absolute value of resistors and the resistor temperature coefficients. The "matched-amplifier" design of Series 5520/7520 circuits depends on resistor ratios rather than absolute values. In this design, excellent stability of the threshold level can be maintained despite component variations and changes in bias levels. The capability of multiple-input amplifiers increases the versatility of the design.

The basic circuit is used to implement several sense-amplifier designs. Additional logic circuitry is added to the strobe-gate output to provide versatile sensing functions. The outputs of two or more input amplifiers can be combined to implement multiple-input amplifiers, a function not previously available in integrated form. The d-c coupled design eliminates many of the problems associated with overload recovery time and threshold shift (with high input repetition rates) usually encountered in sense amplifier designs that use reactive coupling components.

### circuit operation

The basic Series 5520/7520 sense amplifier strobe and threshold circuit is shown in Figure A. The design uses a "matched-amplifier" concept that takes advantage of the inherent excellent component matching and thermal tracking characteristics of monolithic integrated circuits. A reference amplifier is used to generate the collector reference voltage that is distributed to the input amplifiers. Application of an external reference voltage,  $V_{ref}$ , establishes the input-amplifier threshold voltage level,  $V_T$ . The design is such that there is 1:1 correspondence between the applied reference voltage,  $V_{ref}$ , and the nominal threshold voltage level,  $V_T$ . The reference and input amplifiers use identical circuit configurations; therefore, changes in bias levels introduced into the input amplifier through changes in temperature or power-supply voltage levels are compensated by similar changes in the reference amplifier.

The collector reference voltage, supplied by the reference amplifier, can be used to control the threshold-voltage level of more than one input amplifier, thereby establishing equal threshold levels to all of the input sense channels simultaneously.



## SERIES 5520/7520 SENSE AMPLIFIERS

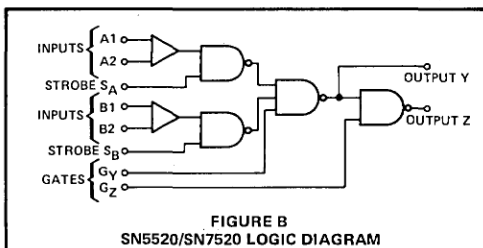
### circuit operation (continued)

The second stage of the input amplifier is a TTL gate. This gate provides the threshold action for the input sense channel and provides a convenient point in the circuit to accomplish the strobe function. The differential-input sense signal switches the output of the TTL gate only when the strobe input voltage is higher than the logic input threshold voltage. The strobe input, therefore, provides the sense amplifier with the capability of time discrimination, allowing the input signal to be detected when the signal-to-noise ratio is at a maximum.

The logic inputs (i.e., gate and strobe) of Series 5520/7520 sense amplifiers are designed to be compatible with Series 54/74 TTL digital integrated circuits. The multiple-emitter transistors are utilized to provide inherent switching-time advantages over other saturated-logic schemes. The same noise margin and logic threshold voltage as guaranteed for Series 54/74 are assured for each of the gate and strobe inputs. This is accomplished by testing each logic input under standard Series 54/74 test conditions, i.e., 2 volts for high-level input condition and 0.8 volt for low-level input conditions. Since the guaranteed minimum high-level output voltage is 2.4 volts and the guaranteed maximum low-level output voltage is 0.4 volt, a minimum noise margin of 0.4 volt is assured at each input.

### SN5520/SN7520 circuit

This circuit is a dual-channel sense amplifier with the preamplifiers connected to a common output stage and a complementary output stage. The output circuit is composed of two cascaded NAND gates, each with external gate inputs. External connection of the Z output and the  $G_Y$  input results in a flip-flop

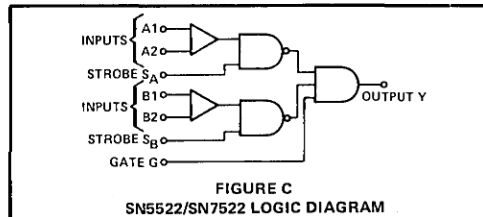


$$\begin{aligned}\text{logic: } Y &= \bar{G}_Y + A \cdot S_A + B \cdot S_B \\ Z &= \bar{G}_Z + \bar{Y} \\ Z &= \bar{G}_Z + G_Y (\bar{A} + \bar{S}_A) (\bar{B} + \bar{S}_B)\end{aligned}$$

or register that is set by signals at the differential-input terminals. Reset of the register is performed at the  $G_Z$  input. Capacitive coupling from output Z to  $G_Y$  results in output pulse stretching. With either connection, complementary output levels are available. The gate and strobe inputs and the outputs are compatible with standard TTL logic. The input function of SN5520/SN7520 can be expanded by connecting the Y output of SN5522/SN7522 to the  $G_Y$  input of the circuit being expanded.

### SN5522/SN7522 circuit

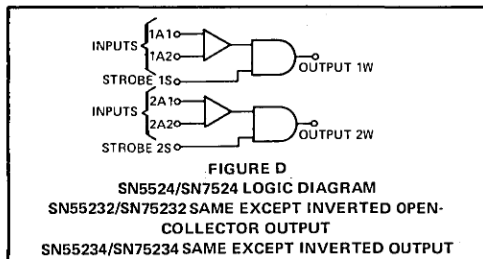
This circuit is a dual-channel sense amplifier with the preamplifiers connected to a common output stage. The output circuit features an open-collector output that permits two or more of these outputs to be connected in the wire-AND configuration. Each package includes a load resistor that may be used as the output pull-up resistor. High sink-current capability is a feature of this design, and a separate ground terminal is used for the output circuitry. These devices can also be used as input expanders for the SN5520/SN7520 circuit.



$$\text{logic: } Y = G (\bar{A} + \bar{S}_A) (\bar{B} + \bar{S}_B)$$

### SN5524/SN7524 circuit

This circuit features two completely independent sense amplifiers in a single package. Each amplifier features high fan-out capability.

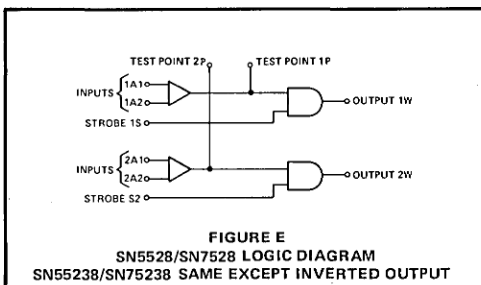


$$\begin{aligned}\text{logic: } W &= AS \text{ for SN5524 and SN7524} \\ W &= \bar{A}\bar{S} \text{ for SN55232, SN75232, SN55234, and SN75234}\end{aligned}$$

## SERIES 5520/7520 SENSE AMPLIFIERS

### SN5528/SN7528 circuit

This circuit features two separate single-preamplifier sense amplifiers in a single package. The output of each preamplifier is available as a test point. These test points can be used to observe the amplified core signal to facilitate accurate strobe timing. When using this device, care should be taken to avoid coupling the strobe signal or other stray signals to the test point. Excessive loading of the test point is also to be avoided. The result of either coupling or loading will be a change in the threshold voltage of the device. The output circuit of each channel features a simple TTL gate configuration with a high fan-out capability.



logic:  $W = AS$  for SN5528 and SN7528  
 $W = \overline{AS}$  for SN55238 and SN75238

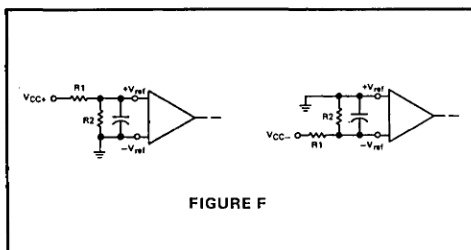
### SN55232, SN75232, SN55234, SN75234, SN55238, and SN75238 circuits

The SN55234, SN75234, SN55238, and SN75238 dual sense amplifier circuits are the same as SN5524, SN7524, SN5528, and SN7528, respectively, except that an additional stage has been added to the output gate to provide an inverted output and internal compensation has been added. Compared to using a separate gate for inversion, not only is package count reduced, but less propagation delay is added. The need for an external roll-off capacitor has been eliminated. SN55232 and SN75232 are identical to the SN55234 and SN75234, respectively, except that their output gates each have an open-collector output. This permits two or more outputs to be connected in wire-AND configuration.

### reference voltage considerations

These sense amplifiers feature a variable-threshold voltage level with simultaneous adjustment of both sense channels or both sense amplifiers by a single reference voltage. The operating threshold voltage level of the input amplifiers is established by and is approximately equal to the applied reference input voltage,  $V_{ref}$ . These sense amplifiers are recommended for use in systems requiring threshold voltage levels of  $\pm 15$  to  $\pm 40$  mV.

A simple method of generating the reference voltage is the use of a resistor voltage divider from either the positive ( $V_{CC+}$ ) or negative ( $V_{CC-}$ ) voltage supplies. See Figure F. This type of voltage divider may be used to supply an individual reference amplifier or to supply a number of paralleled reference amplifiers. The bias current required at the reference amplifier input is low (nominally  $30 \mu A$ ); therefore, voltage dividers of this type may normally be operated with very low current requirements. In noisy environments, the use of a filter capacitor across the inputs is recommended. By locating the capacitor as close to the device terminals as possible, noise and stray signals will be presented common-mode to the reference amplifier and thus be rejected.



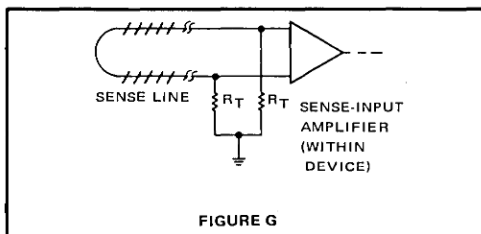
### input line layout considerations

Input sensitivity and device speed require adequate precautions in the routing of signal input and reference lines to prevent noise pickup. Bypassing of supply and reference inputs at the device with low-inductance disc ceramic capacitors and use of a good ground plane to separate strobe and output lines from sense and reference input lines are recommended.

# SERIES 5520/7520 SENSE AMPLIFIERS

## sense-input termination resistor considerations

Termination resistors are intentionally omitted from the sense-input terminals so the designer may select resistor values that will be compatible with the particular application. Matched termination resistors, ( $R_T$ , Figure G), normally in the range of 25  $\Omega$  to 200  $\Omega$  each, are required not only to terminate the sense line in a desired impedance but also to provide a d-c path for the sense-input bias currents. Careful matching of the resistor pairs should be observed or effective common-mode rejection will be reduced.



## output drive capability

The output circuits of these sense amplifiers feature the ability to sink or supply load current. This capability permits direct use with both TTL- and DTL-type loads. The open-collector output of the SN5522/SN7522 circuit may be connected to similar outputs to perform the wire-AND function. Load currents (out of the output terminal) are specified as negative values. Arrows on the d-c test circuit indicate the actual direction of current flow.

## logic input current requirements

Logic input current requirements are specified at worst-case power-supply conditions over the recommended operating free-air temperature range. The logic input currents are identical to those of, and compatible with, Series 54/74 TTL digital integrated circuits. Each logic input of the multiple-emitter input transistors requires no more than a 1.6-mA flow out of the input at a low logic level. Each input emitter requires current into the input when it is at a high-logic level. This current is 40  $\mu$ A maximum. Currents into the input terminals are specified as positive values. Arrows on the d-c test circuits indicate the actual direction of current flow.

## absolute maximum ratings over operating free-air temperature range (unless otherwise noted)

|                                                                |                |
|----------------------------------------------------------------|----------------|
| Supply voltages (see Note 1)                                   |                |
| $V_{CC+}$                                                      | 7 V            |
| $V_{CC-}$                                                      | -7 V           |
| Differential input voltage, $V_{ID}$ or $V_{ref}$              | $\pm 5$ V      |
| Voltage from any input to ground (see Note 2)                  | 5.5 V          |
| Off-state voltage applied to open-collector outputs            | 5.5 V          |
| Operating free-air temperature range: SN55' circuits           | -55°C to 125°C |
| SN75' circuits                                                 | 0°C to 70°C    |
| Storage temperature range                                      | -65°C to 150°C |
| Lead temperature 1/16 inch from case for 60 seconds: J package | 300°C          |
| Lead temperature 1/16 inch from case for 10 seconds: N package | 260°C          |

NOTES: 1. Voltage values, except differential voltages are with respect to network ground terminal.  
2. Strobe and gate input voltages must be zero or positive with respect to network ground terminal.

## recommended operating conditions

|           | MIN   | NOM | MAX   | UNIT |
|-----------|-------|-----|-------|------|
| $V_{CC+}$ | 4.75  | 5   | 5.25  | V    |
| $V_{CC-}$ | -4.75 | -5  | -5.25 | V    |
| $V_{ref}$ | 15    |     | 40    | mV   |

# TYPES SN5520, SN7520

## DUAL-CHANNEL SENSE AMPLIFIERS WITH COMPLEMENTARY OUTPUTS

FUNCTION TABLE

| INPUTS |   |                |                |                |                | OUTPUTS |             |
|--------|---|----------------|----------------|----------------|----------------|---------|-------------|
| A      | B | G <sub>Y</sub> | G <sub>Z</sub> | S <sub>A</sub> | S <sub>B</sub> | Y       | Z           |
| X      | X | L              | X              | X              | X              | H       | $\bar{G}_Z$ |
| H      | X | X              | X              | H              | X              | H       | $\bar{G}_Z$ |
| X      | H | X              | X              | X              | H              | H       | $\bar{G}_Z$ |
| L      | L | H              | X              | X              | X              | L       | H           |
| L      | X | H              | X              | X              | L              | L       | H           |
| X      | L | H              | X              | L              | X              | L       | H           |
| X      | X | H              | X              | L              | L              | L       | H           |
| X      | X | X              | L              | X              | X              | X       | H           |

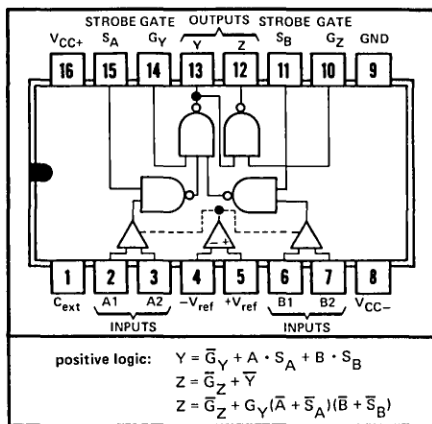
definition of logic levels

| INPUT      | H                          | L                          | X          |
|------------|----------------------------|----------------------------|------------|
| A or B†    | $V_{ID} > V_T \text{ max}$ | $V_{ID} < V_T \text{ min}$ | Irrelevant |
| Any G or S | $V_I > V_{IH} \text{ min}$ | $V_I < V_{IL} \text{ max}$ | Irrelevant |

†A and B are differential voltages ( $V_{ID}$ ) between A1 and A2 or B1 and B2, respectively. For these circuits,  $V_{ID}$  is considered positive regardless of which terminal of each pair is positive with respect to the other.

JORN

DUAL-IN-LINE PACKAGE (TOP VIEW)



electrical characteristics over recommended operating free-air temperature-range,  $V_{CC+} = 5 \text{ V}$ ,  $V_{CC-} = -5 \text{ V}$  (unless otherwise noted)

| PARAMETER                                                  | TEST FIGURE | TEST CONDITIONS                                                                                                                                            |                                                               | MIN       | TYP <sup>‡</sup> | MAX | UNIT          |
|------------------------------------------------------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|-----------|------------------|-----|---------------|
| $V_T$ Differential-input threshold voltage                 | 1           | $V_{ref} = 15 \text{ mV}$                                                                                                                                  | $T_A = 0^\circ\text{C to } 70^\circ\text{C}$                  | 11        | 15               | 19  | mV            |
|                                                            |             |                                                                                                                                                            | $T_A = -55^\circ\text{C to } 125^\circ\text{C}$ , SN5520 only | 10        | 15               | 20  |               |
|                                                            |             | $V_{ref} = 40 \text{ mV}$                                                                                                                                  | $T_A = 0^\circ\text{C to } 70^\circ\text{C}$                  | 36        | 40               | 44  |               |
|                                                            |             |                                                                                                                                                            | $T_A = -55^\circ\text{C to } 125^\circ\text{C}$ , SN5520 only | 35        | 40               | 45  |               |
| $V_{ICF}$ Common-mode input firing voltage (see Note 3)    | none        | $V_{ref} = 40 \text{ mV}$ , $V_{I(S)} = V_{IH}$<br>Common-mode input pulse:<br>$t_r \leq 15 \text{ ns}$ , $t_f \leq 15 \text{ ns}$ , $t_W = 50 \text{ ns}$ |                                                               | $\pm 2.5$ |                  |     | V             |
| $I_{IB}$ Differential-input bias current                   | 2           | $V_{CC+} = 5.25 \text{ V}$ ,<br>$V_{CC-} = -5.25 \text{ V}$ ,<br>$V_{ID} = 0$                                                                              | $T_A = -55^\circ\text{C to } 0^\circ\text{C}$ , SN5520 only   |           |                  | 100 | $\mu\text{A}$ |
|                                                            |             |                                                                                                                                                            | $T_A = 0^\circ\text{C to } 70^\circ\text{C}$                  |           | 30               | 75  |               |
|                                                            |             |                                                                                                                                                            | $T_A = 70^\circ\text{C to } 125^\circ\text{C}$ , SN5520 only  |           |                  | 75  |               |
| $I_{IO}$ Differential-input offset current                 | 2           | $V_{CC+} = 5.25 \text{ V}$ , $V_{CC-} = -5.25 \text{ V}$ , $V_{ID} = 0$                                                                                    |                                                               | 0.5       |                  |     | $\mu\text{A}$ |
| $V_{IH}$ High-level input voltage (strobe and gate inputs) | 3           |                                                                                                                                                            |                                                               | 2         |                  |     | V             |
| $V_{IL}$ Low-level input voltage (strobe and gate inputs)  | 3           |                                                                                                                                                            |                                                               | 0.8       |                  |     | V             |
| $V_{OH}$ High-level output voltage                         | 3           | $V_{CC+} = 4.75 \text{ V}$ , $V_{CC-} = -4.75 \text{ V}$ , $I_{OH} = -400 \mu\text{A}$                                                                     |                                                               | 2.4       | 4                |     | V             |
| $V_{OL}$ Low-level output voltage                          | 3           | $V_{CC+} = 4.75 \text{ V}$ , $V_{CC-} = -4.75 \text{ V}$ , $I_{OL} = 16 \text{ mA}$                                                                        |                                                               | 0.25      | 0.4              |     | V             |
| $I_{IH}$ High-level input current (strobe and gate inputs) | 4           | $V_{CC+} = 5.25 \text{ V}$ , $V_{CC-} = -5.25 \text{ V}$ , $V_{IH} = 2.4 \text{ V}$                                                                        |                                                               | 40        |                  |     | $\mu\text{A}$ |
| $I_{IL}$ Low-level input current (strobe and gate inputs)  | 4           | $V_{CC+} = 5.25 \text{ V}$ , $V_{CC-} = -5.25 \text{ V}$ , $V_{IL} = 0.4 \text{ V}$                                                                        |                                                               | -1        | -1.6             |     | mA            |
| $I_{OS(Y)}$ Short-circuit output current into Y            | 5           | $V_{CC+} = 5.25 \text{ V}$ , $V_{CC-} = -5.25 \text{ V}$ , $T_A = 25^\circ\text{C}$                                                                        |                                                               | -3        | -5               |     | mA            |
| $I_{OS(Z)}$ Short-circuit output current into Z            | 5           | $V_{CC+} = 5.25 \text{ V}$ , $V_{CC-} = -5.25 \text{ V}$ , $T_A = 25^\circ\text{C}$                                                                        |                                                               | -2.1      | -3.5             |     | mA            |
| $I_{CC+}$ Supply current from $V_{CC+}$                    | 6           | $V_{CC+} = 5.25 \text{ V}$ , $V_{CC-} = -5.25 \text{ V}$ , $T_A = 25^\circ\text{C}$                                                                        |                                                               | 28        | 40               |     | mA            |
| $I_{CC-}$ Supply current from $V_{CC-}$                    | 6           | $V_{CC+} = 5.25 \text{ V}$ , $V_{CC-} = -5.25 \text{ V}$ , $T_A = 25^\circ\text{C}$                                                                        |                                                               | -14       | -20              |     | mA            |

<sup>‡</sup>All typical values are at  $V_{CC+} = 5 \text{ V}$ ,  $V_{CC-} = -5 \text{ V}$ ,  $T_A = 25^\circ\text{C}$ .

NOTE 3: Common-mode input firing voltage is the minimum common-mode voltage that will exceed the dynamic range of the input at the specified conditions and cause the logic output to switch. The common-mode input signal is applied when the strobe is high.



# TYPES SN5520, SN7520

## DUAL-CHANNEL SENSE AMPLIFIERS WITH COMPLEMENTARY OUTPUTS

switching characteristics,  $V_{CC+} = 5\text{ V}$ ,  $V_{CC-} = -5\text{ V}$ ,  $C_{ext} \geq 100\text{ pF}$ ,  $T_A = 25^\circ\text{C}$

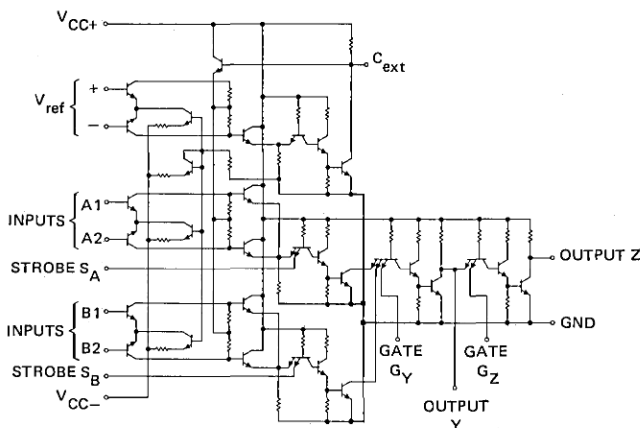
| PROPAGATION DELAY TIMES |                |           | TEST FIGURE | TEST CONDITIONS                            | MIN | TYP | MAX | UNIT |
|-------------------------|----------------|-----------|-------------|--------------------------------------------|-----|-----|-----|------|
| SYMBOL                  | FROM INPUT     | TO OUTPUT |             |                                            |     |     |     |      |
| $t_{PLH}(DY)$           | A1-A2 OR B1-B2 | Y         | 28          | $C_L = 15\text{ pF}$ , $R_L = 288\ \Omega$ | 25  | 40  |     | ns   |
| $t_{PHL}(DY)$           |                |           |             |                                            | 20  |     |     |      |
| $t_{PLH}(DZ)$           | A1-A2 OR B1-B2 | Z         | 28          | $C_L = 15\text{ pF}$ , $R_L = 288\ \Omega$ | 30  |     |     | ns   |
| $t_{PHL}(DZ)$           |                |           |             |                                            | 35  | 55  |     |      |
| $t_{PLH}(SY)$           | STROBE A OR B  | Y         | 28          | $C_L = 15\text{ pF}$ , $R_L = 288\ \Omega$ | 15  | 30  |     | ns   |
| $t_{PHL}(SY)$           |                |           |             |                                            | 20  |     |     |      |
| $t_{PLH}(SZ)$           | STROBE A OR B  | Z         | 28          | $C_L = 15\text{ pF}$ , $R_L = 288\ \Omega$ | 30  |     |     | ns   |
| $t_{PHL}(SZ)$           |                |           |             |                                            | 35  | 55  |     |      |
| $t_{PLH}(GY, Y)$        | GATE $G_Y$     | Y         | 29          | $C_L = 15\text{ pF}$ , $R_L = 288\ \Omega$ | 15  | 25  |     | ns   |
| $t_{PHL}(GY, Y)$        |                |           |             |                                            | 10  |     |     |      |
| $t_{PLH}(GY, Z)$        | GATE $G_Y$     | Z         | 29          | $C_L = 15\text{ pF}$ , $R_L = 288\ \Omega$ | 15  |     |     | ns   |
| $t_{PHL}(GY, Z)$        |                |           |             |                                            | 20  | 30  |     |      |
| $t_{PLH}(GZ, Z)$        | GATE $G_Z$     | Z         | 30          | $C_L = 15\text{ pF}$ , $R_L = 288\ \Omega$ | 15  |     |     | ns   |
| $t_{PHL}(GZ, Z)$        |                |           |             |                                            | 10  | 20  |     |      |

typical recovery and cycle times,  $V_{CC+} = 5\text{ V}$ ,  $V_{CC-} = -5\text{ V}$ ,  $C_{ext} \geq 100\text{ pF}$ ,  $T_A = 25^\circ\text{C}$

| PARAMETER      | TEST CONDITIONS                                                                                                                                       | MIN | TYP | MAX | UNIT |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|------|
| $t_{orD}$      | Differential-input overload recovery time<br>(see Note 4)<br><i>Differential Input Pulse:</i><br>$V_{ID} = 2\text{ V}$ , $t_r = t_f = 20\text{ ns}$   |     | 20  |     | ns   |
| $t_{orC}$      | Common-mode-input overload recovery time<br>(see Note 5)<br><i>Common-Mode Input Pulse:</i><br>$V_{IC} = \pm 2\text{ V}$ , $t_r = t_f = 20\text{ ns}$ |     | 20  |     | ns   |
| $t_{cyc(min)}$ | Minimum cycle time                                                                                                                                    |     | 200 |     | ns   |

- NOTES: 4. Differential-input overload recovery time is the time necessary for the device to recover from the specified differential-input overload signal prior to the strobe-enable signal.
5. Common-mode-input overload recovery time is the time necessary for the device to recover from the specified common-mode-input overload signal prior to the strobe-enable signal.

### schematic



# TYPES SN5522, SN7522

## DUAL-CHANNEL SENSE AMPLIFIERS

FUNCTION TABLE

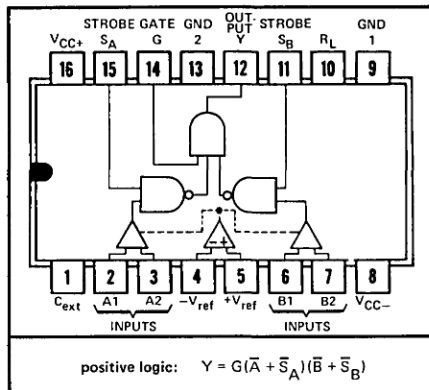
| INPUTS |   |   |                |                | OUTPUT |
|--------|---|---|----------------|----------------|--------|
| A      | B | G | S <sub>A</sub> | S <sub>B</sub> | Y      |
| L      | L | H | X              | X              | H      |
| L      | X | H | X              | L              | H      |
| X      | L | H | L              | X              | H      |
| X      | X | H | L              | L              | H      |
| X      | X | L | X              | X              | L      |
| H      | X | X | H              | X              | L      |
| X      | H | X | X              | H              | L      |

definition of logic levels

| INPUT      | H                             | L                             | X          |
|------------|-------------------------------|-------------------------------|------------|
| A or B†    | $V_{ID} \geq V_T \text{ max}$ | $V_{ID} \leq V_T \text{ min}$ | Irrelevant |
| Any G or S | $V_I \geq V_{IH} \text{ min}$ | $V_I \leq V_{IL} \text{ max}$ | Irrelevant |

†A and B are differential voltages ( $V_{ID}$ ) between A1 and A2 or B1 and B2, respectively. For these circuits,  $V_{ID}$  is considered positive regardless of which terminal of each pair is positive with respect to the other.

J OR N  
DUAL-IN-LINE PACKAGE (TOP VIEW)



electrical characteristics over recommended operating free-air temperature range,  $V_{CC+} = 5 \text{ V}$ ,  $V_{CC-} = -5 \text{ V}$  (unless otherwise noted)

| PARAMETER        |                                                   | TEST FIGURE | TEST CONDITIONS                                                                                                                           |                                              |                           | MIN  | TYP† | MAX | UNIT |    |
|------------------|---------------------------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|---------------------------|------|------|-----|------|----|
| V <sub>T</sub>   | Differential-input threshold voltage              | 7           | V <sub>ref</sub> = 15 mV                                                                                                                  | T <sub>A</sub> = 0°C to 70°C                 |                           | 11   | 15   | 19  | mV   |    |
|                  |                                                   |             | T <sub>A</sub> = −55°C to 125°C, SN5522 only                                                                                              |                                              | 10                        | 15   | 20   |     |      |    |
|                  |                                                   |             | V <sub>ref</sub> = 40 mV                                                                                                                  | T <sub>A</sub> = 0°C to 70°C                 |                           | 36   | 40   | 44  |      |    |
|                  |                                                   |             |                                                                                                                                           | T <sub>A</sub> = −55°C to 125°C, SN5522 only |                           | 35   | 40   | 45  |      |    |
| V <sub>ICF</sub> | Common-mode input firing voltage (see Note 3)     | none        | V <sub>I(S)</sub> = V <sub>IH</sub><br>Common-mode input pulse:<br>t <sub>r</sub> ≤ 15 ns, t <sub>f</sub> ≤ 15 ns, t <sub>w</sub> = 50 ns |                                              |                           | ±2.5 |      |     | V    |    |
| I <sub>IB</sub>  | Differential-input bias current                   | 2           | V <sub>CC+</sub> = 5.25 V,                                                                                                                | T <sub>A</sub> = −55°C to 0°C, SN5522 only   |                           | 100  |      |     | μA   |    |
|                  |                                                   |             | V <sub>CC−</sub> = −5.25 V,                                                                                                               | T <sub>A</sub> = 0°C to 70°C                 |                           | 30   | 75   |     |      |    |
|                  |                                                   |             | V <sub>ID</sub> = 0                                                                                                                       | T <sub>A</sub> = 70°C to 125°C, SN5522 only  |                           |      | 75   |     |      |    |
| I <sub>IO</sub>  | Differential-input offset current                 | 2           | V <sub>CC+</sub> = 5.25 V,                                                                                                                | V <sub>CC−</sub> = −5.25 V,                  | V <sub>ID</sub> = 0       | 0.5  |      |     | μA   |    |
| V <sub>IH</sub>  | High-level input voltage (strobe and gate inputs) | 8           |                                                                                                                                           |                                              |                           | 2    | V    |     |      |    |
| V <sub>IL</sub>  | Low-level input voltage (strobe and gate inputs)  | 8           |                                                                                                                                           |                                              |                           | 0.8  |      |     | V    |    |
| V <sub>OH</sub>  | High-level output voltage                         | 8           | V <sub>CC+</sub> = 4.75 V,                                                                                                                | V <sub>CC−</sub> = −4.75 V,                  | I <sub>OH</sub> = −400 μA | 2.4  | 4    |     | V    |    |
| V <sub>OL</sub>  | Low-level output voltage                          | 8           | V <sub>CC+</sub> = 4.75 V,                                                                                                                | V <sub>CC−</sub> = −4.75 V,                  | I <sub>OL</sub> = 16 mA   | 0.25 | 0.4  |     | V    |    |
| I <sub>IH</sub>  | High-level input current (strobe and gate inputs) | 9           | V <sub>CC+</sub> = 5.25 V,                                                                                                                | V <sub>CC−</sub> = −5.25 V,                  | V <sub>IH</sub> = 2.4 V   | 40   |      |     | μA   |    |
|                  |                                                   |             | V <sub>CC+</sub> = 5.25 V,                                                                                                                | V <sub>CC−</sub> = −5.25 V,                  | V <sub>IH</sub> = 5.25 V  | 1    |      |     | mA   |    |
| I <sub>IL</sub>  | Low-level input current (strobe and gate inputs)  | 9           | V <sub>CC+</sub> = 5.25 V,                                                                                                                | V <sub>CC−</sub> = −5.25 V,                  | V <sub>IL</sub> = 0.4 V   | −1   | −1.6 |     | mA   |    |
| I <sub>OH</sub>  | High-level output current                         | 10          | V <sub>CC+</sub> = 4.75 V,                                                                                                                | V <sub>CC−</sub> = −4.75 V,                  | V <sub>O</sub> = 5.25 V   | 250  |      |     | μA   |    |
| I <sub>OS</sub>  | Short-circuit output current                      | 11          | V <sub>CC+</sub> = 5.25 V,                                                                                                                | V <sub>CC−</sub> = −5.25 V,                  | T <sub>A</sub> = 25°C     | −2.1 | −3.5 |     |      | mA |
| I <sub>CC+</sub> | Supply current from V <sub>CC+</sub>              | 6           | V <sub>CC+</sub> = 5.25 V,                                                                                                                | V <sub>CC−</sub> = −5.25 V,                  | T <sub>A</sub> = 25°C     | 27   |      |     | 40   | mA |
| I <sub>CC−</sub> | Supply current from V <sub>CC−</sub>              | 6           | V <sub>CC+</sub> = 5.25 V,                                                                                                                | V <sub>CC−</sub> = −5.25 V,                  | T <sub>A</sub> = 25°C     | −15  |      |     | −20  | mA |

†All typical values are at  $V_{CC+} = 5 \text{ V}$ ,  $V_{CC-} = -5 \text{ V}$ ,  $T_A = 25^\circ\text{C}$ .

NOTE 3: Common-mode input firing voltage is the minimum common-mode voltage that will exceed the dynamic range of the input at the specified conditions and cause the logic output to switch. The common-mode input signal is applied when the strobe is high.

# TYPES SN5522, SN7522

## DUAL-CHANNEL SENSE AMPLIFIERS

switching characteristics,  $V_{CC+} = 5\text{ V}$ ,  $V_{CC-} = -5\text{ V}$ ,  $C_{ext} \geq 100\text{ pF}$ ,  $T_A = 25^\circ\text{C}$

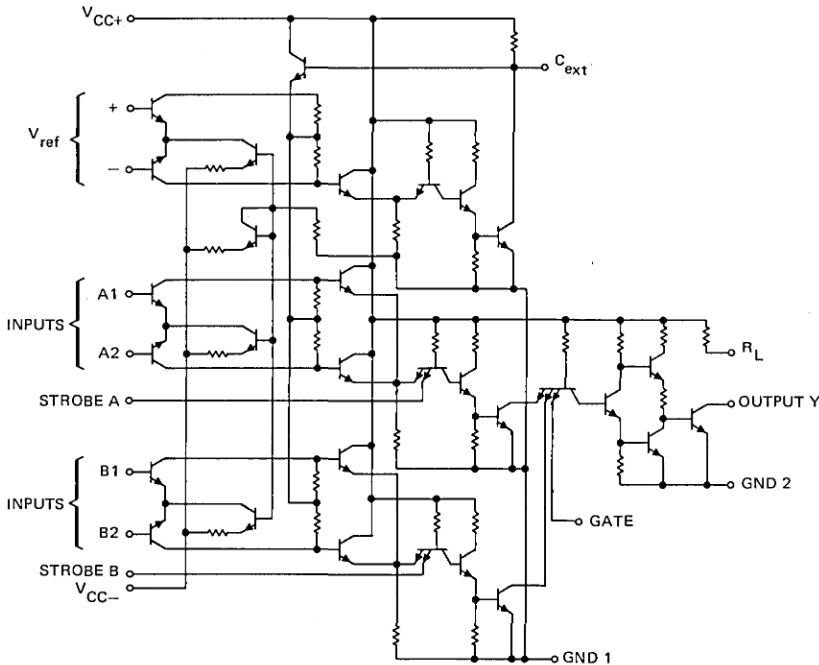
| PROPAGATION DELAY TIMES |                |           | TEST FIGURE | TEST CONDITIONS                            | MIN | TYP | MAX | UNIT |
|-------------------------|----------------|-----------|-------------|--------------------------------------------|-----|-----|-----|------|
| SYMBOL                  | FROM INPUT     | TO OUTPUT |             |                                            |     |     |     |      |
| $t_{PLH(D)}$            | A1-A2 OR B1-B2 | Y         | 31          | $C_L = 15\text{ pF}$ , $R_L = 288\ \Omega$ | 20  |     |     | ns   |
| $t_{PHL(D)}$            |                |           |             |                                            | 30  | 45  |     |      |
| $t_{PLH(S)}$            | STROBE A OR B  | Y         | 31          | $C_L = 15\text{ pF}$ , $R_L = 288\ \Omega$ | 20  |     |     | ns   |
| $t_{PHL(S)}$            |                |           |             |                                            | 20  | 40  |     |      |
| $t_{PLH(G)}$            | GATE           | Y         | 32          | $C_L = 15\text{ pF}$ , $R_L = 288\ \Omega$ | 10  |     |     | ns   |
| $t_{PHL(G)}$            |                |           |             |                                            | 15  | 25  |     |      |

typical recovery and cycle times,  $V_{CC+} = 5\text{ V}$ ,  $V_{CC-} = -5\text{ V}$ ,  $C_{ext} \geq 100\text{ pF}$ ,  $T_A = 25^\circ\text{C}$

| PARAMETER      |                                                           | TEST CONDITIONS                                                                           | MIN | TYP | MAX | UNIT |
|----------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------|-----|-----|-----|------|
|                |                                                           |                                                                                           |     |     |     |      |
| $t_{orD}$      | Differential-input overload recovery time<br>(see Note 4) | <i>Differential Input Pulse:</i><br>$V_{ID} = 2\text{ V}$ , $t_r = t_f = 20\text{ ns}$    |     | 20  |     | ns   |
| $t_{orC}$      | Common-mode-input overload recovery time<br>(see Note 5)  | <i>Common-Mode Input Pulse:</i><br>$V_{IC} = \pm 2\text{ V}$ , $t_r = t_f = 20\text{ ns}$ |     | 20  |     | ns   |
| $t_{cyc(min)}$ | Minimum cycle time                                        |                                                                                           |     | 200 |     | ns   |

- NOTES: 4. Differential-input overload recovery time is the time necessary for the device to recover from the specified differential input-overload signal prior to the strobe-enable signal.  
5. Common-mode-input overload recovery time is the time necessary for the device to recover from the specified common-mode input overload signal prior to the strobe-enable signal.

### schematic



# TYPES SN5524, SN7524

## DUAL SENSE AMPLIFIERS

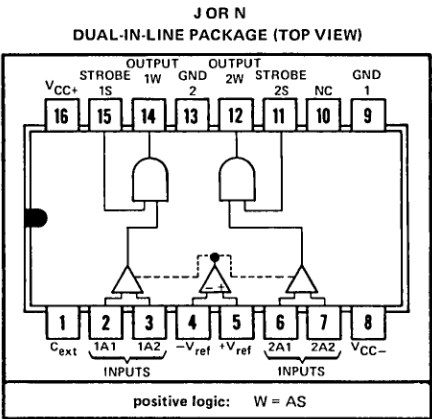
FUNCTION TABLE

| INPUTS |   | OUTPUT |
|--------|---|--------|
| A      | S | W      |
| H      | H | H      |
| L      | X | L      |
| X      | L | L      |

definition of logic levels

| INPUT | H                        | L                     | X          |
|-------|--------------------------|-----------------------|------------|
| A†    | $V_{ID} \geq V_{T \max}$ | $V_{ID} < V_{T \min}$ | Irrelevant |
| S     | $V_I \geq V_{IH \min}$   | $V_I < V_{IL \max}$   | Irrelevant |

†A is a differential voltage ( $V_{ID}$ ) between A1 and A2. For these circuits,  $V_{ID}$  is considered positive regardless of which terminal is positive with respect to the other.



NC—No internal connection

electrical characteristics over recommended operating free-air temperature range,  $V_{CC+} = 5\text{ V}$ ,  $V_{CC-} = -5\text{ V}$  (unless otherwise noted)

| PARAMETER        |                                               | TEST FIGURE | TEST CONDITIONS                                                                                                                                                     |                                              | MIN  | TYP† | MAX  | UNIT |
|------------------|-----------------------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|------|------|------|------|
| V <sub>T</sub>   | Differential-input threshold voltage          | 12          | V <sub>ref</sub> = 15 mV                                                                                                                                            | T <sub>A</sub> = 0°C to 70°C                 | 11   | 15   | 19   | mV   |
|                  |                                               |             |                                                                                                                                                                     | T <sub>A</sub> = −55°C to 125°C, SN5524 only | 10   | 15   | 20   |      |
|                  |                                               |             | V <sub>ref</sub> = 40 mV                                                                                                                                            | T <sub>A</sub> = 0°C to 70°C                 | 36   | 40   | 44   |      |
|                  |                                               |             |                                                                                                                                                                     | T <sub>A</sub> = −55°C to 125°C, SN5524 only | 35   | 40   | 45   |      |
| V <sub>ICF</sub> | Common-mode input firing voltage (see Note 3) | none        | V <sub>ref</sub> = 40 mV, V <sub>I(S)</sub> = V <sub>IH</sub><br>Common-mode input pulse:<br>t <sub>r</sub> ≤ 15 ns, t <sub>f</sub> ≤ 15 ns, t <sub>w</sub> = 50 ns |                                              | ±2.5 |      |      | V    |
| I <sub>IB</sub>  | Differential-input bias current               | 2           | V <sub>CC+</sub> = 5.25 V, V <sub>CC−</sub> = −5.25 V, V <sub>ID</sub> = 0                                                                                          | T <sub>A</sub> = −55°C to 0°C, SN5524 only   |      |      | 100  | μA   |
|                  |                                               |             |                                                                                                                                                                     | T <sub>A</sub> = 0°C to 70°C                 |      | 30   | 75   |      |
|                  |                                               |             |                                                                                                                                                                     | T <sub>A</sub> = 70°C to 125°C, SN5524 only  |      |      | 75   |      |
|                  |                                               |             |                                                                                                                                                                     |                                              |      |      |      |      |
| I <sub>IO</sub>  | Differential-input offset current             | 2           | V <sub>CC+</sub> = 5.25 V, V <sub>CC−</sub> = −5.25 V, V <sub>ID</sub> = 0                                                                                          |                                              | 0.5  |      |      | μA   |
| V <sub>IH</sub>  | High-level input voltage (strobe inputs)      | 13          |                                                                                                                                                                     |                                              | 2    |      |      | V    |
| V <sub>IL</sub>  | Low-level input voltage (strobe inputs)       | 13          |                                                                                                                                                                     |                                              |      |      | 0.8  | V    |
| V <sub>OH</sub>  | High-level output voltage                     | 13          | V <sub>CC+</sub> = 4.75 V, V <sub>CC−</sub> = −4.75 V, I <sub>OH</sub> = −400 μA                                                                                    |                                              | 2.4  | 4    |      | V    |
| V <sub>OL</sub>  | Low-level output voltage                      | 13          | V <sub>CC+</sub> = 4.75 V, V <sub>CC−</sub> = −4.75 V, I <sub>OL</sub> = 16 mA                                                                                      |                                              |      | 0.25 | 0.4  | V    |
| I <sub>IH</sub>  | High-level input current (strobe inputs)      | 14          | V <sub>CC+</sub> = 5.25 V, V <sub>CC−</sub> = −5.25 V, V <sub>IH</sub> = 2.4 V                                                                                      |                                              |      |      | 40   | μA   |
|                  |                                               |             | V <sub>CC+</sub> = 5.25 V, V <sub>CC−</sub> = −5.25 V, V <sub>IH</sub> = 5.25 V                                                                                     |                                              |      |      | 1    | mA   |
| I <sub>IL</sub>  | Low-level input current (strobe inputs)       | 14          | V <sub>CC+</sub> = 5.25 V, V <sub>CC−</sub> = −5.25 V, V <sub>IL</sub> = 0.4 V                                                                                      |                                              |      | −1   | −1.6 | mA   |
| I <sub>OS</sub>  | Short-circuit output current                  | 15          | V <sub>CC+</sub> = 5.25 V, V <sub>CC−</sub> = −5.25 V, T <sub>A</sub> = 25°C                                                                                        |                                              | −2.1 |      | −3.5 | mA   |
| I <sub>CC+</sub> | Supply current from V <sub>CC+</sub>          | 6           | V <sub>CC+</sub> = 5.25 V, V <sub>CC−</sub> = −5.25 V, T <sub>A</sub> = 25°C                                                                                        |                                              |      | 25   | 40   | mA   |
| I <sub>CC−</sub> | Supply current from V <sub>CC−</sub>          | 6           | V <sub>CC+</sub> = 5.25 V, V <sub>CC−</sub> = −5.25 V, T <sub>A</sub> = 25°C                                                                                        |                                              |      | −15  | −20  | mA   |

†All typical values are at  $V_{CC+} = 5\text{ V}$ ,  $V_{CC-} = -5\text{ V}$ ,  $T_A = 25^\circ\text{C}$ .

NOTE 3: Common-mode input firing voltage is the minimum common-mode voltage that will exceed the dynamic range of the input at the specified conditions and cause the logic output to switch. The common-mode input signal is applied when the strobe is high.

# TYPES SN5524, SN7524 DUAL SENSE AMPLIFIERS

switching characteristics,  $V_{CC+} = 5\text{ V}$ ,  $V_{CC-} = -5\text{ V}$ ,  $C_{ext} \geq 100\text{ pF}$ ,  $T_A = 25^\circ\text{C}$

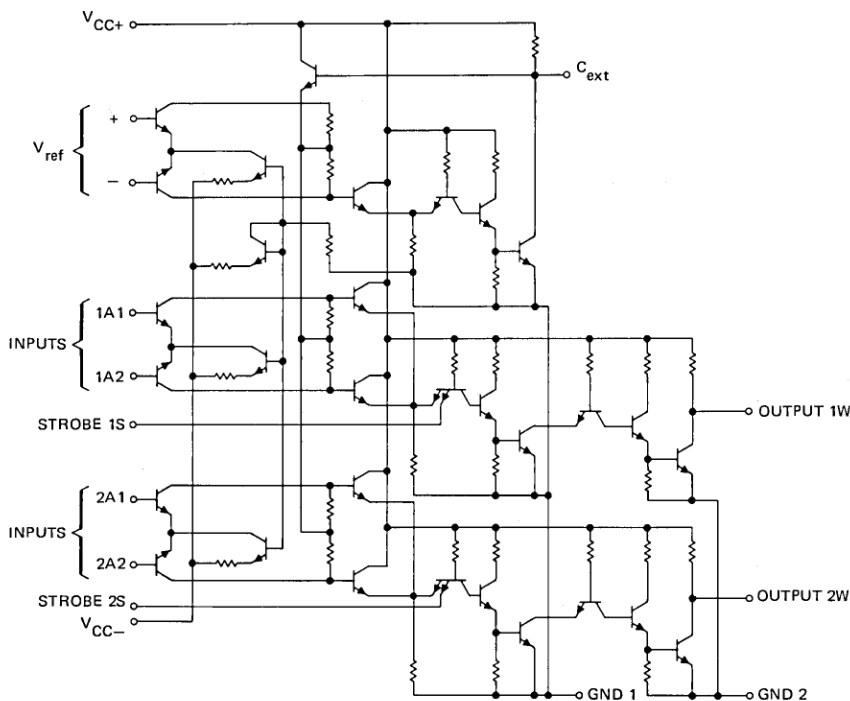
| PROPAGATION DELAY TIMES |            |           | TEST FIGURE | TEST CONDITIONS                            | MIN | TYP | MAX | UNIT |
|-------------------------|------------|-----------|-------------|--------------------------------------------|-----|-----|-----|------|
| SYMBOL                  | FROM INPUT | TO OUTPUT |             |                                            |     |     |     |      |
| $t_{PLH}(D)$            | A1-A2      | W         | 33          | $C_L = 15\text{ pF}$ , $R_L = 288\ \Omega$ | 25  | 40  |     | ns   |
| $t_{PHL}(D)$            |            |           |             |                                            |     |     |     |      |
| $t_{PLH}(S)$            | STROBE     | W         | 33          | $C_L = 15\text{ pF}$ , $R_L = 288\ \Omega$ | 15  | 30  |     | ns   |
| $t_{PHL}(S)$            |            |           |             |                                            |     |     |     |      |

typical recovery and cycle times,  $V_{CC+} = 5\text{ V}$ ,  $V_{CC-} = -5\text{ V}$ ,  $C_{ext} \geq 100\text{ pF}$ ,  $T_A = 25^\circ\text{C}$

| PARAMETER             |                                                        | TEST CONDITIONS                                                                           |  | MIN | TYP | MAX | UNIT |
|-----------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------------|--|-----|-----|-----|------|
| $t_{orD}$             | Differential-input overload recovery time (see Note 4) | <i>Differential Input Pulse:</i><br>$V_{ID} = 2\text{ V}$ , $t_r = t_f = 20\text{ ns}$    |  |     | 20  |     | ns   |
| $t_{orC}$             | Common-mode-input overload recovery time (see Note 5)  | <i>Common-Mode Input Pulse:</i><br>$V_{IC} = \pm 2\text{ V}$ , $t_r = t_f = 20\text{ ns}$ |  |     | 20  |     | ns   |
| $t_{cyc}(\text{min})$ | Minimum cycle time                                     |                                                                                           |  |     | 200 |     | ns   |

- NOTES: 4. Differential-input overload recovery time is the time necessary for the device to recover from the specified differential-input-overload signal prior to the strobe-enable signal.  
5. Common-mode-input overload recovery time is the time necessary for the device to recover from the specified common-mode-input overload signal prior to the strobe-enable signal.

## schematic



# TYPES SN5528, SN7528

## DUAL SENSE AMPLIFIERS WITH PREAMPLIFIER TEST POINTS

FUNCTION TABLE

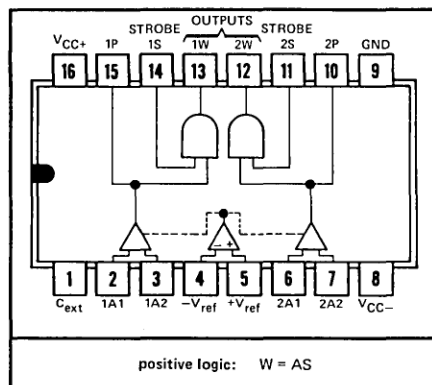
| INPUTS |   | OUTPUT |
|--------|---|--------|
| A      | S | W      |
| H      | H | H      |
| L      | X | L      |
| X      | L | L      |

definition of logic levels

| INPUT | H                     | L                     | X          |
|-------|-----------------------|-----------------------|------------|
| A†    | $V_{ID} > V_{T \max}$ | $V_{ID} < V_{T \min}$ | Irrelevant |
| S     | $V_I > V_{IH \min}$   | $V_I < V_{IL \max}$   | Irrelevant |

†A is a differential voltage ( $V_{ID}$ ) between A1 and A2. For these circuits,  $V_{ID}$  is considered positive regardless of which terminal is positive with respect to the other.

JORN  
DUAL-IN-LINE PACKAGE (TOP VIEW)



electrical characteristics over recommended operating free-air temperature range,  $V_{CC+} = 5\text{ V}$ ,  $V_{CC-} = -5\text{ V}$  (unless otherwise noted)

| PARAMETER        |                                               | TEST FIGURE | TEST CONDITIONS                                                                                                                                                     |                                              |       | MIN   | TYP‡ | MAX | UNIT |
|------------------|-----------------------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|-------|-------|------|-----|------|
| V <sub>T</sub>   | Differential-input threshold voltage          | 16          | V <sub>ref</sub> = 15 mV                                                                                                                                            | T <sub>A</sub> = 0°C to 70°C                 | 11    | 15    | 19   | mV  |      |
|                  |                                               |             |                                                                                                                                                                     | T <sub>A</sub> = -55°C to 125°C, SN5528 only | 10    | 15    | 20   |     |      |
|                  |                                               |             | V <sub>ref</sub> = 40 mV                                                                                                                                            | T <sub>A</sub> = 0°C to 70°C                 | 36    | 40    | 44   |     |      |
|                  |                                               |             |                                                                                                                                                                     | T <sub>A</sub> = -55°C to 125°C, SN5528 only | 35    | 40    | 45   |     |      |
| V <sub>ICF</sub> | Common-mode input firing voltage (see Note 3) | none        | V <sub>ref</sub> = 40 mV, V <sub>I(S)</sub> = V <sub>IH</sub><br>Common-mode input pulse:<br>t <sub>r</sub> ≤ 15 ns, t <sub>f</sub> ≤ 15 ns, t <sub>w</sub> = 50 ns |                                              |       | ±2.5  |      | V   |      |
| I <sub>IB</sub>  | Differential-input bias current               | 2           | V <sub>CC+</sub> = 5.25 V,<br>V <sub>CC-</sub> = -5.25 V,<br>V <sub>ID</sub> = 0                                                                                    | T <sub>A</sub> = -55°C to 0°C, SN5528 only   | 100   |       |      | μA  |      |
|                  |                                               |             |                                                                                                                                                                     | T <sub>A</sub> = 0°C to 70°C                 | 30 75 |       |      |     |      |
|                  |                                               |             |                                                                                                                                                                     | T <sub>A</sub> = 70°C to 125°C, SN5528 only  | 75    |       |      |     |      |
| I <sub>IO</sub>  | Differential-input offset current             | 2           | V <sub>CC+</sub> = 5.25 V, V <sub>CC-</sub> = -5.25 V, V <sub>ID</sub> = 0                                                                                          | 0.5                                          |       |       | μA   |     |      |
| V <sub>IH</sub>  | High-level input voltage (strobe inputs)      | 17          |                                                                                                                                                                     |                                              |       | 2     | V    |     |      |
| V <sub>IL</sub>  | Low-level input voltage (strobe inputs)       | 17          |                                                                                                                                                                     |                                              |       | 0.8 V |      |     |      |
| V <sub>OH</sub>  | High-level output voltage                     | 17          | V <sub>CC+</sub> = 4.75 V, V <sub>CC-</sub> = -4.75 V, I <sub>OH</sub> = -400 μA                                                                                    | 2.4                                          |       | 4     | V    |     |      |
| V <sub>OL</sub>  | Low-level output voltage                      | 17          | V <sub>CC+</sub> = 4.75 V, V <sub>CC-</sub> = -4.75 V, I <sub>OL</sub> = 16 mA                                                                                      | 0.25                                         |       | 0.4   | V    |     |      |
| I <sub>IH</sub>  | High-level input current (strobe inputs)      | 18          | V <sub>CC+</sub> = 5.25 V, V <sub>CC-</sub> = -5.25 V, V <sub>IH</sub> = 2.4 V                                                                                      | 40                                           |       |       | μA   |     |      |
|                  |                                               |             | V <sub>CC+</sub> = 5.25 V, V <sub>CC-</sub> = -5.25 V, V <sub>IH</sub> = 5.25 V                                                                                     | 1                                            |       |       | mA   |     |      |
| I <sub>IL</sub>  | Low-level input current (strobe inputs)       | 18          | V <sub>CC+</sub> = 5.25 V, V <sub>CC-</sub> = -5.25 V, V <sub>IL</sub> = 0.4 V                                                                                      | -1                                           |       | -1.6  | mA   |     |      |
| I <sub>OS</sub>  | Short-circuit output current                  | 19          | V <sub>CC+</sub> = 5.25 V, V <sub>CC-</sub> = -5.25 V, T <sub>A</sub> = 25°C                                                                                        | -2.1                                         |       | -3.5  | mA   |     |      |
| I <sub>CC+</sub> | Supply current from V <sub>CC+</sub>          | 6           | V <sub>CC+</sub> = 5.25 V, V <sub>CC-</sub> = -5.25 V, T <sub>A</sub> = 25°C                                                                                        | 25                                           |       | 40    | mA   |     |      |
| I <sub>CC-</sub> | Supply current from V <sub>CC-</sub>          | 6           | V <sub>CC+</sub> = 5.25 V, V <sub>CC-</sub> = -5.25 V, T <sub>A</sub> = 25°C                                                                                        | -15                                          |       | -20   | mA   |     |      |

‡All typical values are at  $V_{CC+} = 5\text{ V}$ ,  $V_{CC-} = -5\text{ V}$ ,  $T_A = 25^\circ\text{C}$ .

NOTE 3: Common-mode input firing voltage is the minimum common-mode voltage that will exceed the dynamic range of the input at the specified conditions and cause the logic output to switch. The common-mode input signal is applied when the strobe is high.

# TYPES SN5528, SN7528

## DUAL SENSE AMPLIFIERS WITH PREAMPLIFIER TEST POINTS

switching characteristics,  $V_{CC+} = 5\text{ V}$ ,  $V_{CC-} = -5\text{ V}$ ,  $T_A = 25^\circ\text{C}$

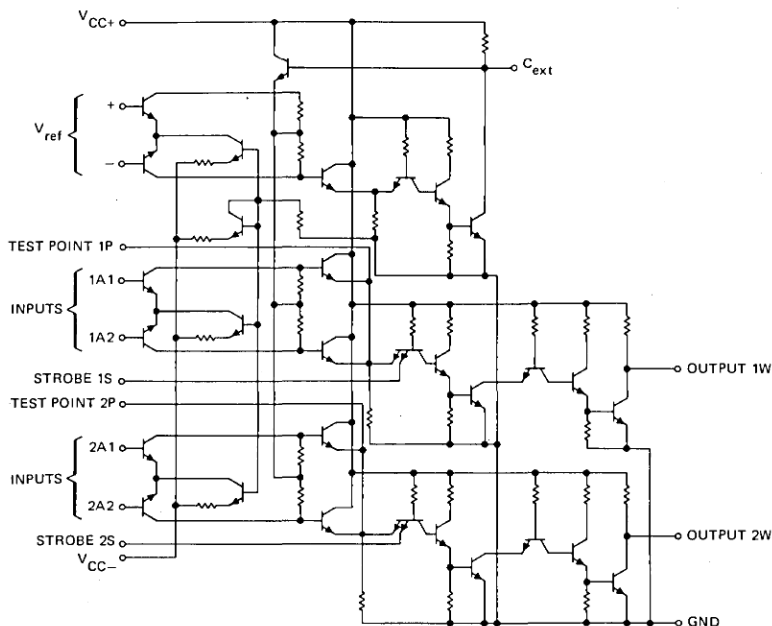
| PROPAGATION DELAY TIMES |            |           | TEST FIGURE | TEST CONDITIONS                            | MIN | TYP | MAX | UNIT |
|-------------------------|------------|-----------|-------------|--------------------------------------------|-----|-----|-----|------|
| SYMBOL                  | FROM INPUT | TO OUTPUT |             |                                            |     |     |     |      |
| $t_{PLH(D)}$            | A1–A2      | W         | 34          | $C_L = 15\text{ pF}$ , $R_L = 288\ \Omega$ | 25  | 40  |     | ns   |
| $t_{PHL(D)}$            |            |           |             |                                            | 20  |     |     | ns   |
| $t_{PLH(S)}$            | STROBE     | W         | 34          | $C_L = 15\text{ pF}$ , $R_L = 288\ \Omega$ | 15  | 30  |     | ns   |
| $t_{PHL(S)}$            |            |           |             |                                            | 20  |     |     | ns   |

typical recovery and cycle times,  $V_{CC+} = 5\text{ V}$ ,  $V_{CC-} = -5\text{ V}$ ,  $T_A = 25^\circ\text{C}$

| PARAMETER      | TEST CONDITIONS                                                                                                                                       | MIN | TYP | MAX | UNIT |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|------|
| $t_{orD}$      | Differential-input overload recovery time<br>(see Note 4)<br><i>Differential Input Pulse:</i><br>$V_{ID} = 2\text{ V}$ , $t_f = 20\text{ ns}$         |     | 20  |     | ns   |
| $t_{orC}$      | Common-mode-input overload recovery time<br>(see Note 5)<br><i>Common-Mode Input Pulse:</i><br>$V_{IC} = \pm 2\text{ V}$ , $t_r = t_f = 20\text{ ns}$ |     | 20  |     | ns   |
| $t_{cyc(min)}$ | Minimum cycle time                                                                                                                                    |     | 200 |     | ns   |

- NOTES: 4. Differential-input overload recovery time is the time necessary for the device to recover from the specified differential-input overload signal prior to the strobe-enable signal.  
5. Common-mode-input overload recovery time is the time necessary for the device to recover from the specified common-mode-input overload signal prior to the strobe-enable signal.

schematic



# TYPES SN55232, SN75232

## DUAL SENSE AMPLIFIERS

FUNCTION TABLE

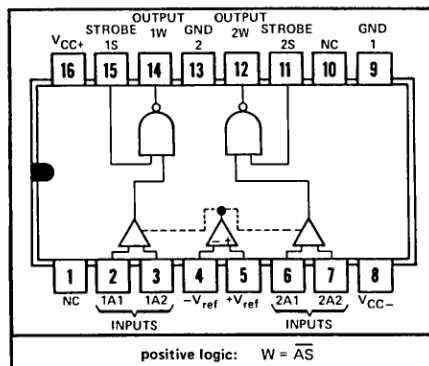
| INPUTS |   | OUTPUT |
|--------|---|--------|
| A      | S | W      |
| H      | H | L      |
| L      | X | H      |
| X      | L | H      |

definition of logic levels

| INPUT | H                          | L                          | X          |
|-------|----------------------------|----------------------------|------------|
| A1    | $V_{ID} > V_T \text{ max}$ | $V_{ID} < V_T \text{ min}$ | Irrelevant |
| S     | $V_I > V_{IH} \text{ min}$ | $V_I < V_{IL} \text{ max}$ | Irrelevant |

$V_T$  is a differential voltage ( $V_{ID}$ ) between A1 and A2. For these circuits,  $V_{ID}$  is considered positive regardless of which terminal is positive with respect to the other.

JORN  
DUAL-IN-LINE PACKAGE (TOP VIEW)



NC—No internal connection

electrical characteristics over recommended operating free-air temperature range,  $V_{CC+} = 5 \text{ V}$ ,  $V_{CC-} = -5 \text{ V}$  (unless otherwise noted)

| PARAMETER                                               | TEST FIGURE | TEST CONDITIONS                                                                      | MIN | TYP <sup>‡</sup> | MAX  | UNIT          |
|---------------------------------------------------------|-------------|--------------------------------------------------------------------------------------|-----|------------------|------|---------------|
| $V_T$ Differential-input threshold voltage              | 20          | $V_{ref} = 15 \text{ mV}$                                                            |     | 11               | 15   | 19            |
|                                                         |             | $T_A = 0^\circ\text{C to } 70^\circ\text{C}$                                         |     |                  |      |               |
|                                                         |             | $T_A = -55^\circ\text{C to } 125^\circ\text{C}$ , SN55232 only                       |     | 10               | 15   | 20            |
|                                                         |             | $V_{ref} = 40 \text{ mV}$                                                            |     | 36               | 40   | 44            |
| $V_{ICF}$ Common-mode input firing voltage (see Note 3) | none        | $V_{ref} = 40 \text{ mV}$ , $V_I(S) = V_{IH}$                                        |     |                  |      |               |
|                                                         |             | Common-mode input pulse:                                                             |     |                  |      |               |
|                                                         |             | $t_r \leq 15 \text{ ns}$ , $t_f \leq 15 \text{ ns}$ , $t_w = 50 \text{ ns}$          |     | $\pm 2.5$        |      | V             |
|                                                         |             | $T_A = -55^\circ\text{C to } 0^\circ\text{C}$ , SN55232 only                         |     |                  | 100  |               |
| $I_{IB}$ Differential-input bias current                | 2           | $V_{CC+} = 5.25 \text{ V}$ , $V_{CC-} = -5.25 \text{ V}$ , $V_{ID} = 0$              |     |                  | 30   | 75            |
|                                                         |             | $T_A = 0^\circ\text{C to } 70^\circ\text{C}$                                         |     |                  |      |               |
|                                                         |             | $T_A = 70^\circ\text{C to } 125^\circ\text{C}$ , SN55232 only                        |     |                  | 75   | $\mu\text{A}$ |
| $I_{IO}$ Differential-input offset current              | 2           | $V_{CC+} = 5.25 \text{ V}$ , $V_{CC-} = -5.25 \text{ V}$ , $V_{ID} = 0$              |     | 0.5              |      | $\mu\text{A}$ |
| $V_{IH}$ High-level input voltage (strobe inputs)       | 21          |                                                                                      |     | 2                |      | V             |
| $V_{IL}$ Low-level input voltage (strobe inputs)        | 21          |                                                                                      |     |                  | 0.8  | V             |
| $I_{OH}$ High-level output current                      | 21          | $V_{CC+} = 4.75 \text{ V}$ , $V_{CC-} = -4.75 \text{ V}$ , $V_{OH} = 5.25 \text{ V}$ |     |                  | 250  | $\mu\text{A}$ |
| $I_{OL}$ Low-level output voltage                       | 21          | $V_{CC+} = 4.75 \text{ V}$ , $V_{CC-} = -4.75 \text{ V}$ , $I_{OL} = 16 \text{ mA}$  |     | 0.25             | 0.4  | V             |
| $I_{IH}$ High-level input current (strobe inputs)       | 22          | $V_{CC+} = 5.25 \text{ V}$ , $V_{CC-} = -5.25 \text{ V}$ , $V_{IH} = 2.4 \text{ V}$  |     |                  | 40   | $\mu\text{A}$ |
|                                                         |             | $V_{CC+} = 5.25 \text{ V}$ , $V_{CC-} = -5.25 \text{ V}$ , $V_{IH} = 5.25 \text{ V}$ |     |                  | 1    | $\text{mA}$   |
| $I_{IL}$ Low-level input current (strobe inputs)        | 22          | $V_{CC+} = 5.25 \text{ V}$ , $V_{CC-} = -5.25 \text{ V}$ , $V_{IL} = 0.4 \text{ V}$  |     | -1               | -1.6 | $\text{mA}$   |
|                                                         |             |                                                                                      |     |                  |      |               |
| $I_{CC+}$ Supply current from $V_{CC+}$                 | 6           | $V_{CC+} = 5.25 \text{ V}$ , $V_{CC-} = -5.25 \text{ V}$ , $T_A = 25^\circ\text{C}$  |     | 25               | 40   | $\text{mA}$   |
| $I_{CC-}$ Supply current from $V_{CC-}$                 | 6           | $V_{CC+} = 5.25 \text{ V}$ , $V_{CC-} = -5.25 \text{ V}$ , $T_A = 25^\circ\text{C}$  |     | -15              | -20  | $\text{mA}$   |

<sup>‡</sup>All typical values are at  $V_{CC+} = 5 \text{ V}$ ,  $V_{CC-} = -5 \text{ V}$ ,  $T_A = 25^\circ\text{C}$ .

NOTE 3: Common-mode input firing voltage is the minimum common-mode voltage that will exceed the dynamic range of the input at the specified conditions and cause the logic output to switch. The common-mode input signal is applied when the strobe is high.



# TYPES SN55232, SN75232

## DUAL SENSE AMPLIFIERS

switching characteristics,  $V_{CC+} = 5\text{ V}$ ,  $V_{CC-} = -5\text{ V}$ ,  $T_A = 25^\circ\text{C}$

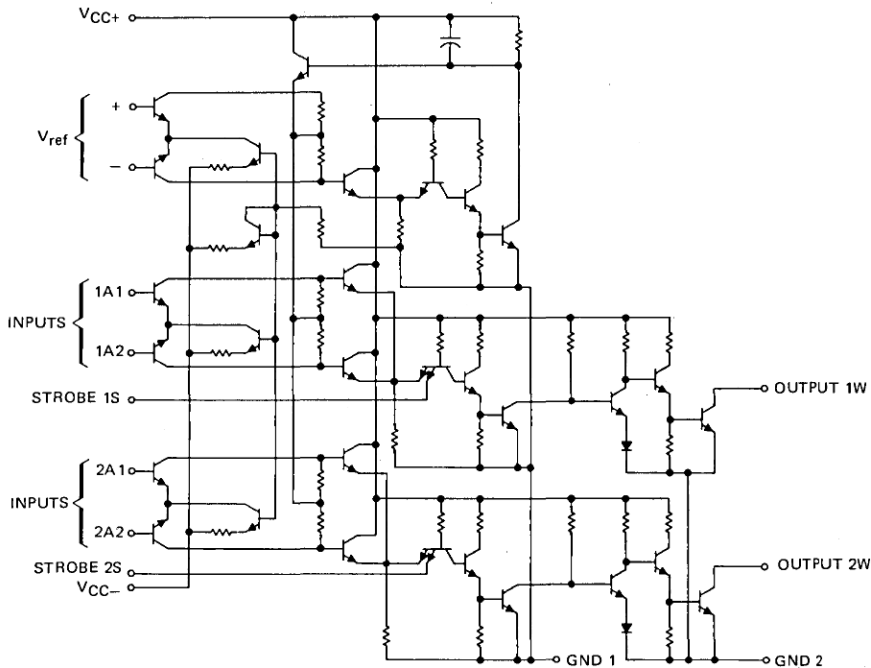
| PROPAGATION DELAY TIMES |            |           | TEST FIGURE | TEST CONDITIONS                            | MIN | TYP | MAX | UNIT |
|-------------------------|------------|-----------|-------------|--------------------------------------------|-----|-----|-----|------|
| SYMBOL                  | FROM INPUT | TO OUTPUT |             |                                            |     |     |     |      |
| $t_{PLH(D)}$            | A1–A2      | W         | 35          | $C_L = 15\text{ pF}$ , $R_L = 288\ \Omega$ |     | 25  |     | ns   |
| $t_{PHL(D)}$            |            |           |             |                                            |     | 25  |     |      |
| $t_{PLH(S)}$            | STROBE     | W         | 35          | $C_L = 15\text{ pF}$ , $R_L = 288\ \Omega$ |     | 25  |     | ns   |
| $t_{PHL(S)}$            |            |           |             |                                            |     | 15  |     |      |

typical recovery and cycle times,  $V_{CC+} = 5\text{ V}$ ,  $V_{CC-} = -5\text{ V}$ ,  $T_A = 25^\circ\text{C}$

| PARAMETER      | TEST CONDITIONS                                                                                                                                       | MIN | TYP | MAX | UNIT |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|------|
| $t_{orD}$      | Differential-input overload recovery time<br>(see Note 4)<br><i>Differential Input Pulse:</i><br>$V_{ID} = 2\text{ V}$ , $t_r = t_f = 20\text{ ns}$   |     | 20  |     | ns   |
| $t_{orC}$      | Common-mode-input overload recovery time<br>(see Note 5)<br><i>Common-Mode Input Pulse:</i><br>$V_{IC} = \pm 2\text{ V}$ , $t_r = t_f = 20\text{ ns}$ |     | 20  |     | ns   |
| $t_{cyc(min)}$ | Minimum cycle time                                                                                                                                    |     | 200 |     | ns   |

- NOTES: 4. Differential-input overload recovery time is the time necessary for the device to recover from the specified differential-input-overload signal prior to the strobe-enable signal.  
5. Common-mode-input overload recovery time is the time necessary for the device to recover from the specified common-mode-input overload signal prior to the strobe-enable signal.

schematic



**TYPES SN55234, SN75234**  
**DUAL SENSE AMPLIFIERS**

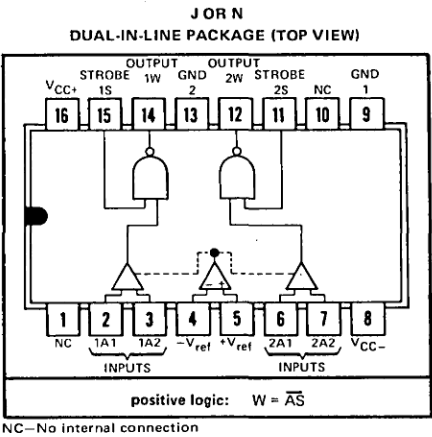
FUNCTION TABLE

| INPUTS |   | OUTPUT |
|--------|---|--------|
| A      | S | W      |
| H      | H | L      |
| L      | X | L      |
| X      | L | H      |

definition of logic levels

| INPUT | H                        | L                        | X          |
|-------|--------------------------|--------------------------|------------|
| A†    | $V_{ID} \geq V_{T \max}$ | $V_{ID} \leq V_{T \min}$ | Irrelevant |
| S     | $V_I \geq V_{IH \min}$   | $V_I \leq V_{IL \max}$   | Irrelevant |

†A is a differential voltage ( $V_{ID}$ ) between A1 and A2. For these circuits,  $V_{ID}$  is considered positive regardless of which terminal is positive with respect to the other.



electrical characteristics over recommended operating free-air temperature range,  $V_{CC+} = 5\text{ V}$ ,  $V_{CC-} = -5\text{ V}$  (unless otherwise noted)

| PARAMETER                                               | TEST FIGURE | TEST CONDITIONS                                                                                                                                        | MIN                                                            | TYP†      | MAX  | UNIT          |
|---------------------------------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|-----------|------|---------------|
| $V_T$ Differential-input threshold voltage              | 20          | $V_{ref} = 15\text{ mV}$                                                                                                                               | $T_A = 0^\circ\text{C to } 70^\circ\text{C}$                   | 11        | 15   | 19            |
|                                                         |             |                                                                                                                                                        | $T_A = -55^\circ\text{C to } 125^\circ\text{C}$ , SN55234 only | 10        | 15   | 20            |
|                                                         |             | $V_{ref} = 40\text{ mV}$                                                                                                                               | $T_A = 0^\circ\text{C to } 70^\circ\text{C}$                   | 36        | 40   | 44            |
|                                                         |             |                                                                                                                                                        | $T_A = -55^\circ\text{C to } 125^\circ\text{C}$ , SN55234 only | 35        | 40   | 45            |
| $V_{ICF}$ Common-mode input firing voltage (see Note 3) | none        | $V_{ref} = 40\text{ mV}$ , $V_{I(S)} = V_{IH}$<br>Common-mode input pulse:<br>$t_r \leq 15\text{ ns}$ , $t_f \leq 15\text{ ns}$ , $t_w = 50\text{ ns}$ |                                                                | $\pm 2.5$ |      | V             |
| $I_{IB}$ Differential-input bias current                | 2           | $V_{CC+} = 5.25\text{ V}$ , $V_{CC-} = -5.25\text{ V}$ , $V_{ID} = 0$                                                                                  | $T_A = -55^\circ\text{C to } 0^\circ\text{C}$ , SN55234 only   |           | 100  | $\mu\text{A}$ |
|                                                         |             |                                                                                                                                                        | $T_A = 0^\circ\text{C to } 70^\circ\text{C}$                   | 30        | 75   |               |
|                                                         |             |                                                                                                                                                        | $T_A = 70^\circ\text{C to } 125^\circ\text{C}$ , SN55234 only  |           | 75   |               |
| $I_{IO}$ Differential-input offset current              | 2           | $V_{CC+} = 5.25\text{ V}$ , $V_{CC-} = -5.25\text{ V}$ , $V_{ID} = 0$                                                                                  |                                                                | 0.5       |      | $\mu\text{A}$ |
| $V_{IH}$ High-level input voltage (strobe inputs)       | 21          |                                                                                                                                                        |                                                                | 2         |      | V             |
| $V_{IL}$ Low-level input voltage (strobe inputs)        | 21          |                                                                                                                                                        |                                                                |           | 0.8  | V             |
| $V_{OH}$ High-level output voltage                      | 21          | $V_{CC+} = 4.75\text{ V}$ , $V_{CC-} = -4.75\text{ V}$ , $I_{OH} = -400\text{ }\mu\text{A}$                                                            |                                                                | 2.4       | 4    | V             |
| $V_{OL}$ Low-level output voltage                       | 21          | $V_{CC+} = 4.75\text{ V}$ , $V_{CC-} = -4.75\text{ V}$ , $I_{OL} = 16\text{ mA}$                                                                       |                                                                | 0.25      | 0.4  | V             |
| $I_{IH}$ High-level input current (strobe inputs)       | 22          | $V_{CC+} = 5.25\text{ V}$ , $V_{CC-} = -5.25\text{ V}$ , $V_{IH} = 2.4\text{ V}$                                                                       |                                                                |           | 40   | $\mu\text{A}$ |
|                                                         |             | $V_{CC+} = 5.25\text{ V}$ , $V_{CC-} = -5.25\text{ V}$ , $V_{IH} = 5.25\text{ V}$                                                                      |                                                                |           | 1    | $\text{mA}$   |
| $I_{IL}$ Low-level input current (strobe inputs)        | 22          | $V_{CC+} = 5.25\text{ V}$ , $V_{CC-} = -5.25\text{ V}$ , $V_{IL} = 0.4\text{ V}$                                                                       |                                                                | -1        | -1.6 | $\text{mA}$   |
| $I_{OS}$ Short-circuit output current                   | 23          | $V_{CC+} = 5.25\text{ V}$ , $V_{CC-} = -5.25\text{ V}$ , $T_A = 25^\circ\text{C}$                                                                      | -2.1                                                           |           | -3.5 | $\text{mA}$   |
| $I_{CC+}$ Supply current from $V_{CC+}$                 | 6           | $V_{CC+} = 5.25\text{ V}$ , $V_{CC-} = -5.25\text{ V}$ , $T_A = 25^\circ\text{C}$                                                                      |                                                                | 25        | 40   | $\text{mA}$   |
| $I_{CC-}$ Supply current from $V_{CC-}$                 | 6           | $V_{CC+} = 5.25\text{ V}$ , $V_{CC-} = -5.25\text{ V}$ , $T_A = 25^\circ\text{C}$                                                                      | -15                                                            |           | -20  | $\text{mA}$   |

†All typical values are at  $V_{CC+} = 5\text{ V}$ ,  $V_{CC-} = -5\text{ V}$ ,  $T_A = 25^\circ\text{C}$ .

NOTE 3: Common-mode input firing voltage is the minimum common-mode voltage that will exceed the dynamic range of the input at the specified conditions and cause the logic output to switch. The common-mode input signal is applied when the strobe is high.

# TYPES SN55234, SN75234 DUAL SENSE AMPLIFIERS

switching characteristics,  $V_{CC+} = 5\text{ V}$ ,  $V_{CC-} = -5\text{ V}$ ,  $T_A = 25^\circ\text{C}$

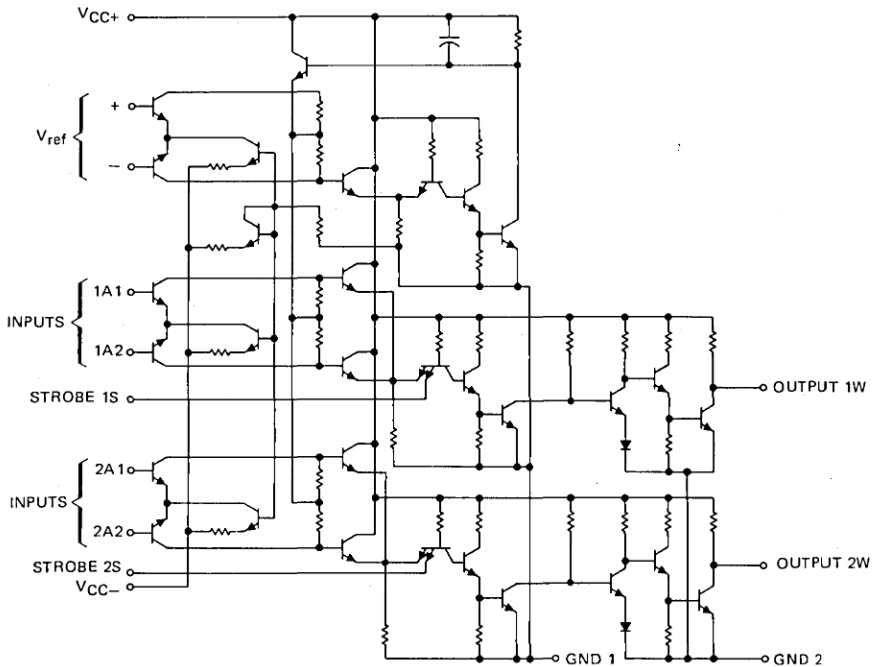
| PROPAGATION DELAY TIMES |            |           | TEST FIGURE | TEST CONDITIONS                            | MIN | TYP | MAX | UNIT |
|-------------------------|------------|-----------|-------------|--------------------------------------------|-----|-----|-----|------|
| SYMBOL                  | FROM INPUT | TO OUTPUT |             |                                            |     |     |     |      |
| $t_{PLH(D)}$            | A1–A2      | W         | 35          | $C_L = 15\text{ pF}$ , $R_L = 288\ \Omega$ | 25  | 25  | 40  | ns   |
| $t_{PLH(S)}$            |            |           |             |                                            |     |     |     |      |
| $t_{PLH(S)}$            | STROBE     | W         | 35          | $C_L = 15\text{ pF}$ , $R_L = 288\ \Omega$ | 15  | 15  | 30  | ns   |
| $t_{PLH(S)}$            |            |           |             |                                            |     |     |     |      |

typical recovery and cycle times,  $V_{CC+} = 5\text{ V}$ ,  $V_{CC-} = -5\text{ V}$ ,  $T_A = 25^\circ\text{C}$

| PARAMETER             |                                                           | TEST CONDITIONS                                                                                         | MIN | TYP | MAX | UNIT |
|-----------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----|-----|-----|------|
| t <sub>orD</sub>      | Differential-input overload recovery time<br>(see Note 4) | <i>Differential Input Pulse:</i><br>V <sub>ID</sub> = 2 V,      t <sub>r</sub> = t <sub>f</sub> = 20 ns |     | 20  |     | ns   |
| t <sub>orC</sub>      | Common-mode-input overload recovery time<br>(see Note 5)  | <i>Common-Mode Input Pulse:</i><br>V <sub>IC</sub> = ±2 V,      t <sub>r</sub> = t <sub>f</sub> = 20 ns |     | 20  |     | ns   |
| t <sub>cyc(min)</sub> | Minimum cycle time                                        |                                                                                                         |     | 200 |     | ns   |

- NOTES: 4. Differential-input overload recovery time is the time necessary for the device to recover from the specified differential-input-overload signal prior to the strobe-enable signal.  
5. Common-mode-input overload recovery time is the time necessary for the device to recover from the specified common-mode-input overload signal prior to the strobe-enable signal.

schematic



# TYPES SN55238, SN75238

## DUAL SENSE AMPLIFIERS WITH PREAMPLIFIER TEST POINTS

FUNCTION TABLE

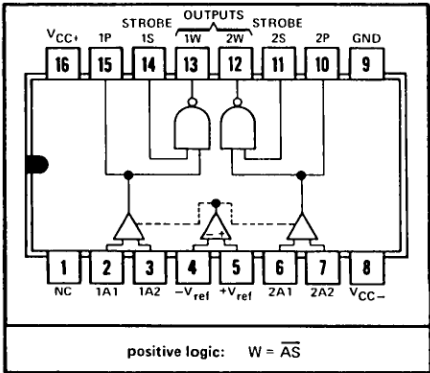
| INPUTS |   | OUTPUT |
|--------|---|--------|
| A      | S | W      |
| H      | H | L      |
| L      | X | H      |
| X      | L | H      |

definition of logic levels

| INPUT | H                       | L                       | X          |
|-------|-------------------------|-------------------------|------------|
| A†    | $V_{ID} \geq V_{T\max}$ | $V_{ID} \leq V_{T\min}$ | Irrelevant |
| S     | $V_I \geq V_{IH\min}$   | $V_I \leq V_{IL\max}$   | Irrelevant |

†A is a differential voltage ( $V_{ID}$ ) between A1 and A2. For these circuits,  $V_{ID}$  is considered positive regardless of which terminal is positive with respect to the other.

J OR N  
DUAL-IN-LINE PACKAGE (TOP VIEW)



NC—No internal connection

electrical characteristics over recommended operating free-air temperature range,  $V_{CC+} = 5\text{ V}$ ,  $V_{CC-} = -5\text{ V}$  (unless otherwise noted)

| PARAMETER                                               | TEST FIGURE | TEST CONDITIONS                                                                             | MIN  | TYP <sup>‡</sup> | MAX  | UNIT          |
|---------------------------------------------------------|-------------|---------------------------------------------------------------------------------------------|------|------------------|------|---------------|
| $V_T$ Differential-input threshold voltage              | 24          | $V_{ref} = 15\text{ mV}$                                                                    |      | 11               | 15   | mV            |
|                                                         |             | $T_A = 0^\circ\text{C to } 70^\circ\text{C}$                                                |      |                  |      |               |
|                                                         |             | $T_A = -55^\circ\text{C to } 125^\circ\text{C}$ , SN55238 only                              |      | 10               | 15   |               |
|                                                         |             | $V_{ref} = 40\text{ mV}$                                                                    |      | 36               | 40   |               |
| $V_{ICF}$ Common-mode input firing voltage (see Note 3) | none        | $V_{ref} = 40\text{ mV}$ , $V_I(S) = V_{IH}$                                                |      |                  |      | V             |
|                                                         |             | Common-mode input pulse:                                                                    |      |                  |      |               |
|                                                         |             | $t_r \leq 15\text{ ns}$ , $t_f \leq 15\text{ ns}$ , $t_w = 50\text{ ns}$                    |      | $\pm 2.5$        |      |               |
|                                                         |             | $V_{CC+} = 5.25\text{ V}$ , $T_A = -55^\circ\text{C to } 0^\circ\text{C}$ , SN55238 only    |      |                  | 100  |               |
| $I_{IB}$ Differential-input bias current                | 2           | $V_{CC-} = -5.25\text{ V}$ , $T_A = 0^\circ\text{C to } 70^\circ\text{C}$                   |      | 30               | 75   | $\mu\text{A}$ |
|                                                         |             | $V_{ID} = 0$ , $T_A = 70^\circ\text{C to } 125^\circ\text{C}$ , SN55238 only                |      |                  | 75   |               |
|                                                         |             | $V_{CC+} = 5.25\text{ V}$ , $V_{CC-} = -5.25\text{ V}$ , $V_{ID} = 0$                       |      | 0.5              |      |               |
| $I_{IO}$ Differential-input offset current              | 2           | $V_{CC+} = 5.25\text{ V}$ , $V_{CC-} = -5.25\text{ V}$ , $V_{ID} = 0$                       |      |                  |      | $\mu\text{A}$ |
| $V_{IH}$ High-level input voltage (strobe inputs)       | 25          |                                                                                             |      | 2                |      | V             |
| $V_{IL}$ Low-level input voltage (strobe inputs)        | 25          |                                                                                             |      |                  | 0.8  | V             |
| $V_{OH}$ High-level output voltage                      | 25          | $V_{CC+} = 4.75\text{ V}$ , $V_{CC-} = -4.75\text{ V}$ , $I_{OH} = -400\text{ }\mu\text{A}$ | 2.4  | 4                |      | V             |
| $V_{OL}$ Low-level output voltage                       | 25          | $V_{CC+} = 4.75\text{ V}$ , $V_{CC-} = -4.75\text{ V}$ , $I_{OL} = 16\text{ mA}$            |      | 0.25             | 0.4  | V             |
| $I_{IH}$ High-level input current (strobe inputs)       | 26          | $V_{CC+} = 5.25\text{ V}$ , $V_{CC-} = -5.25\text{ V}$ , $V_{IH} = 2.4\text{ V}$            |      |                  | 40   | $\mu\text{A}$ |
|                                                         |             | $V_{CC+} = 5.25\text{ V}$ , $V_{CC-} = -5.25\text{ V}$ , $V_{IH} = 5.25\text{ V}$           |      |                  | 1    | mA            |
| $I_{IL}$ Low-level input current (strobe inputs)        | 26          | $V_{CC+} = 5.25\text{ V}$ , $V_{CC-} = -5.25\text{ V}$ , $V_{IL} = 0.4\text{ V}$            | -1   | -1.6             |      | mA            |
| $I_{OS}$ Short-circuit output current                   | 27          | $V_{CC+} = 5.25\text{ V}$ , $V_{CC-} = -5.25\text{ V}$ , $T_A = 25^\circ\text{C}$           | -2.1 |                  | -3.5 | mA            |
| $I_{CC+}$ Supply current from $V_{CC+}$                 | 6           | $V_{CC+} = 5.25\text{ V}$ , $V_{CC-} = -5.25\text{ V}$ , $T_A = 25^\circ\text{C}$           |      | 25               | 40   | mA            |
| $I_{CC-}$ Supply current from $V_{CC-}$                 | 6           | $V_{CC+} = 5.25\text{ V}$ , $V_{CC-} = -5.25\text{ V}$ , $T_A = 25^\circ\text{C}$           | -15  |                  | -20  | mA            |

<sup>‡</sup>All typical values are at  $V_{CC+} = 5\text{ V}$ ,  $V_{CC-} = -5\text{ V}$ ,  $T_A = 25^\circ\text{C}$ .

NOTE 3: Common-mode input firing voltage is the minimum common-mode voltage that will exceed the dynamic range of the input at the specified conditions and cause the logic output to switch. The common-mode input signal is applied when the strobe is high.

# TYPES SN55238, SN75238

## DUAL SENSE AMPLIFIERS WITH PREAMPLIFIER TEST POINTS

switching characteristics,  $V_{CC+} = 5\text{ V}$ ,  $V_{CC-} = -5\text{ V}$ ,  $T_A = 25^\circ\text{C}$

| PROPAGATION DELAY TIMES |            |           | TEST<br>FIGURE | TEST CONDITIONS                            | MIN | TYP | MAX | UNIT |
|-------------------------|------------|-----------|----------------|--------------------------------------------|-----|-----|-----|------|
| SYMBOL                  | FROM INPUT | TO OUTPUT |                |                                            |     |     |     |      |
| $t_{PLH(D)}$            | A1–A2      | W         | 36             | $C_L = 15\text{ pF}$ , $R_L = 288\ \Omega$ | 25  | 25  | 40  | ns   |
| $t_{PHL(D)}$            |            |           |                |                                            |     |     |     |      |
| $t_{PLH(S)}$            | STROBE     | W         | 36             | $C_L = 15\text{ pF}$ , $R_L = 288\ \Omega$ | 25  | 15  | 30  | ns   |
| $t_{PHL(S)}$            |            |           |                |                                            |     |     |     |      |

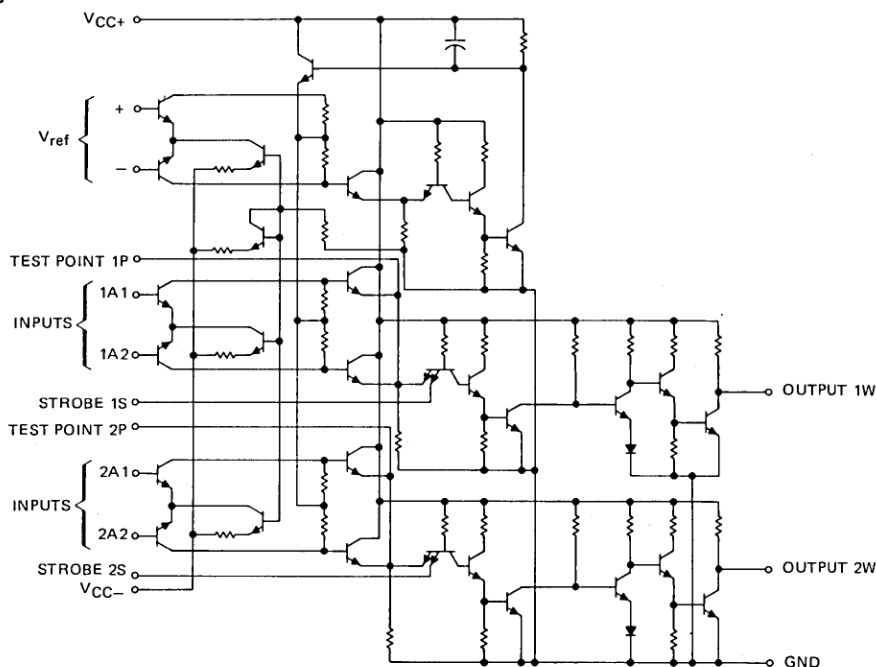
typical recovery and cycle times,  $V_{CC+} = 5\text{ V}$ ,  $V_{CC-} = -5\text{ V}$ ,  $T_A = 25^\circ\text{C}$

| PARAMETER      |                                                           | TEST CONDITIONS                                                                    | MIN | TYP | MAX | UNIT |
|----------------|-----------------------------------------------------------|------------------------------------------------------------------------------------|-----|-----|-----|------|
| $t_{orD}$      | Differential-input overload recovery time<br>(see Note 4) | Differential Input Pulse:<br>$V_{ID} = 2\text{ V}$ , $t_f = 20\text{ ns}$          |     | 20  |     | ns   |
| $t_{orC}$      | Common-mode-input overload recovery time<br>(see Note 5)  | Common-Mode Input Pulse:<br>$V_{IC} = \pm 2\text{ V}$ , $t_r = t_f = 20\text{ ns}$ |     | 20  |     | ns   |
| $t_{cyc(min)}$ | Minimum cycle time                                        |                                                                                    |     | 200 |     | ns   |

NOTES: 4. Differential-input overload recovery time is the time necessary for the device to recover from the specified differential-input overload signal prior to the strobe-enable signal.

5. Common-mode-input overload recovery time is the time necessary for the device to recover from the specified common-mode-input overload signal prior to the strobe-enable signal.

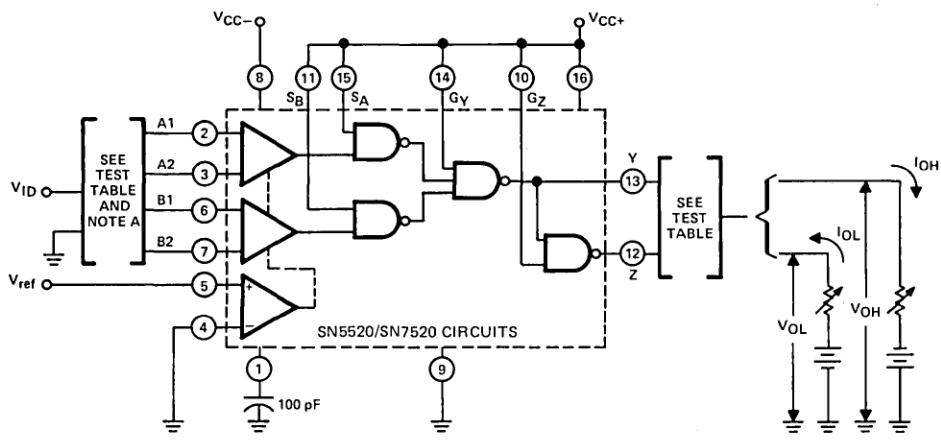
schematic



**SERIES 5520/7520**  
**SENSE AMPLIFIERS**

**PARAMETER MEASUREMENT INFORMATION**

d-c test circuits†



**TEST TABLE**

| CIRCUIT TYPE      | INPUTS         | V <sub>ref</sub> | V <sub>ID</sub> | OUTPUT Y       |                 |                 | OUTPUT Z       |                 |                 |
|-------------------|----------------|------------------|-----------------|----------------|-----------------|-----------------|----------------|-----------------|-----------------|
|                   |                |                  |                 | V <sub>O</sub> | I <sub>OH</sub> | I <sub>OL</sub> | V <sub>O</sub> | I <sub>OH</sub> | I <sub>OL</sub> |
| SN5520/<br>SN7520 | A1-A2 or B1-B2 | 15 mV            | <11 mV          | <0.4 V         |                 | 16 mA           | >2.4 V         | -400 μA         |                 |
|                   | A1-A2 or B1-B2 | 15 mV            | >19 mV          | >2.4 V         | -400 μA         |                 | <0.4 V         |                 | 16 mA           |
|                   | A1-A2 or B1-B2 | 40 mV            | <36 mV          | <0.4 V         |                 | 16 mA           | >2.4 V         | -400 μA         |                 |
|                   | A1-A2 or B1-B2 | 40 mV            | >44 mV          | >2.4 V         | -400 μA         |                 | <0.4 V         |                 | 16 mA           |

NOTE A: Each pair of differential inputs is tested separately with the other pair grounded.

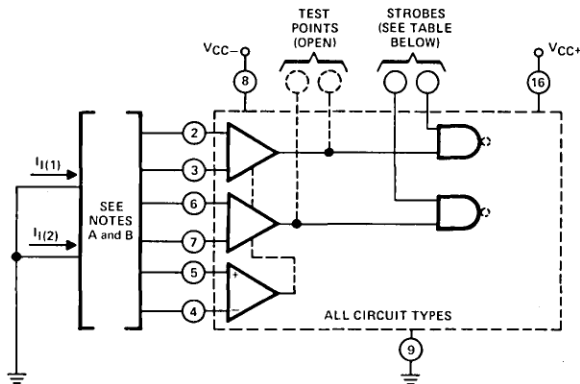
**FIGURE 1-V<sub>T</sub>**

† Arrows indicate actual direction of current flow. Current into a terminal is a positive value.

**SERIES 5520/7520**  
**SENSE AMPLIFIERS**

**PARAMETER MEASUREMENT INFORMATION**

d-c test circuits<sup>†</sup> (continued)



- NOTES: A. Each preamplifier is tested separately. Inputs not under test are grounded.  
B.  $I_{IB} = I_{I(1)}$  or  $I_{I(2)}$  (limit applies to each);  $I_{IO} = I_{I(1)} - I_{I(2)}$ ;  $I_{I(1)}$  and  $I_{I(2)}$  are the currents into the two inputs of the pair under test.

**PIN CONNECTIONS (OTHER THAN THOSE SHOWN ABOVE)**

| CIRCUIT TYPES                         | 100 pF to GND  | APPLY VCC+        | APPLY GND                   | LEAVE OPEN                  | OTHER           |
|---------------------------------------|----------------|-------------------|-----------------------------|-----------------------------|-----------------|
| SN5520, SN7520                        | $C_{ext}$<br>① | $G_Y, G_Z$<br>⑭ ⑩ | $S_A, S_B$<br>⑮ ⑪           | $Y, Z$<br>⑬ ⑫               |                 |
| SN5522, SN7522                        | $C_{ext}$<br>① | $G$<br>⑭          | $S_A, S_B, GND\ 2$<br>⑮ ⑪ ⑬ |                             | $R_L, Y$<br>⑩ ⑫ |
| SN5524, SN7524                        | $C_{ext}$<br>① |                   | $1S, 2S, GND\ 2$<br>⑮ ⑪ ⑬   | $1W, 2W$<br>⑭ ⑫             |                 |
| SN5528, SN7528                        | $C_{ext}$<br>① |                   | $1S, 2S$<br>⑭ ⑪             | $1P, 2P, 1W, 2W$<br>⑮ ⑩ ⑬ ⑫ |                 |
| SN55232, SN75232,<br>SN55234, SN75234 |                |                   | $1S, 2S, GND\ 2$<br>⑮ ⑪ ⑬   | $1W, 2W$<br>⑭ ⑫             |                 |
| SN55238, SN75238                      |                |                   | $1S, 2S$<br>⑭ ⑪             | $1P, 2P, 1W, 2W$<br>⑮ ⑩ ⑬ ⑫ |                 |

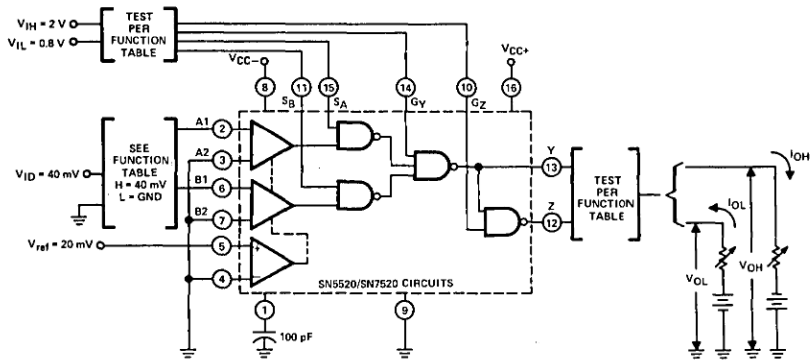
**FIGURE 2— $I_{IB}, I_{IO}$**

<sup>†</sup>Arrows indicate actual direction of current flow. Current into a terminal is a positive value.

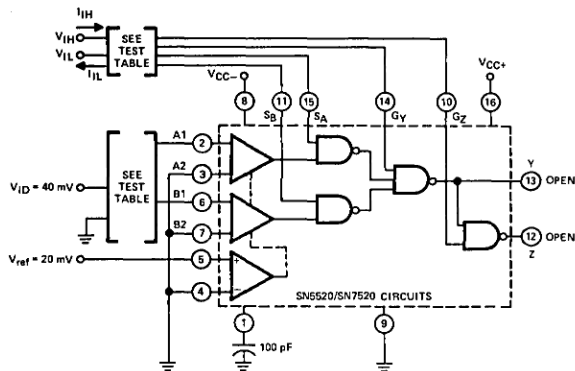
**SERIES 5520/7520**  
**SENSE AMPLIFIERS**

**PARAMETER MEASUREMENT INFORMATION**

d-c test circuits† (continued)



**FIGURE 3— $V_{IH}$ ,  $V_{IL}$ ,  $V_{OH}$ ,  $V_{OL}$**



**TEST TABLE**

| TEST                              | INPUT<br>A1 | INPUT<br>B1 | STROBE<br>S <sub>A</sub> | STROBE<br>S <sub>B</sub> | GATE<br>G <sub>Y</sub> | GATE<br>G <sub>Z</sub> |
|-----------------------------------|-------------|-------------|--------------------------|--------------------------|------------------------|------------------------|
| $I_{IH}$ at STROBE S <sub>A</sub> | GND         | GND         | $V_{IH}$                 | $V_{IL}$                 | $V_{IL}$               | $V_{IL}$               |
| $I_{IH}$ at STROBE S <sub>B</sub> | GND         | GND         | $V_{IL}$                 | $V_{IH}$                 | $V_{IL}$               | $V_{IL}$               |
| $I_{IH}$ at GATE G <sub>Y</sub>   | $V_{ID}$    | $V_{ID}$    | $V_{IH}$                 | $V_{IH}$                 | $V_{IH}$               | $V_{IL}$               |
| $I_{IH}$ at GATE G <sub>Z</sub>   | GND         | GND         | $V_{IL}$                 | $V_{IL}$                 | $V_{IH}$               | $V_{IH}$               |
| $I_{IL}$ at STROBE S <sub>A</sub> | $V_{ID}$    | GND         | $V_{IL}$                 | $V_{IL}$                 | $V_{IL}$               | $V_{IL}$               |
| $I_{IL}$ at STROBE S <sub>B</sub> | GND         | $V_{ID}$    | $V_{IL}$                 | $V_{IL}$                 | $V_{IL}$               | $V_{IL}$               |
| $I_{IL}$ at GATE G <sub>Y</sub>   | GND         | GND         | $V_{IL}$                 | $V_{IL}$                 | $V_{IL}$               | $V_{IL}$               |
| $I_{IL}$ at GATE G <sub>Z</sub>   | GND         | GND         | $V_{IL}$                 | $V_{IL}$                 | $V_{IL}$               | $V_{IL}$               |

**FIGURE 4— $I_{IH}$ ,  $I_{IL}$**

† Arrows indicate actual direction of current flow. Current into a terminal is a positive value.



# SERIES 5520/7520 SENSE AMPLIFIERS

## PARAMETER MEASUREMENT INFORMATION

d-c test circuits† (continued)

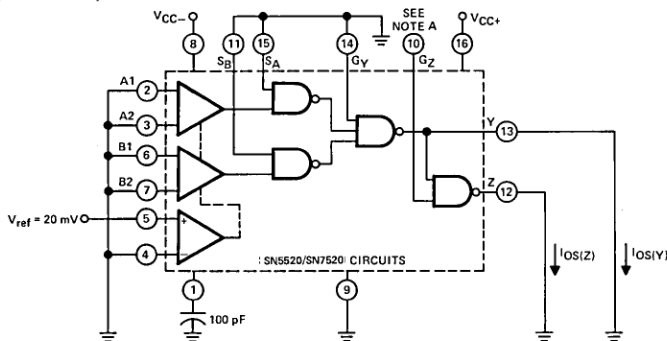
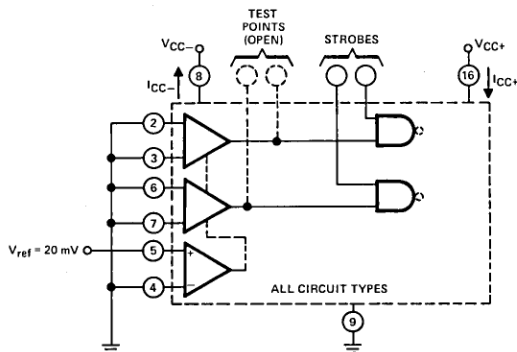


FIGURE 5— $I_{OS}$

NOTE A: When testing  $I_{OS}(Y)$ , Pin 10 is open; when testing  $I_{OS}(Z)$ , Pin 10 is grounded.



PIN CONNECTIONS (OTHER THAN THOSE SHOWN ABOVE)

| CIRCUIT TYPES                         | 100 pF to GND    | APPLY GND                                   | LEAVE OPEN                              |
|---------------------------------------|------------------|---------------------------------------------|-----------------------------------------|
| SN5520, SN7520                        | $C_{ext}$<br>(1) | $G_Y, G_Z, S_A, S_B$<br>(14) (10) (15) (11) | $Y, Z$<br>(13) (12)                     |
| SN5522, SN7522                        | $C_{ext}$<br>(1) | $G, S_A, S_B, GND 2$<br>(14) (15) (11) (13) | $R_L, Y$<br>(10) (12)                   |
| SN5524, SN7524                        | $C_{ext}$<br>(1) | $1S, 2S, GND 2$<br>(15) (11) (13)           | $1W, 2W$<br>(14) (12)                   |
| SN5528, SN7528                        | $C_{ext}$<br>(1) | $1S, 2S$<br>(14) (11)                       | $1P, 2P, 1W, 2W$<br>(15) (10) (13) (12) |
| SN55232, SN75232,<br>SN55234, SN75234 |                  | $1S, 2S, GND 2$<br>(15) (11) (13)           | $1W, 2W$<br>(14) (12)                   |
| SN55238, SN75238                      |                  | $1S, 2S$<br>(14) (11)                       | $1P, 2P, 1W, 2W$<br>(15) (10) (13) (12) |

FIGURE 6— $I_{CC+}, I_{CC-}$

† Arrows indicate actual direction of current flow. Current into a terminal is a positive value.

## 4

## d-c test circuits† (continued)



**NOTE A:** Each pair of differential inputs is tested separately with the other pair grounded.

<sup>†</sup>Arrows indicate actual direction of current flow. Current into a terminal is a positive value.

# SERIES 5520/7520 SENSE AMPLIFIERS

## PARAMETER MEASUREMENT INFORMATION

d-c test circuits<sup>†</sup> (continued)

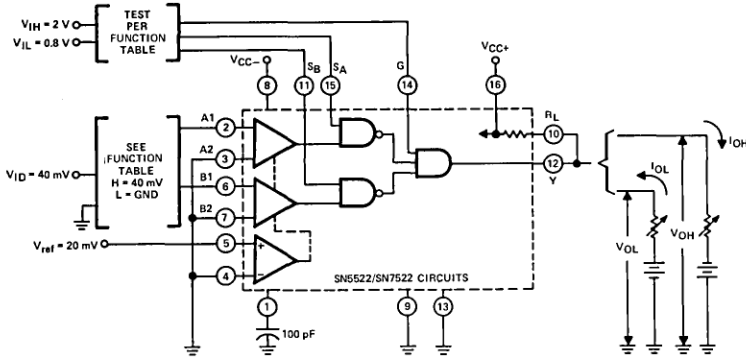
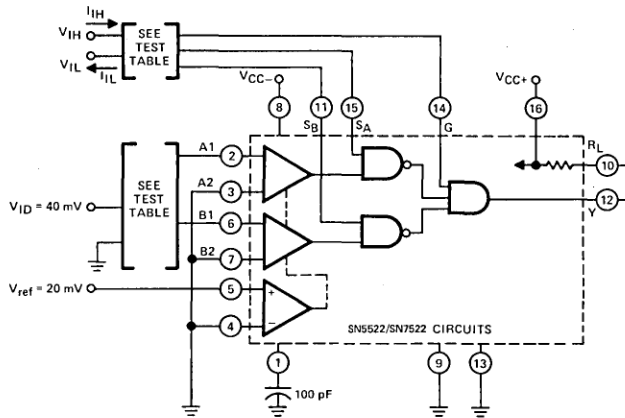


FIGURE 8— $V_{IH}$ ,  $V_{IL}$ ,  $V_{OH}$ ,  $V_{OL}$



TEST TABLE

| TEST                              | INPUT<br>A1 | INPUT<br>B1 | STROBE<br>S <sub>A</sub> | STROBE<br>S <sub>B</sub> | GATE<br>G |
|-----------------------------------|-------------|-------------|--------------------------|--------------------------|-----------|
| $I_{IH}$ at STROBE S <sub>A</sub> | GND         | GND         | $V_{IH}$                 | $V_{IL}$                 | $V_{IH}$  |
| $I_{IH}$ at STROBE S <sub>B</sub> | GND         | GND         | $V_{IL}$                 | $V_{IH}$                 | $V_{IH}$  |
| $I_{IH}$ at GATE                  | $V_{ID}$    | $V_{ID}$    | $V_{IH}$                 | $V_{IH}$                 | $V_{IH}$  |
| $I_{IL}$ at STROBE S <sub>A</sub> | $V_{ID}$    | GND         | $V_{IL}$                 | $V_{IL}$                 | $V_{IH}$  |
| $I_{IL}$ at STROBE S <sub>B</sub> | GND         | $V_{ID}$    | $V_{IL}$                 | $V_{IL}$                 | $V_{IH}$  |
| $I_{IL}$ at GATE                  | GND         | GND         | $V_{IL}$                 | $V_{IL}$                 | $V_{IL}$  |

FIGURE 9— $I_{IH}$ ,  $I_{IL}$

<sup>†</sup> Arrows indicate actual direction of current flow. Current into a terminal is a positive value.

## 4

## d-c test circuits† (continued)

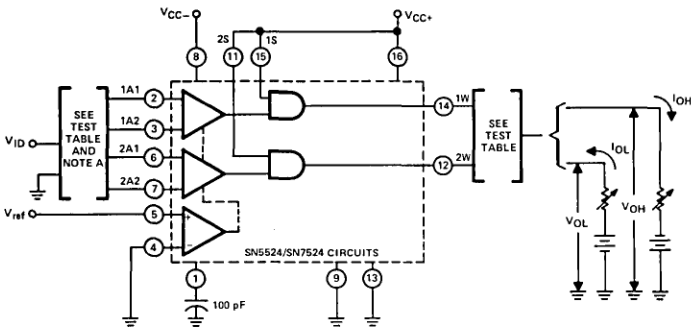


<sup>†</sup> Arrows indicate actual direction of current flow. Current into a terminal is a positive value.

# SERIES 5520/7520 SENSE AMPLIFIERS

## PARAMETER MEASUREMENT INFORMATION

d-c test circuits† (continued)



TEST TABLE

| CIRCUIT TYPE      | INPUTS | V <sub>ref</sub> | V <sub>ID</sub> | OUTPUT         |                 |                 |
|-------------------|--------|------------------|-----------------|----------------|-----------------|-----------------|
|                   |        |                  |                 | V <sub>O</sub> | I <sub>OH</sub> | I <sub>OL</sub> |
| SN5524/<br>SN7524 | A1-A2  | 15 mV            | ≤11 mV          | ≤0.4 V         |                 | 16 mA           |
|                   | A1-A2  | 15 mV            | ≥19 mV          | ≥2.4 V         | -400 μA         |                 |
|                   | A1-A2  | 40 mV            | ≤36 mV          | ≤0.4 V         |                 | 16 mA           |
|                   | A1-A2  | 40 mV            | ≥44 mV          | ≥2.4 V         | -400 μA         |                 |

NOTE A: Each pair of differential inputs is tested separately with its corresponding output.

FIGURE 12-V<sub>T</sub>

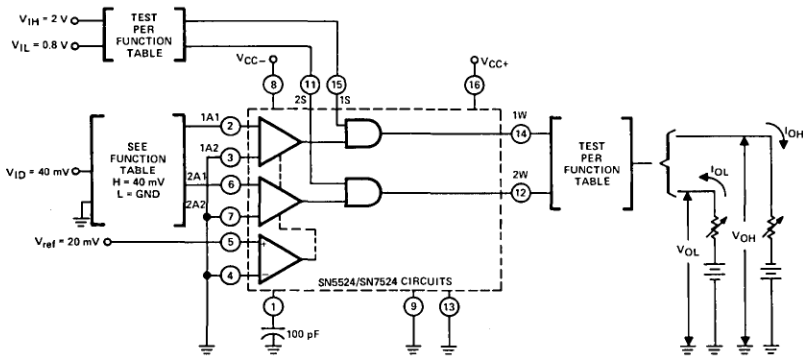


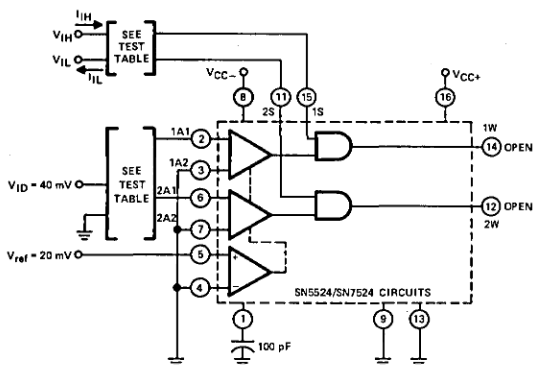
FIGURE 13-V<sub>IH</sub>, V<sub>IL</sub>, V<sub>OH</sub>, V<sub>OL</sub>

†Arrows indicate actual direction of current flow. Current into a terminal is a positive value.

# SERIES 5520/7520 SENSE AMPLIFIERS

## PARAMETER MEASUREMENT INFORMATION

d-c test circuits<sup>†</sup> (continued)



TEST TABLE

| TEST                  | INPUT 1A1 | INPUT 2A1 | STROBE 1S | STROBE 2S |
|-----------------------|-----------|-----------|-----------|-----------|
| $I_{IH}$ at STROBE 1S | GND       | GND       | $V_{IH}$  | $V_{IL}$  |
| $I_{IH}$ at STROBE 2S | GND       | GND       | $V_{IL}$  | $V_{IH}$  |
| $I_{IL}$ at STROBE 1S | $V_{ID}$  | GND       | $V_{IL}$  | $V_{IL}$  |
| $I_{IL}$ at STROBE 2S | GND       | $V_{ID}$  | $V_{IL}$  | $V_{IL}$  |

FIGURE 14— $I_{IH}$ ,  $I_{IL}$

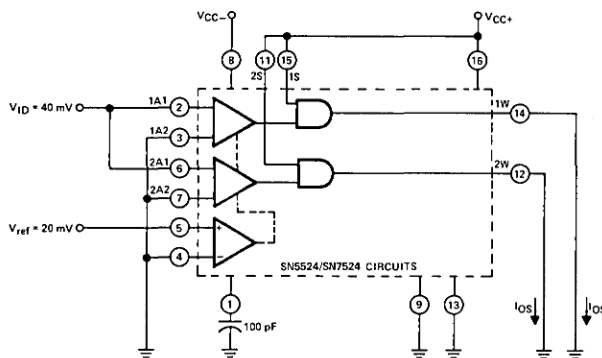


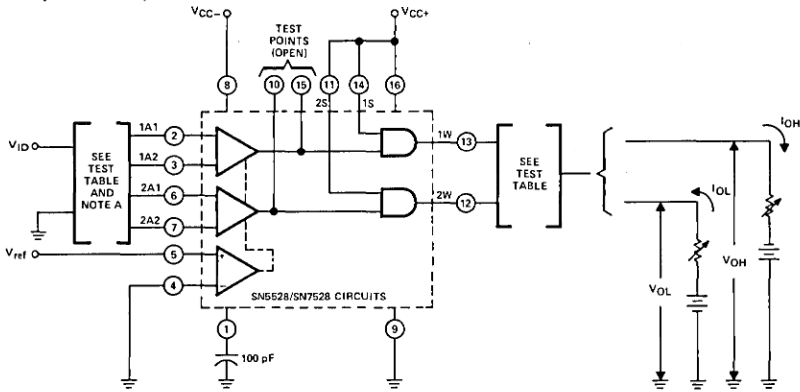
FIGURE 15— $I_{OS}$

<sup>†</sup> Arrows indicate actual direction of current flow. Current into a terminal is a positive value.

# SERIES 5520/7520 SENSE AMPLIFIERS

## PARAMETER MEASUREMENT INFORMATION

d-c test circuits† (continued)



TEST TABLE

| CIRCUIT TYPE      | INPUTS | $V_{ref}$ | $V_{ID}$     | OUTPUT       |              |          |
|-------------------|--------|-----------|--------------|--------------|--------------|----------|
|                   |        |           |              | $V_O$        | $I_{OH}$     | $I_{OL}$ |
| SN5528/<br>SN7528 | A1-A2  | 15 mV     | $\leq 11$ mV | $\leq 0.4$ V |              | 16 mA    |
|                   | A1-A2  | 15 mV     | $\geq 19$ mV | $\geq 2.4$ V | $-400 \mu A$ |          |
|                   | A1-A2  | 40 mV     | $\leq 36$ mV | $\leq 0.4$ V |              | 16 mA    |
|                   | A1-A2  | 40 mV     | $\geq 44$ mV | $\geq 2.4$ V | $-400 \mu A$ |          |

NOTE A: Each pair of inputs is tested separately with its corresponding output.

FIGURE 16— $V_T$

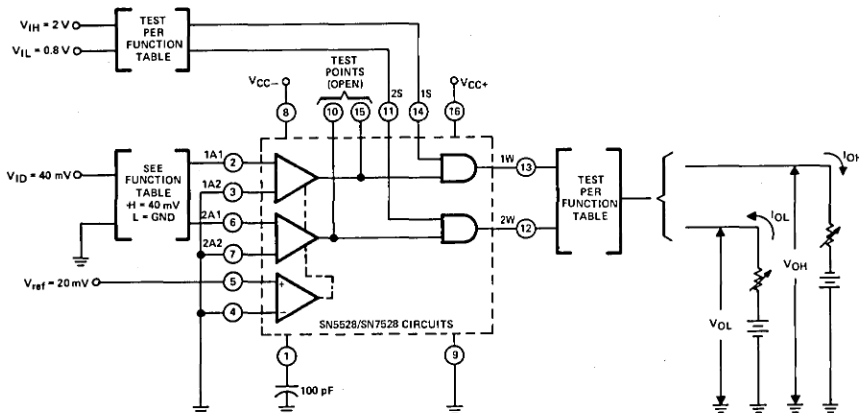


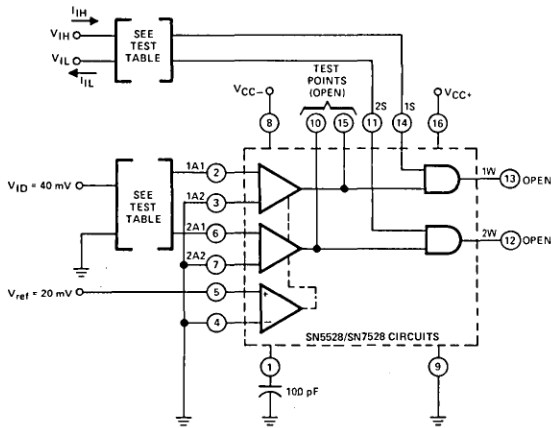
FIGURE 17— $V_{IH}$ ,  $V_{IL}$ ,  $V_{OH}$ ,  $V_{OL}$

† Arrows indicate actual direction of current flow. Current into a terminal is a positive value.

# SERIES 5520/7520 SENSE AMPLIFIERS

## PARAMETER MEASUREMENT INFORMATION

d-c test circuits† (continued)



TEST TABLE

| TEST                  | INPUT 1A1 | INPUT 2A1 | STROBE 1S | STROBE 2S |
|-----------------------|-----------|-----------|-----------|-----------|
| $I_{IH}$ at STROBE 1S | GND       | GND       | $V_{IH}$  | $V_{IL}$  |
| $I_{IH}$ at STROBE 2S | GND       | GND       | $V_{IL}$  | $V_{IH}$  |
| $I_{IL}$ at STROBE 1S | $V_{ID}$  | GND       | $V_{IL}$  | $V_{IL}$  |
| $I_{IL}$ at STROBE 2S | GND       | $V_{ID}$  | $V_{IL}$  | $V_{IL}$  |

FIGURE 18 — $I_{IH}$ ,  $I_{IL}$

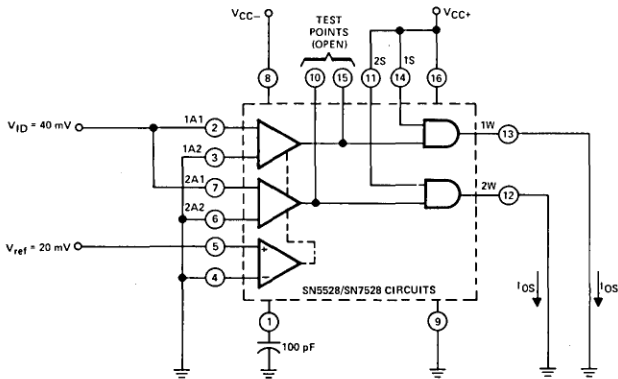


FIGURE 19— $I_{OS}$

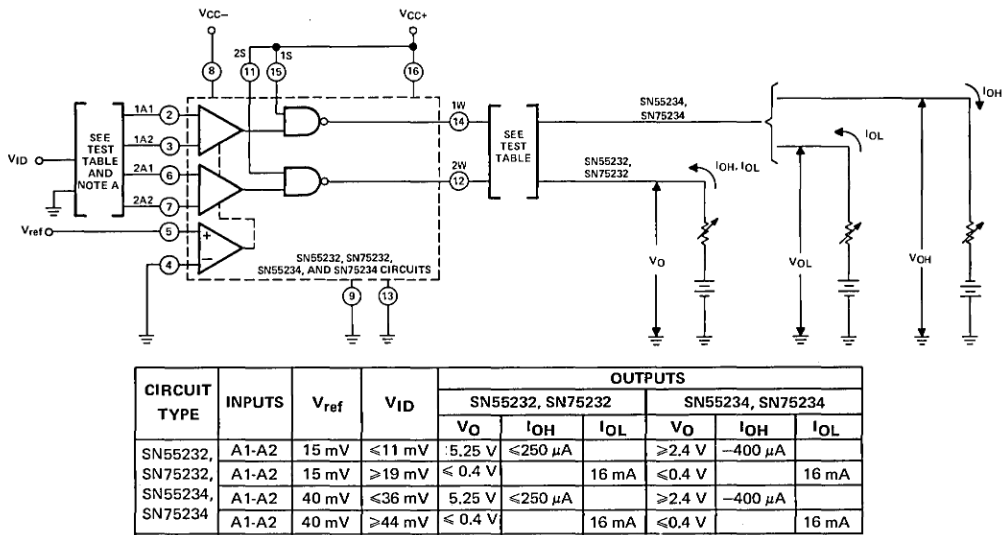
† Arrows indicate actual direction of current flow. Current into a terminal is a positive value.



# SERIES 5520/7520 SENSE AMPLIFIERS

## PARAMETER MEASUREMENT INFORMATION

d-c test circuits† (continued)



NOTE A: Each pair of differential inputs is tested separately with its corresponding output.

FIGURE 20— $V_T$

4

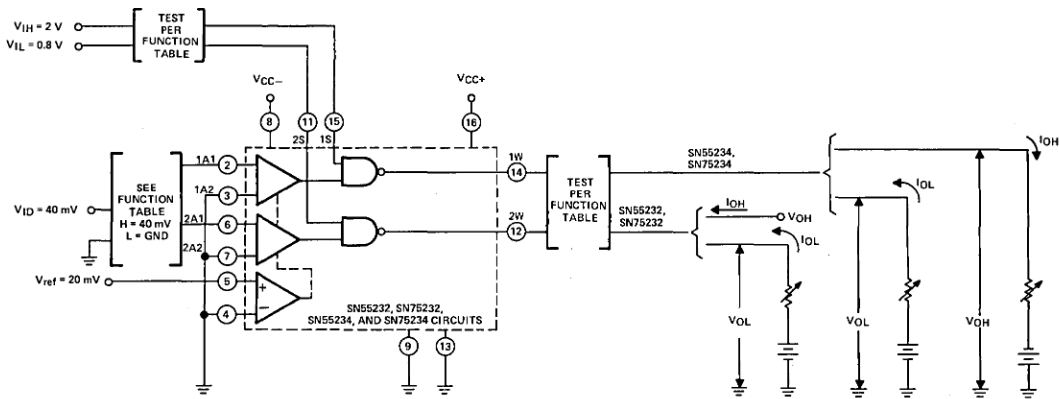


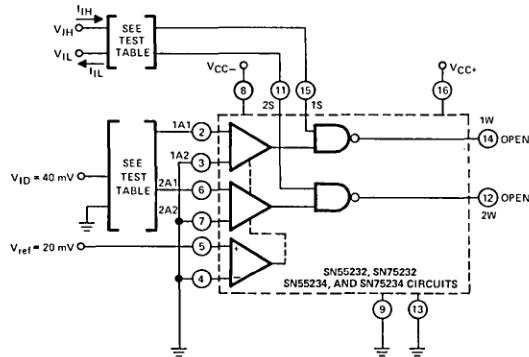
FIGURE 21— $V_{IH}$ ,  $V_{IL}$ ,  $V_{OL}$ ,  $V_{OH}$

† Arrows indicate actual direction of current flow. Current into a terminal is a positive value.

**SERIES 5520/7520**  
**SENSE AMPLIFIERS**

**PARAMETER MEASUREMENT INFORMATION**

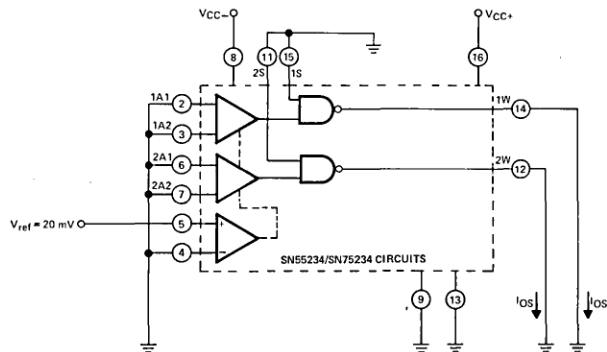
d-c test circuits† (continued)



**TEST TABLE**

| TEST                  | INPUT 1A1 | INPUT 2A1 | STROBE 1S | STROBE 2S |
|-----------------------|-----------|-----------|-----------|-----------|
| $I_{IH}$ at STROBE 1S | GND       | GND       | $V_{IH}$  | $V_{IL}$  |
| $I_{IH}$ at STROBE 2S | GND       | GND       | $V_{IL}$  | $V_{IH}$  |
| $I_{IL}$ at STROBE 1S | $V_{ID}$  | GND       | $V_{IL}$  | $V_{IL}$  |
| $I_{IL}$ at STROBE 2S | GND       | $V_{ID}$  | $V_{IL}$  | $V_{IL}$  |

**FIGURE 22— $I_{IH}$ ,  $I_{IL}$**



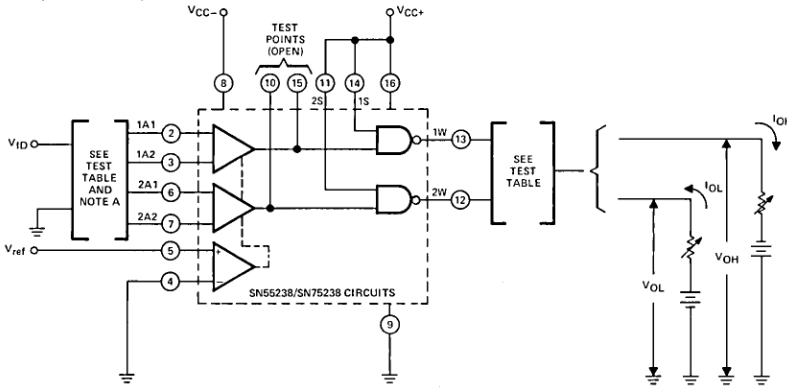
**FIGURE 23— $I_{OS}$**

† Arrows indicate actual direction of current flow. Current into a terminal is a positive value.

# SERIES 5520/7520 SENSE AMPLIFIERS

## PARAMETER MEASUREMENT INFORMATION

d-c test circuits† (continued)



TEST TABLE

| CIRCUIT TYPE        | INPUTS | $V_{ref}$ | $V_{ID}$     | OUTPUT       |              |          |
|---------------------|--------|-----------|--------------|--------------|--------------|----------|
|                     |        |           |              | $V_O$        | $I_{OH}$     | $I_{OL}$ |
| SN55238/<br>SN75238 | A1-A2  | 15 mV     | $\leq 11$ mV | $\geq 2.4$ V | $-400 \mu A$ |          |
|                     | A1-A2  | 15 mV     | $\geq 19$ mV | $\leq 0.4$ V |              | 16 mA    |
|                     | A1-A2  | 40 mV     | $\leq 36$ mV | $\geq 2.4$ V | $-400 \mu A$ |          |
|                     | A1-A2  | 40 mV     | $\geq 44$ mV | $\leq 0.4$ V |              | 16 mA    |

NOTE A: Each pair of inputs is tested separately with its corresponding output.

FIGURE 24- $V_T$

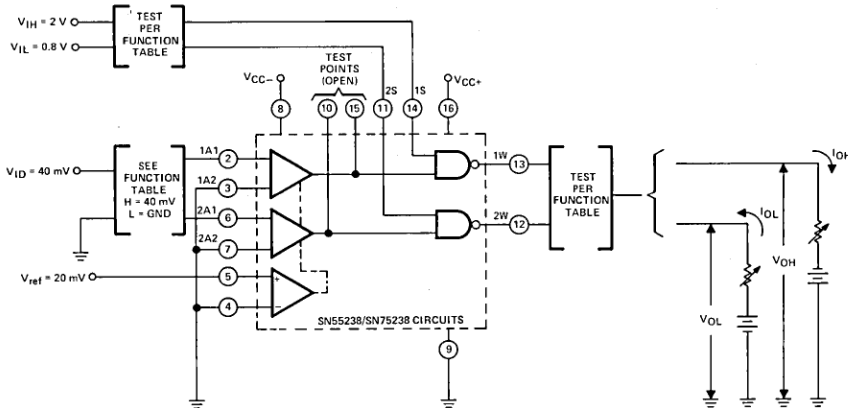


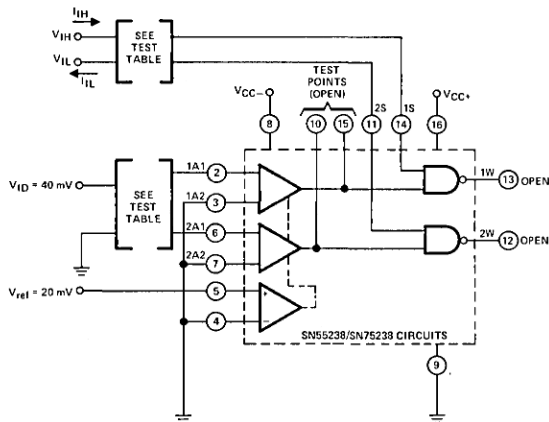
FIGURE 25- $V_{IH}$ ,  $V_{IL}$ ,  $V_{OH}$ ,  $V_{OL}$

† Arrows indicate actual direction of current flow. Current into a terminal is a positive value.

# SERIES 5520/7520 SENSE AMPLIFIERS

## PARAMETER MEASUREMENT INFORMATION

d-c test circuits<sup>†</sup> (continued)



TEST TABLE

| TEST                  | INPUT 1A1 | INPUT 2A1 | STROBE 1S | STROBE 2S |
|-----------------------|-----------|-----------|-----------|-----------|
| $I_{IH}$ at STROBE 1S | GND       | GND       | $V_{IH}$  | $V_{IL}$  |
| $I_{IH}$ at STROBE 2S | GND       | GND       | $V_{IL}$  | $V_{IH}$  |
| $I_{IL}$ at STROBE 1S | $V_{ID}$  | GND       | $V_{IL}$  | $V_{IL}$  |
| $I_{IL}$ at STROBE 2S | GND       | $V_{ID}$  | $V_{IL}$  | $V_{IL}$  |

FIGURE 26— $I_{IH}$ ,  $I_{IL}$

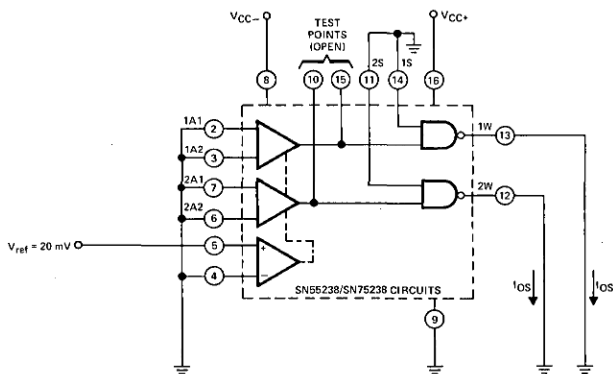
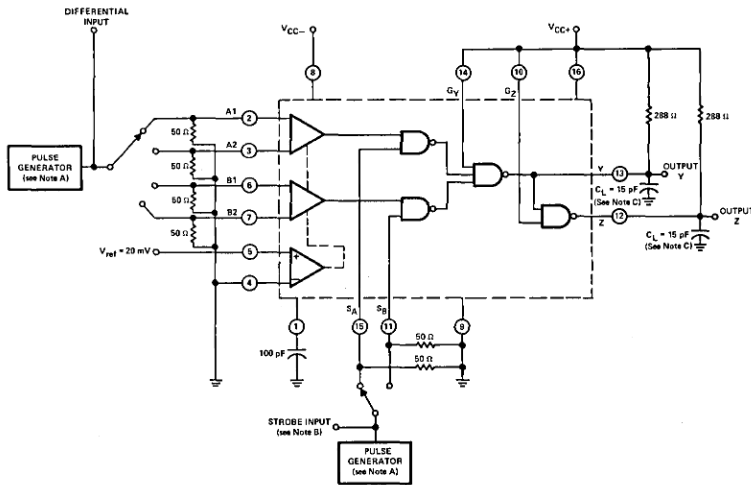


FIGURE 27— $I_{OS}$

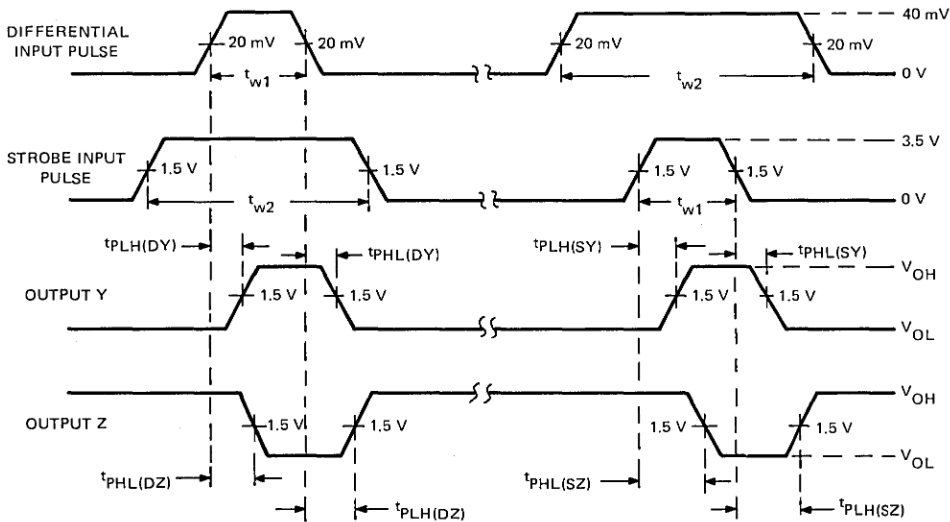
<sup>†</sup>Arrows indicate actual direction of current flow. Current into a terminal is a positive value.

## PARAMETER MEASUREMENT INFORMATION

### switching characteristics



TEST CIRCUIT



VOLTAGE WAVEFORMS

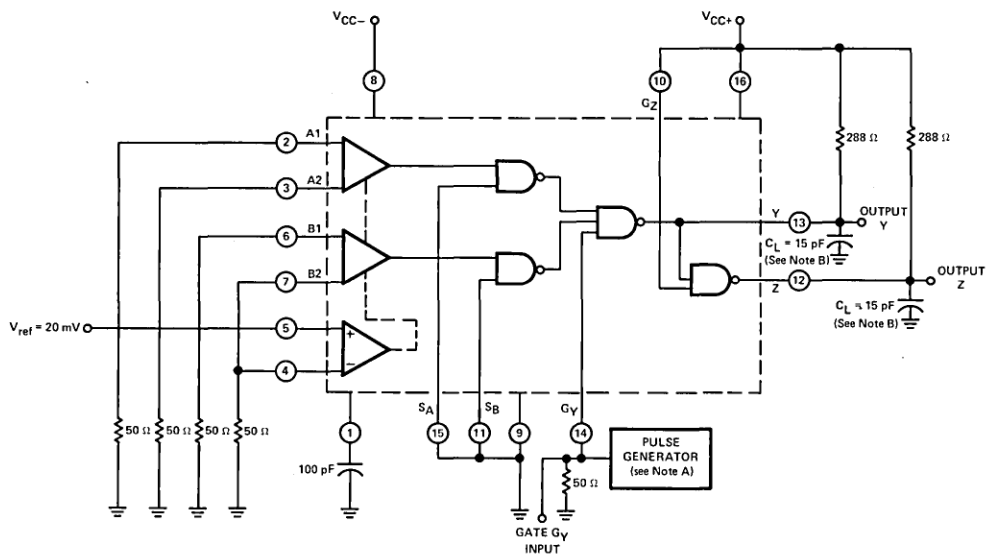
- NOTES: A. The pulse generators have the following characteristics:  $Z_{out} = 50 \Omega$ ,  $t_r = 15 \pm 5 \text{ ns}$ ,  $t_f = 15 \pm 5 \text{ ns}$ ,  $t_{w1} = 100 \text{ ns}$ ,  $t_{w2} = 300 \text{ ns}$ , and  $PRR = 1 \text{ MHz}$ .
- B. The strobe input pulse is applied to Strobe  $S_A$  when inputs A1-A2 are being tested and to Strobe  $S_B$  when inputs B1-B2 are being tested.
- C.  $C_L$  includes probe and jig capacitance.

FIGURE 28—SN5520/SN7520 PROPAGATION DELAY TIMES FROM DIFFERENTIAL AND STROBE INPUTS

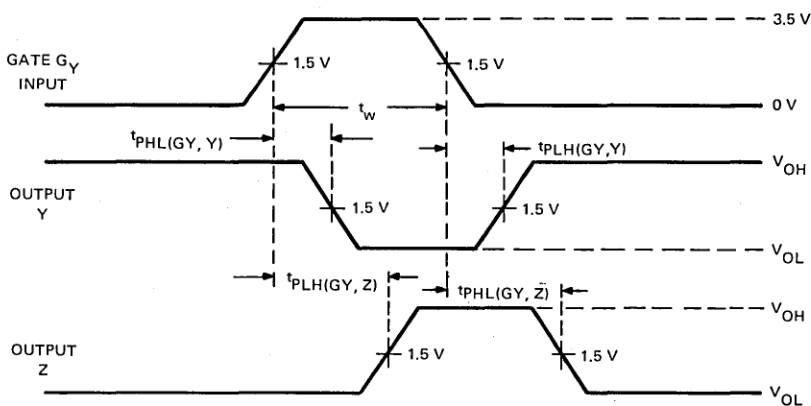
# SERIES 5520/7520 SENSE AMPLIFIERS

## PARAMETER MEASUREMENT INFORMATION

switching characteristics (continued)



TEST CIRCUIT



VOLTAGE WAVEFORMS

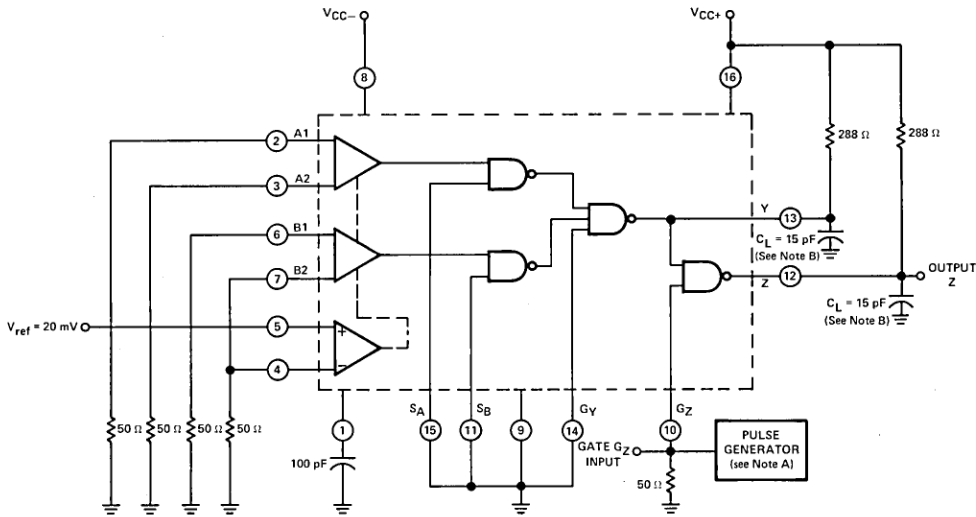
NOTES: A. The pulse generator has the following characteristics:  $Z_{out} = 50 \Omega$ ,  $t_r = 15 \pm 5 \text{ ns}$ ,  $t_f = 15 \pm 5 \text{ ns}$ ,  $t_w = 100 \text{ ns}$ , and  $PRR = 1 \text{ MHz}$ .  
B.  $C_L$  includes probe and jig capacitance.

FIGURE 29—SN5520/SN7520 PROPAGATION DELAY TIMES FROM GATE  $G_Y$

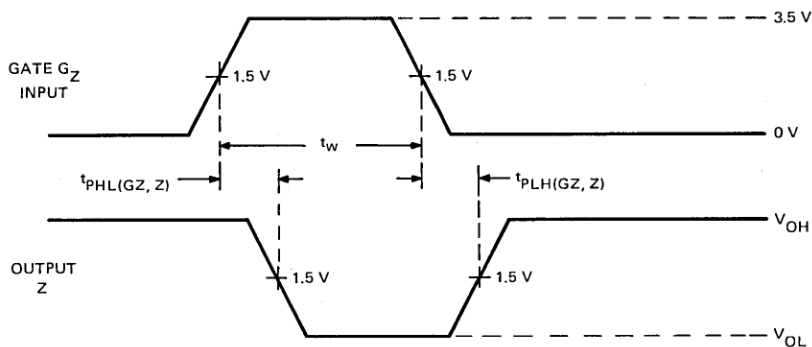
# SERIES 5520/7520 SENSE AMPLIFIERS

## PARAMETER MEASUREMENT INFORMATION

switching characteristics (continued)



TEST CIRCUIT



VOLTAGE WAVEFORMS

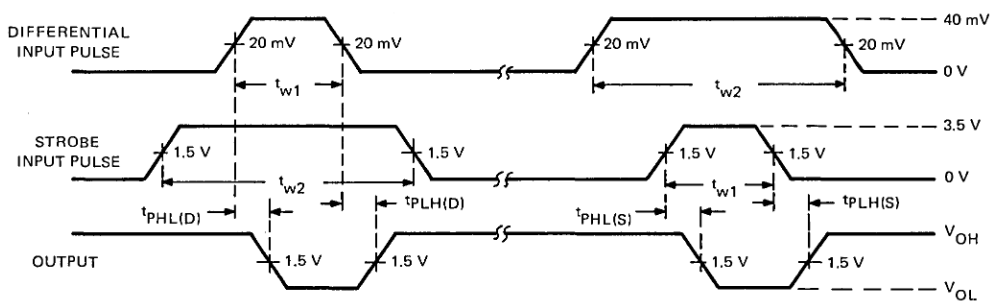
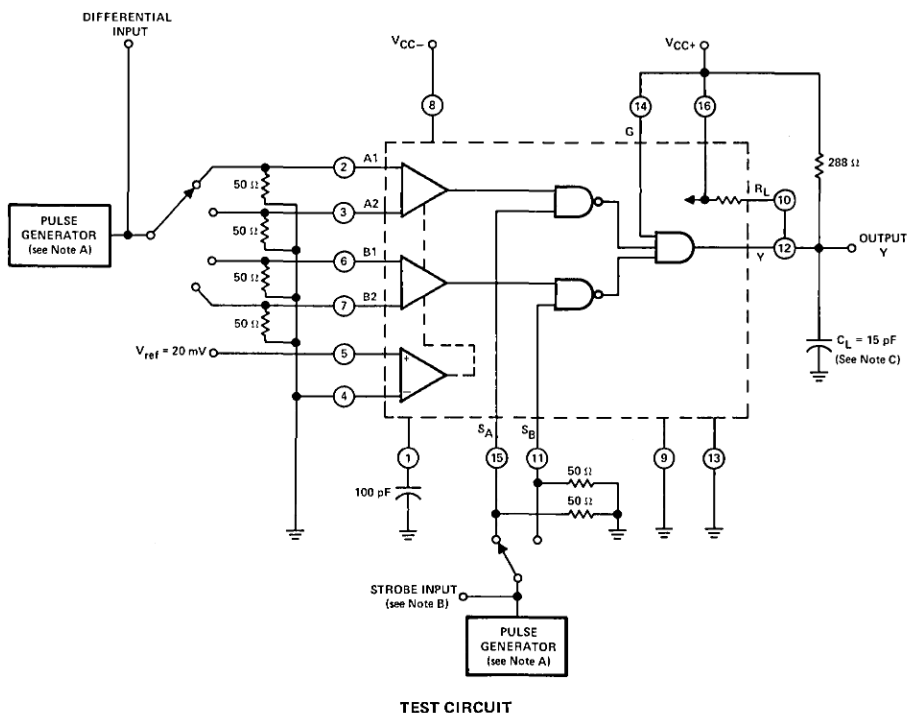
NOTES: A. The pulse generator has the following characteristics:  $Z_{out} = 50 \Omega$ ,  $t_r = 15 \pm 5 \text{ ns}$ ,  $t_f = 15 \pm 5 \text{ ns}$ ,  $t_w = 100 \text{ ns}$ , and  $PRR = 1 \text{ MHz}$ .  
B.  $C_L$  includes probe and jig capacitance.

FIGURE 30—SN5520/SN7520 PROPAGATION DELAY TIMES FROM GATE  $G_Z$

# SERIES 5520/7520 SENSE AMPLIFIERS

## PARAMETER MEASUREMENT INFORMATION

switching characteristics (continued)



- NOTES: A. The pulse generators have the following characteristics:  $Z_{out} = 50 \Omega$ ,  $t_r = t_f = 15 \pm 5$  ns,  $t_{w1} = 100$  ns,  $t_{w2} = 300$  ns,  $PRR = 1$  MHz.  
B. The strobe input pulse is applied to Strobe  $S_A$  when testing inputs A1-A2 and to Strobe  $S_B$  when testing inputs B1-B2.  
C.  $C_L$  includes probe and jig capacitance.

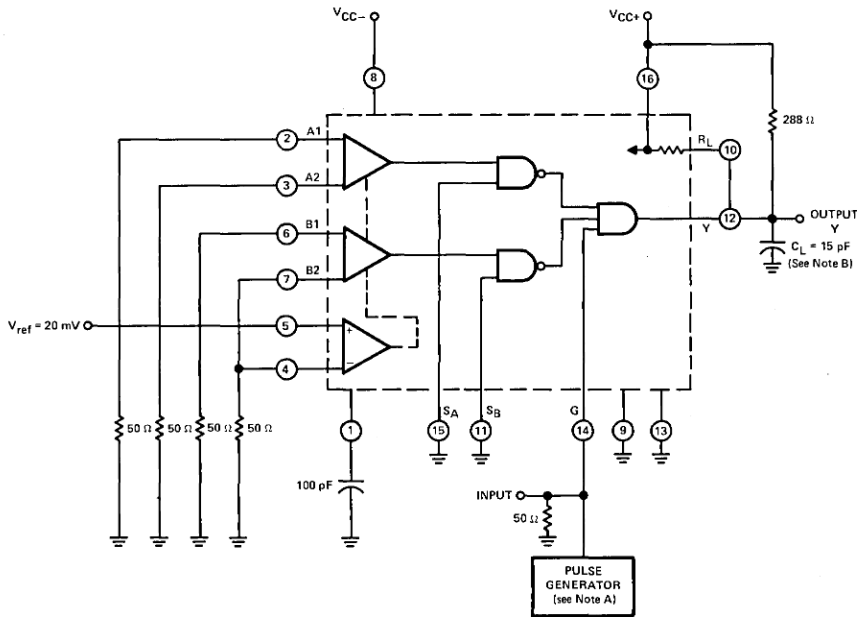
FIGURE 31—SN5522/SN7522 PROPAGATION DELAY TIMES FROM DIFFERENTIAL AND STROBE INPUTS



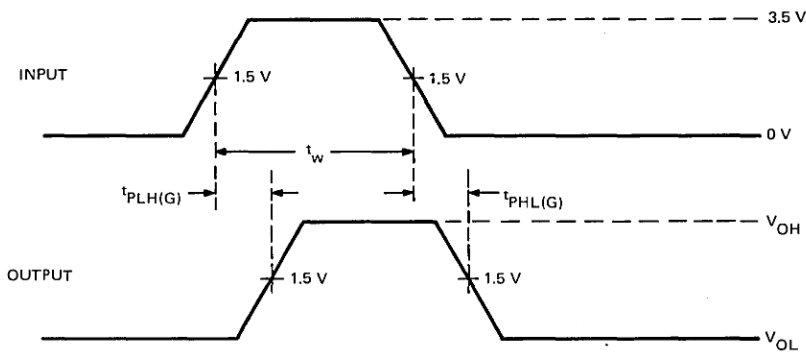
# SERIES 5520/7520 SENSE AMPLIFIERS

## PARAMETER MEASUREMENT INFORMATION

switching characteristics (continued)



TEST CIRCUIT



VOLTAGE WAVEFORMS

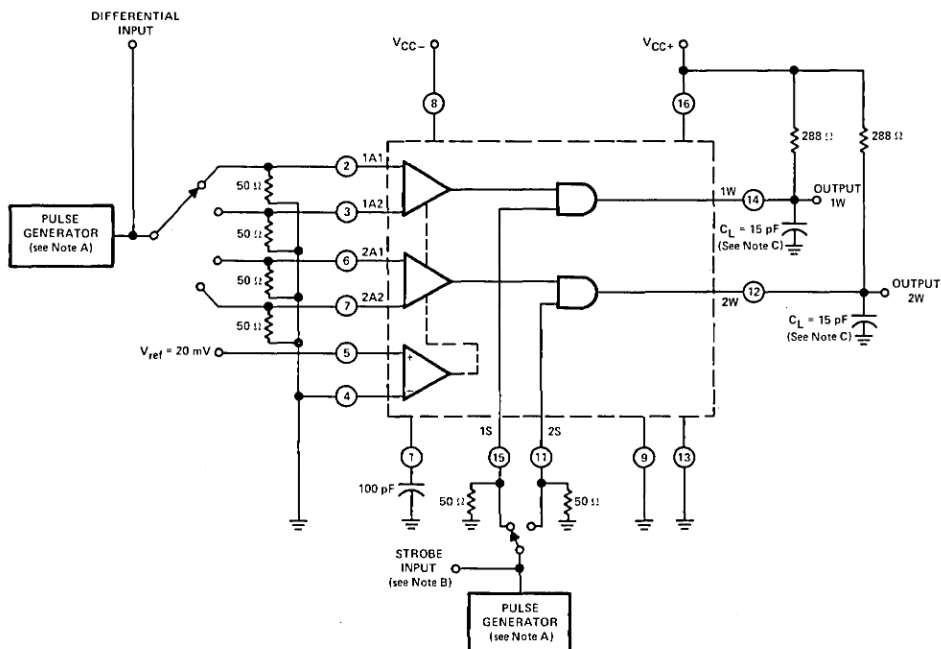
NOTES: A. The pulse generator has the following characteristics:  $Z_{out} = 50 \Omega$ ,  $t_r = 15 \pm 5 \text{ ns}$ ,  $t_f = 15 \pm 5 \text{ ns}$ ,  $t_w = 100 \text{ ns}$ , and  $PRR = 1 \text{ MHz}$ .  
B.  $C_L$  includes probe and jig capacitance.

FIGURE 32—SN5522/SN7522 PROPAGATION DELAY TIMES FROM GATE INPUT

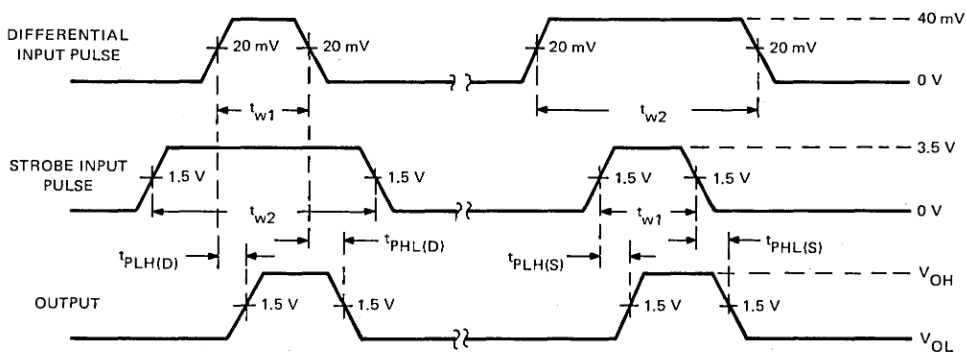
# SERIES 5520/7520 SENSE AMPLIFIERS

## PARAMETER MEASUREMENT INFORMATION

switching characteristics (continued)



TEST CIRCUIT



VOLTAGE WAVEFORMS

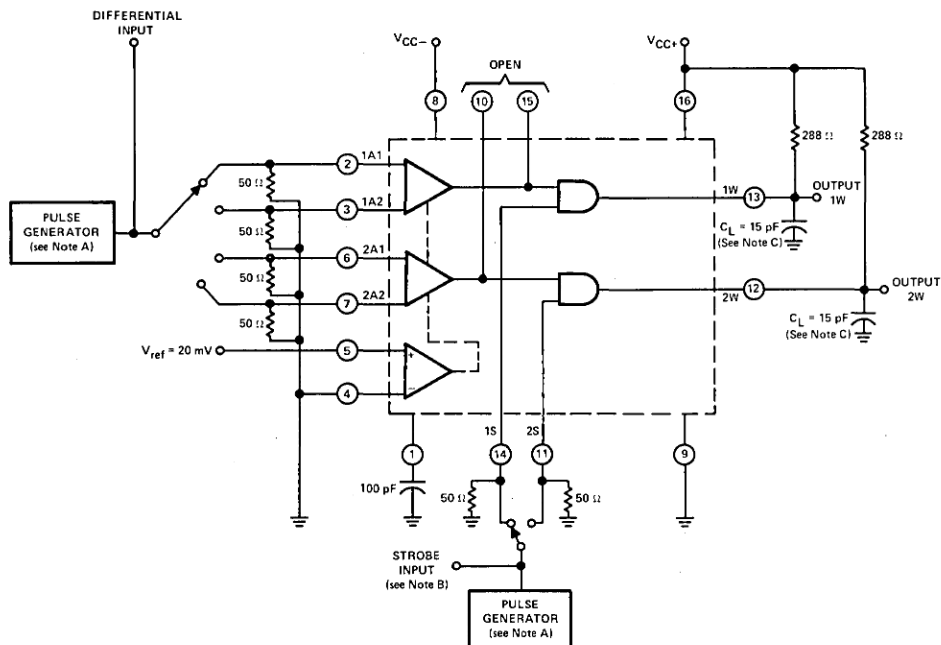
- NOTES:
- A. The pulse generators have the following characteristics:  $Z_{out} = 50 \Omega$ ,  $t_r = 15 \pm 5 \text{ ns}$ ,  $t_f = 15 \pm 5 \text{ ns}$ ,  $t_{w1} = 100 \text{ ns}$ ,  $t_{w2} = 300 \text{ ns}$ , and  $PRR = 1 \text{ MHz}$ .
  - B. The strobe input pulse is applied to Strobe 1S when inputs 1A1-1A2 are being tested and to Strobe 2S when inputs 2A1-2A2 are being tested.
  - C.  $C_L$  includes probe and jig capacitance.

FIGURE 33—SN5524/SN7524 PROPAGATION DELAY TIMES

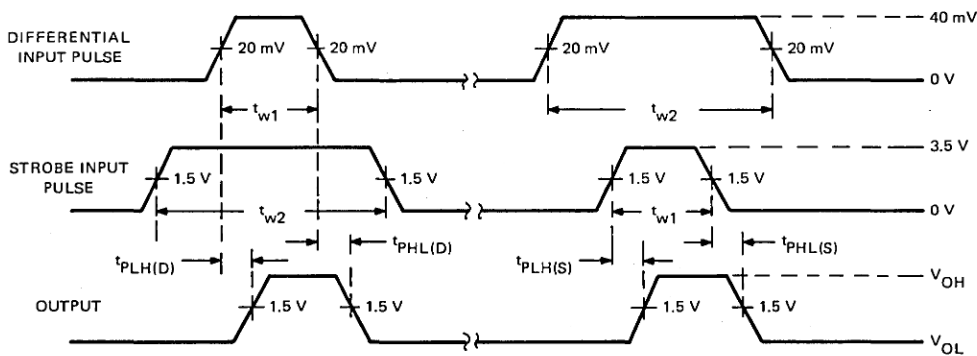
# SERIES 5520/7520 SENSE AMPLIFIERS

## PARAMETER MEASUREMENT INFORMATION

### switching characteristics (continued)



TEST CIRCUIT



VOLTAGE WAVEFORMS

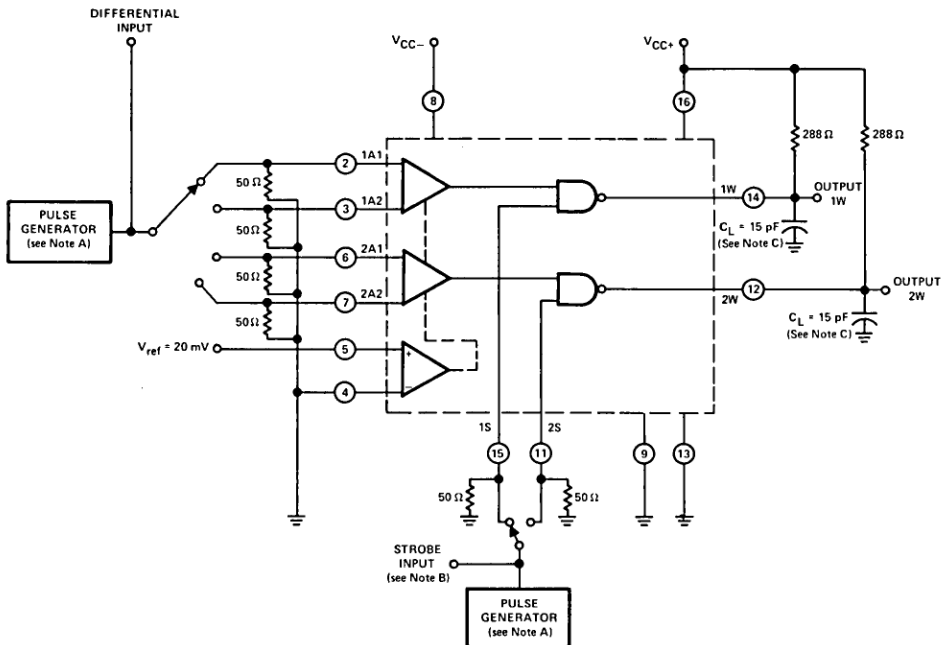
- NOTES: A. The pulse generators have the following characteristics:  $Z_{out} = 50 \Omega$ ,  $t_r = 15 \pm 5 \text{ ns}$ ,  $t_f = 15 \pm 5 \text{ ns}$ ,  $t_{w1} = 100 \text{ ns}$ ,  $t_{w2} = 300 \text{ ns}$ , and  $PRR = 1 \text{ MHz}$ .
- B. The strobe input pulse is applied to Strobe 1S when inputs 1A1-1A2 are being tested and to Strobe 2S when inputs 2A1-2A2 are being tested.
- C.  $C_L$  includes probe and jig capacitance.

FIGURE 34—SN5528/SN7528: PROPAGATION DELAY TIMES

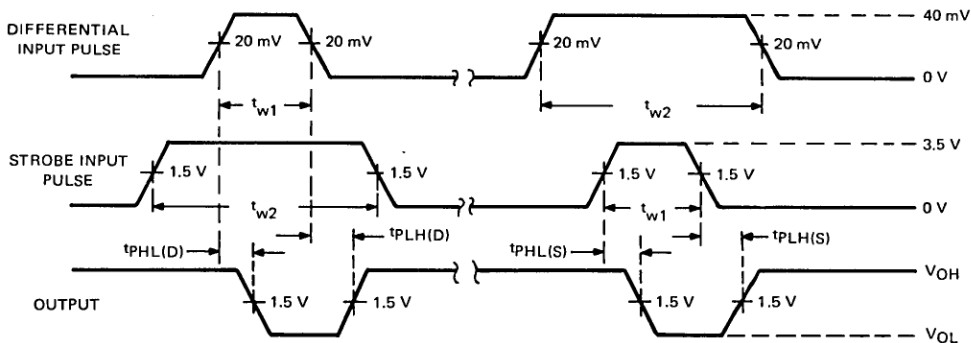
# SERIES 5520/7520 SENSE AMPLIFIERS

## PARAMETER MEASUREMENT INFORMATION

### switching characteristics (continued)



### TEST CIRCUIT



### VOLTAGE WAVEFORMS

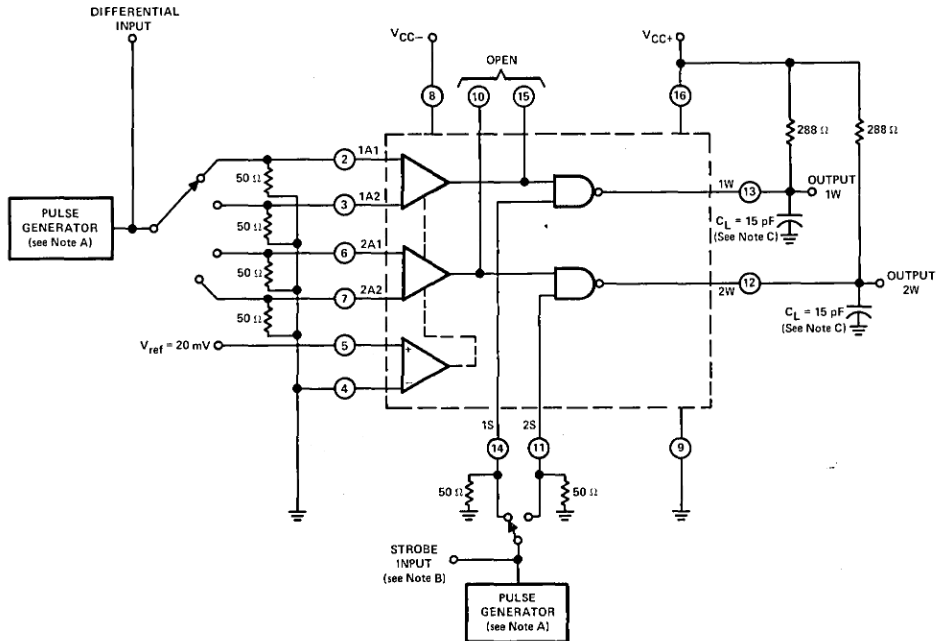
- NOTES: A. The pulse generators have the following characteristics:  $Z_{out} = 50\Omega$ ,  $t_r = 15 \pm 5\text{ ns}$ ,  $t_f = 15 \pm 5\text{ ns}$ ,  $t_{w1} = 100\text{ ns}$ ,  $t_{w2} = 300\text{ ns}$ , and  $PRR = 1\text{ MHz}$ .  
B. The strobe input pulse is applied to Strobe 1S when inputs 1A1-1A2 are being tested and to Strobe 2S when inputs 2A1-2A2 are being tested.  
C.  $C_L$  includes probe and jig capacitance.

FIGURE 35—SN55232, SN75232, SN55234, and SN75234 PROPAGATION DELAY TIMES

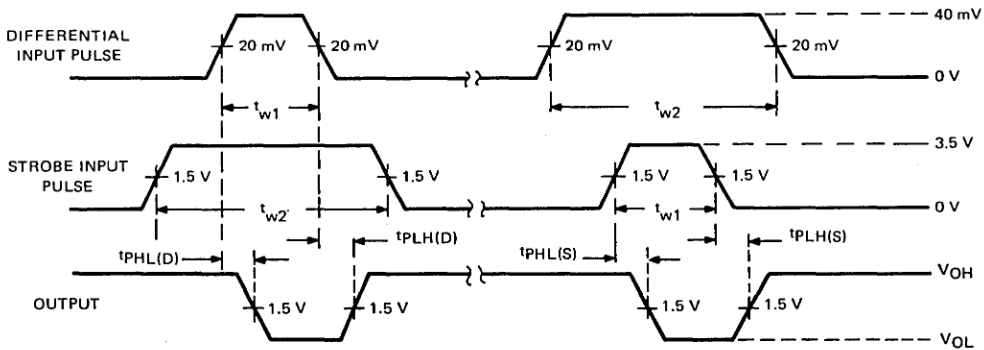
# SERIES 5520/7520 SENSE AMPLIFIERS

## PARAMETER MEASUREMENT INFORMATION

### switching characteristics (continued)



TEST CIRCUIT



VOLTAGE WAVEFORMS

- NOTES: A. The pulse generators have the following characteristics:  $Z_{out} = 50 \Omega$ ,  $t_r = 15 \pm 5$  ns,  $t_f = 15 \pm 5$  ns,  $t_{w1} = 100$  ns,  $t_{w2} = 300$  ns, and PRR = 1 MHz.  
B. The strobe input pulse is applied to Strobe 1S when inputs 1A1-1A2 are being tested and to Strobe 2S when inputs 2A1-2A2 are being tested.  
C.  $C_L$  includes probe and jig capacitance.

FIGURE 36—SN55238/SN75238 PROPAGATION DELAY TIMES

# SERIES 5520/7520 SENSE AMPLIFIERS

## TYPICAL CHARACTERISTICS

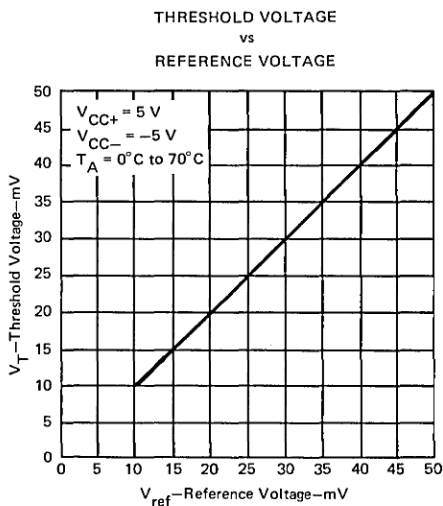


FIGURE 37

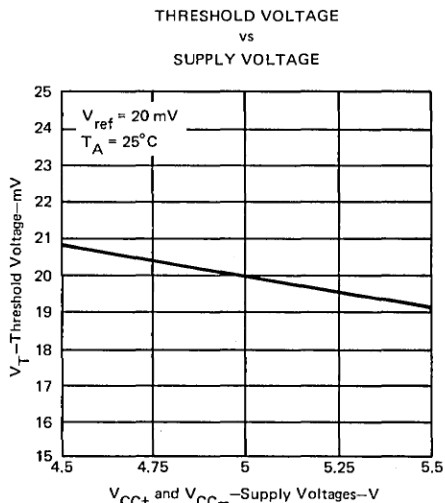


FIGURE 38

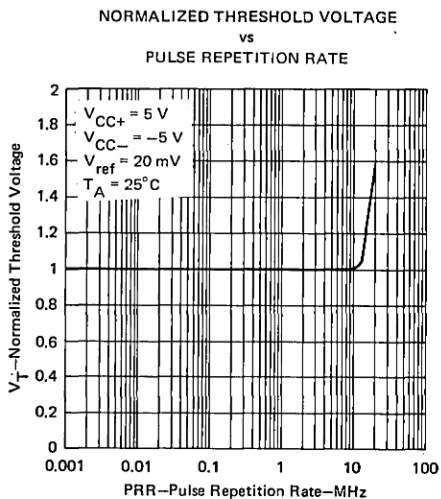


FIGURE 39

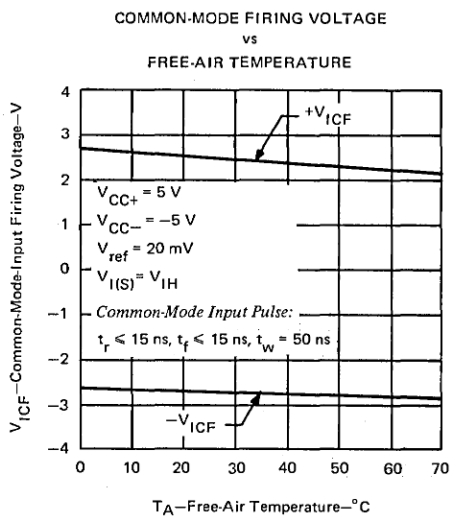


FIGURE 40

# SERIES 5520/7520 SENSE AMPLIFIERS

## TYPICAL CHARACTERISTICS

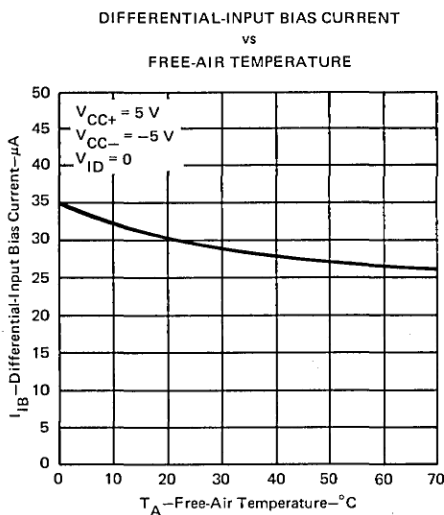


FIGURE 41

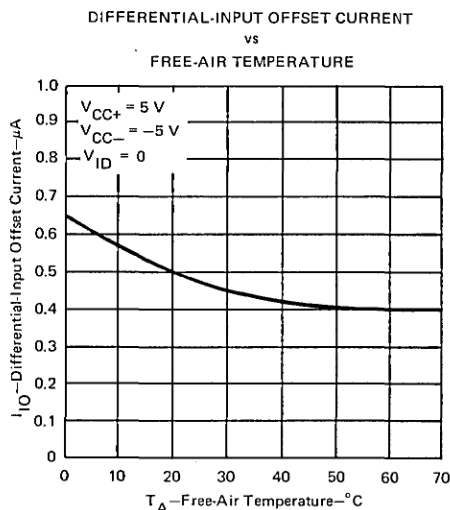


FIGURE 42

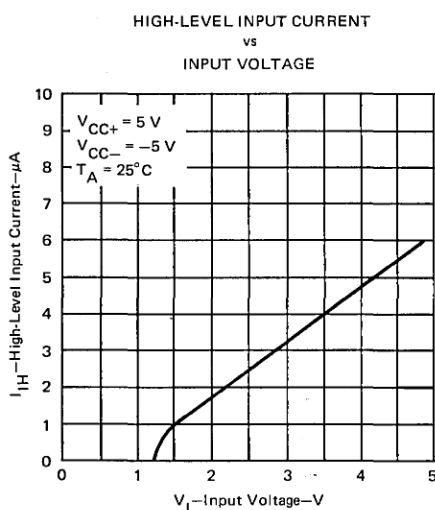


FIGURE 43

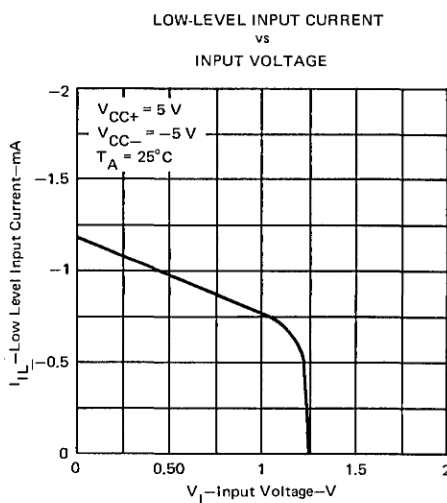


FIGURE 44

# SERIES 5520/7520 SENSE AMPLIFIERS

## TYPICAL CHARACTERISTICS

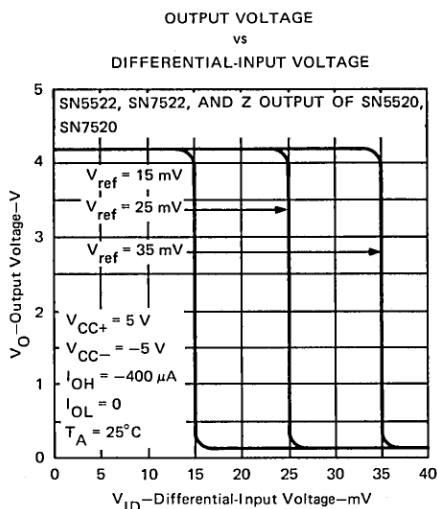


FIGURE 45

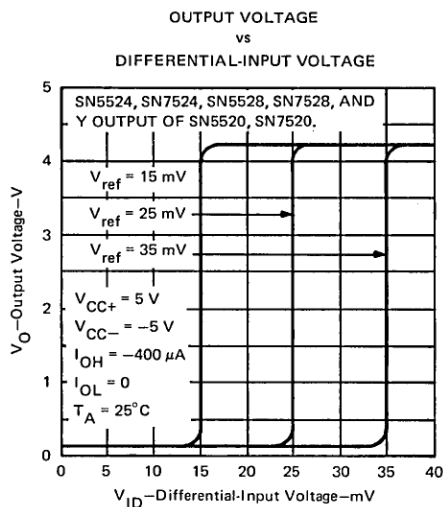


FIGURE 46

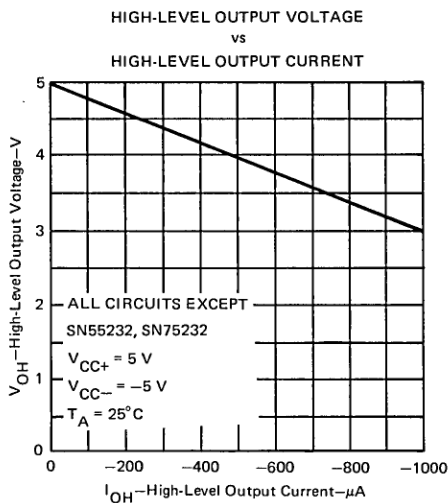


FIGURE 47

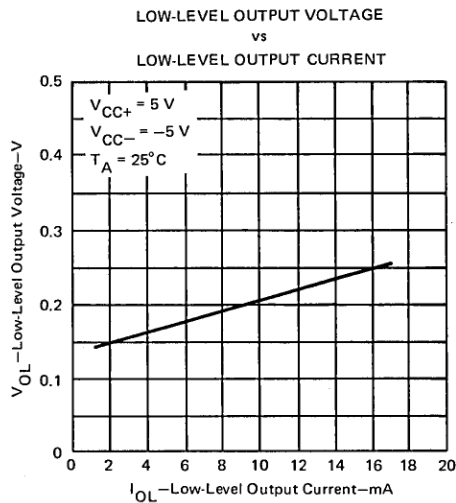


FIGURE 48



**APPLICATION DATA**

**combined fan-out and wire-AND capabilities**

The open-collector TTL gate, when supplied with a proper load resistor ( $R_L$ ), may be paralleled with other similar TTL gates to perform the wire-AND function, and simultaneously, will drive from one to nine Series 54/74 loads. When no other open-collector gates are paralleled, this gate may be used to drive ten Series 54/74 loads. For any of these conditions an appropriate load resistor value must be determined for the desired circuit configuration. A maximum resistor value must be determined which will ensure that sufficient load current (to TTL loads) and off current (through paralleled outputs) will be available while the output is high. A minimum resistor value must be determined which will ensure that current through this resistor and sink current from the TTL loads will not cause the output voltage to rise above the low level even if one of the paralleled outputs is sinking all the current.

In both conditions (low and high level) the value of  $R_L$  is determined by:

$$R_L = \frac{V_{RL}}{I_{RL}}$$

where  $V_{RL}$  is the voltage drop in volts, and  $I_{RL}$  is the current in amperes.

**high-level (off-state) circuit calculations (see figure 49)**

The allowable voltage drop across the load resistor ( $V_{RL}$ ) is the difference between  $V_{CC}$  applied and the  $V_{OH}$  level required at the load:

$$V_{RL} = V_{CC} - V_{OH \text{ min}}$$

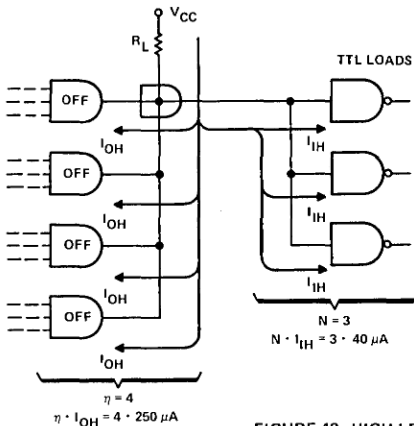
The total current through the load resistor ( $I_{RL}$ ) is the sum of the load currents ( $I_{IH}$ ) and off-state reverse currents ( $I_{OH}$ ) through each of the wire-AND-connected outputs:

$$I_{RL} = \eta \cdot I_{OH} + N \cdot I_{IH} \text{ to TTL loads}$$

Therefore, calculations for the maximum value of  $R_L$  would be:

$$R_{L(max)} = \frac{V_{CC} - V_{OH \text{ min}}}{\eta \cdot I_{OH} + N \cdot I_{IH}}$$

where  $\eta$  = number of gates wire-AND-connected, and  $N$  = number of TTL loads.



Calculation:

$$R_{L(max)} = \frac{V_{CC} - V_{OH \text{ min}}}{\eta \cdot I_{OH} + N \cdot I_{IH}}$$

$$R_{L(max)} = \frac{5 - 2.4}{0.001 + 0.00012} \Omega = \frac{2.6}{0.00112} \Omega = 2321 \Omega$$

**FIGURE 49—HIGH-LEVEL CIRCUIT CONDITIONS**

# SERIES 5520/7520 SENSE AMPLIFIERS

## APPLICATION DATA

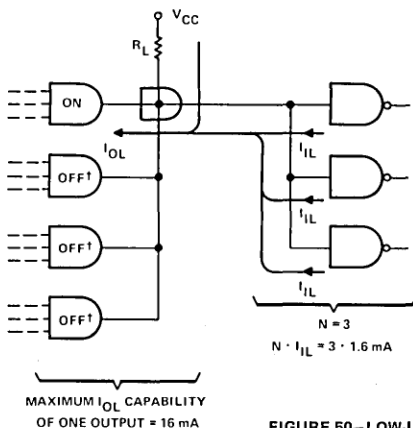
### low-level (on-state) circuit calculations (see figure 50)

The current through the resistor must be limited to the maximum sink-current of one output transistor. Note that if several output transistors are wire-AND connected, the current through  $R_L$  may be shared by those paralleled transistors. However, unless it can be absolutely guaranteed that more than one transistor will be on during low-level periods, the current must be limited to 16 mA, the maximum current which will ensure a low-level maximum of 0.4 volt.

Also, fan-out must be considered. Part of the 16 mA will be supplied from the inputs which are being driven. This reduces the amount of current which can be allowed through  $R_L$ .

Therefore, the equation used to determine the minimum value of  $R_L$  would be:

$$R_{L(\min)} = \frac{V_{CC} - V_{OL \max}}{I_{OL \text{ capability}} - N \cdot I_{IL}}$$



Calculation:

$$R_{L(\min)} = \frac{V_{CC} - V_{OL \max}}{I_{OL \text{ capability}} - N \cdot I_{IL}}$$

$$R_{L(\min)} = \frac{5 - 0.4}{0.016 - 0.0048} \Omega = \frac{4.6}{0.0112} \Omega = 410 \Omega$$

†Current into OFF outputs is negligible at the low logic level.

FIGURE 50—LOW-LEVEL CIRCUIT CONDITIONS

### driving series 54/74 loads and combining outputs

Table 1 provides minimum and maximum resistor values, calculated from equations shown above, for driving one to ten Series 54/74 loads and wire-AND connecting two to seven parallel outputs. Each value shown for one wire-AND output is determined by the fan-out plus the cutoff current of a single output transistor. Extension beyond seven wire-AND connections is permitted with fan-outs of seven or less if a valid minimum and maximum  $R_L$  is possible. When fanning-out to ten Series 54/74 loads, the calculation for the minimum value of  $R_L$  indicates that an infinite resistance should be used ( $V_{RL} \div 0 = \infty$ ); however, the use of a 4-k $\Omega$  resistor in this case will satisfy the high-level condition and limit the low level to less than 0.43 volt.

TABLE 1

| FAN-OUT<br>TO TTL<br>LOADS  | WIRE-AND OUTPUTS |      |      |      |      |      |      |        |
|-----------------------------|------------------|------|------|------|------|------|------|--------|
|                             | 1                | 2    | 3    | 4    | 5    | 6    | 7    | 1 to 7 |
| 1                           | 8985             | 4814 | 3291 | 2500 | 2015 | 1688 | 1452 | 319    |
| 2                           | 7878             | 4482 | 3132 | 2407 | 1954 | 1645 | 1420 | 359    |
| 3                           | 7027             | 4193 | 2988 | 2321 | 1897 | 1604 | 1390 | 410    |
| 4                           | 6341             | 3939 | 2857 | 2241 | 1843 | 1566 | 1361 | 479    |
| 5                           | 5777             | 3714 | 2736 | 2166 | 1793 | 1529 | 1333 | 575    |
| 6                           | 5306             | 3513 | 2626 | 2096 | 1744 | 1494 | 1306 | 718    |
| 7                           | 4905             | 3333 | 2524 | 2031 | 1699 | 1460 | 1280 | 958    |
| 8                           | 4561             | 3170 | 2429 | 1969 | 1656 | X    | X    | 1437   |
| 9                           | 4262             | 3023 | X    | X    | X    | X    | X    | 2875   |
| 10                          | 4000             | X    | X    | X    | X    | X    | X    | 4000§  |
| MAXIMUM                     |                  |      |      |      |      |      |      | MIN    |
| LOAD RESISTOR VALUE IN OHMS |                  |      |      |      |      |      |      |        |

†—All values shown in the table are based on:

High-level conditions:  $V_{CC} = 5 \text{ V}$ ,  $V_{OH \min} = 2.4 \text{ V}$

Low-level conditions:  $V_{CC} = 5 \text{ V}$ ,  $V_{OL \max} = 0.4 \text{ V}$

X—Not recommended or not possible.

§—The theoretical value is  $\infty$ . See explanation in text.

# SERIES 5520/7520 SENSE AMPLIFIERS

## TYPICAL APPLICATIONS

### small memory systems

This application demonstrates an improved method of sensing data from relatively small memory systems. Two individual core planes, usually consisting of 4096 cores each, can be interfaced by each of the dual-channel SN5524 or SN7524 sense amplifiers, see Figure 51. Standard TTL or DTL integrated circuits, driven directly from the compatible sense-amplifier outputs, may be selected to serve as the memory data register (MDR).

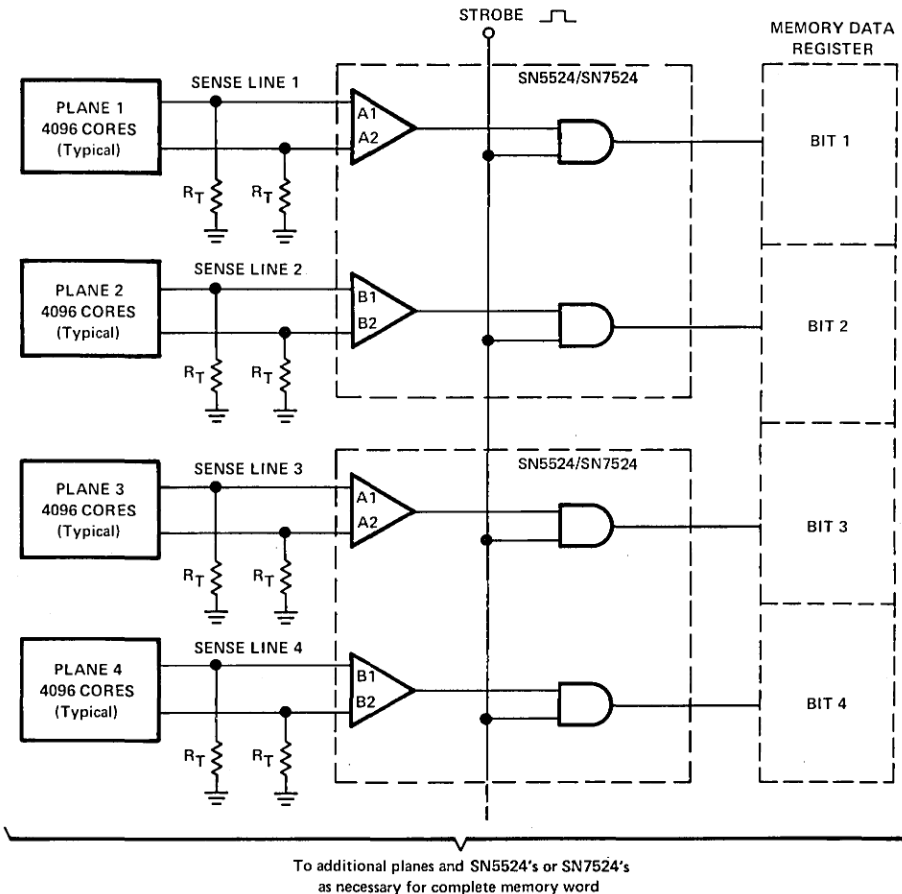


FIGURE 51—SENSING SMALL MEMORY SYSTEMS

## SERIES 5520/7520 SENSE AMPLIFIERS

### TYPICAL APPLICATIONS (continued)

#### large memory systems

This application demonstrates an improved method of sensing data from large memory systems. The signal-to-noise ratio can be increased by sectioning the large core planes as illustrated in Figure 52. Two segments, usually consisting of 4096 cores each, can be interfaced by each of the dual-input channels of the SN5520/SN7520 or SN5522/SN7522 sense amplifiers. The cascaded output gates of the SN5520/SN7520 circuits may be connected to serve as the memory data register (MDR). A number of SN5522/SN7522 sense amplifiers may be wire-AND connected to expand the input function of the MDR to interface all the segments of the plane. Complementary outputs, clear, and preset functions are provided for the MDR. Rules for combined fan-out and wire-AND capabilities must be observed.

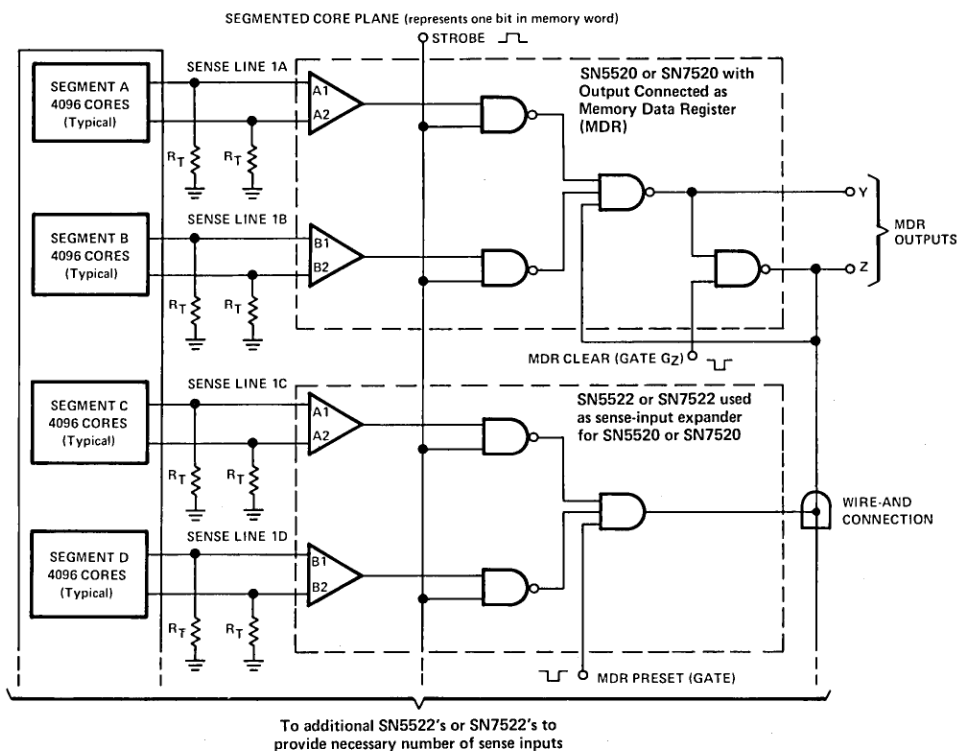


FIGURE 52—SENSING LARGE MEMORY SYSTEMS

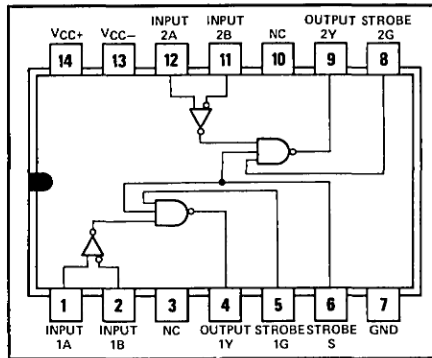
## INTERFACE CIRCUITS

# TYPES SN75207, SN75207B, SN75208, SN75208B DUAL SENSE AMPLIFIERS FOR MOS MEMORIES OR DUAL HIGH-SENSITIVITY LINE RECEIVERS

BULLETIN NO. DL-S 7711793, JULY 1973—REVISED JANUARY 1977

- Plug-in Replacement for SN75107A, SN75107B, SN75108A, SN75108B with Improved Characteristics
- $\pm 10$  mV Guaranteed Input Sensitivity
- TTL Compatible
- Standard Supply Voltages . . .  $\pm 5$  V
- Differential Input Common-Mode Voltage Range of  $\pm 3$  V
- Strobe Inputs for Channel Selection
- '207 and '207B Have Totem-Pole Outputs
- '208 and '208B Have Open-Collector Outputs
- "B" Versions Have Diode-Protected Input Stage for Power-Off Condition
- Sense Amplifier for MOS Memories
- Dual Comparator
- High-Sensitivity Line Receiver

J OR N  
DUAL-IN-LINE PACKAGE  
(TOP VIEW)



NC—No internal connection

### description

The SN75207, SN75207B, SN75208, and SN75208B are pin-for-pin replacements for the SN75107A, SN75107B, SN75108A, and SN75108B, respectively. The improved input sensitivity makes them more suitable for MOS memory sense amplifiers and can result in faster memory cycles. Improved sensitivity also makes them more useful in line receiver applications by allowing use of longer transmission line lengths. The '207 and '207B each features a TTL-compatible active-pull-up output. The '208 and '208B each features an open-collector output that permits wired-AND logic connections with similar output configurations. These devices are designed for operation from  $0^{\circ}\text{C}$  to  $70^{\circ}\text{C}$  and are available in the ceramic dual-in-line (J) package or in the plastic dual-in-line (N) package.

FUNCTION TABLE

| DIFFERENTIAL<br>INPUTS<br>A-B | STROBES |   | OUTPUT<br>Y   |
|-------------------------------|---------|---|---------------|
|                               | G       | S |               |
| $V_{ID} \geq 10$ mV           | X       | X | H             |
| $-10$ mV $< V_{ID} < 10$ mV   | X       | L | H             |
|                               | L       | X | H             |
| $V_{ID} \leq -10$ mV          | H       | H | INDETERMINATE |
|                               | X       | L | H             |
|                               | L       | X | H             |
|                               | H       | H | L             |

H = high level, L = low level, X = irrelevant

The essential difference between the unsuffixed and "B" versions can be seen in the schematics. Input-protection diodes are in series with the collectors of the differential-input transistors of the "B" versions. These diodes are useful in certain "party-line" systems that may have multiple  $V_{CC+}$  power supplies and may be operated with some of the  $V_{CC+}$  supplies turned off. In such a system, if a supply is turned off and allowed to go to ground, the equivalent input circuit connected to that supply would be as follows:



This would be a problem in specific systems that might possibly have the transmission lines biased to some potential greater than 1.4 volts.

# TYPES SN75207, SN75207B, SN75208, SN75208B DUAL SENSE AMPLIFIERS FOR MOS MEMORIES OR DUAL HIGH-SENSITIVITY LINE RECEIVERS

## design characteristics

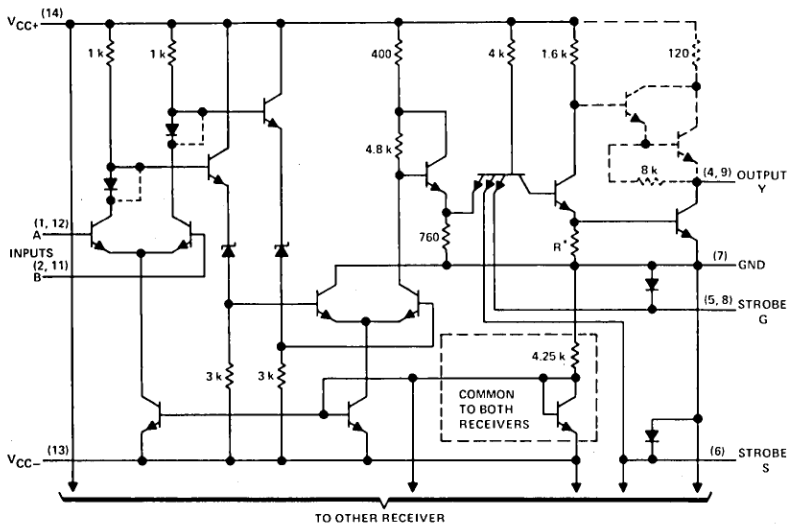
The '207, '207B, '208, and '208B line receivers/sense amplifiers are TTL-compatible dual circuits intended for use in high-speed data-transmission systems or MOS memory systems. They are designed to detect low-level differential signals in the presence of common-mode noise and variations of temperature and supplies. Dc specifications reflect worst-case conditions of temperature, supply voltages, and input voltages.

The input common-mode voltage range is  $\pm 3$  volts. This is adequate for application in most systems. In systems with requirements for greater common-mode voltage range, input attenuators may be used to decrease the noise to an acceptable level at the receiver-input terminals.

The circuits feature individual strobe inputs for each channel and a strobe input common to both channels for logic versatility. The strobe inputs are tested to guarantee 400 millivolts of dc noise margin when interfaced with Series 54/74 TTL.

The circuits feature high input impedance and low input currents, which induce very little loading on the transmission line. This makes these devices especially useful in party-line systems. The excellent input sensitivity (3 millivolts typical) is particularly important when data is to be detected at the end of a long transmission line and the amplitude of the data has deteriorated due to cable losses. The circuits are designed to detect input signals of 10 millivolts (or greater) amplitude and convert the polarity of the signal into appropriate TTL-compatible output logic levels.

## schematic (each receiver)



\*R = 1 k $\Omega$  for '207 and '207B, 750  $\Omega$  for '208 and '208B.

NOTES: A. Resistor values shown are nominal and in ohms.

B. Components shown with dashed lines in the output circuitry are applicable to the '207 and '207B only. Diodes in series with the collectors of the differential input transistors are short-circuited on '207 and '208.

# TYPES SN75207, SN75207B, SN75208, SN75208B

## DUAL SENSE AMPLIFIERS FOR MOS MEMORIES

## OR DUAL HIGH-SENSITIVITY LINE RECEIVERS

absolute maximum ratings over operating free-air temperature range (unless otherwise noted)

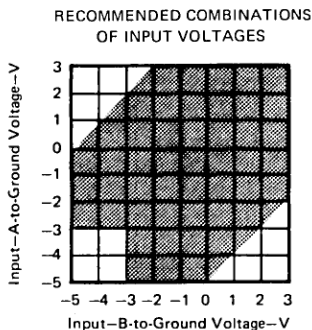
|                                         |                |
|-----------------------------------------|----------------|
| Supply voltage $V_{CC+}$ (see Note 1)   | 7 V            |
| Supply voltage $V_{CC-}$                | -7 V           |
| Differential input voltage (see Note 2) | $\pm 6$ V      |
| Common-mode input voltage (see Note 3)  | $\pm 5$ V      |
| Strobe input voltage                    | 5.5 V          |
| Continuous total dissipation            | 600 mW         |
| Operating free-air temperature range    | 0°C to 70°C    |
| Storage temperature range               | -65°C to 150°C |

recommended operating conditions (see note 4)

|                                                                    | MIN   | NOM | MAX   | UNIT |
|--------------------------------------------------------------------|-------|-----|-------|------|
| Supply voltage $V_{CC+}$                                           | 4.75  | 5   | 5.25  | V    |
| Supply voltage $V_{CC-}$                                           | -4.75 | -5  | -5.25 | V    |
| Low-level output current, $I_{OL}$                                 |       |     | -16   | mA   |
| Differential input voltage, $V_{ID}$ (see Note 5)                  | -5†   |     | 5     | V    |
| Common-mode input voltage, $V_{IC}$ (see Notes 5 and 6)            | -3†   |     | 3     | V    |
| Input voltage range, any differential input to ground (see Note 5) | -5†   |     | 3     | V    |
| Operating free-air temperature                                     | 0     |     | 70    | °C   |

†The algebraic convention where the more positive (less negative) limit is designated as maximum is used in this data sheet for logic voltage levels only.

- NOTES: 1. All voltage values, except differential voltages, are with respect to network ground terminal.
2. Differential voltage values are at the noninverting (A) terminal with respect to the inverting (B) terminal.
3. Common-mode input voltage is the average of the voltages at the A and B inputs.
4. When using only one channel of the line receiver, the strobe G of the unused channel should be grounded and at least one of the differential inputs of the unused receiver should be terminated at some voltage between -3 V and 3 V.
5. The recommended combinations of input voltages fall within the shaded area of the figure at the right.
6. The common-mode voltage may be as low as -4 V provided that one of the two inputs is not more negative than -3 V.



# TYPES SN75207, SN75207B, SN75208, SN75208B

## DUAL SENSE AMPLIFIERS FOR MOS MEMORIES

## OR DUAL HIGH-SENSITIVITY LINE RECEIVERS

definition of input logic levels<sup>†</sup>

|                                                                       | MIN  | MAX   | UNIT |
|-----------------------------------------------------------------------|------|-------|------|
| V <sub>IDH</sub> High-level input voltage between differential inputs | 0.01 | 5     | V    |
| V <sub>IDL</sub> Low-level input voltage between differential inputs  | -5   | -0.01 | V    |
| V <sub>IH(S)</sub> High-level input voltage at strobe inputs          | 2    | 5.5   | V    |
| V <sub>IL(S)</sub> Low-level input voltage at strobe inputs           | 0    | 0.8   | V    |

<sup>†</sup>The algebraic convention, where the more positive (less negative) limit is designated maximum, is used in this data sheet with logic input voltage levels only.

electrical characteristics over recommended free-air temperature range (unless otherwise noted)

| PARAMETER         |                                                     |   | TEST CONDITIONS†                                                                                                                       |                        | '207, '207B |      | '208, '208B |      | UNIT |      |
|-------------------|-----------------------------------------------------|---|----------------------------------------------------------------------------------------------------------------------------------------|------------------------|-------------|------|-------------|------|------|------|
|                   |                                                     |   |                                                                                                                                        |                        | MIN         | TYP‡ | MAX         | MIN  |      | TYP‡ |
| I <sub>IH</sub>   | High-level input current                            | A | V <sub>CC±</sub> = MAX                                                                                                                 | V <sub>ID</sub> = 5 V  |             | 30   | 75          | 30   | 75   | μA   |
|                   |                                                     | B |                                                                                                                                        | V <sub>ID</sub> = -5 V |             | 30   | 75          | 30   | 75   |      |
| I <sub>IL</sub>   | Low-level input current                             | A | V <sub>CC±</sub> = MAX                                                                                                                 | V <sub>ID</sub> = -5 V |             | -10  |             | -10  | μA   |      |
|                   |                                                     | B |                                                                                                                                        | V <sub>ID</sub> = 5 V  |             | -10  |             | -10  |      |      |
| I <sub>IH</sub>   | High-level input current into 1G or 2G              |   | V <sub>CC±</sub> = MAX, V <sub>IH(S)</sub> = 2.4 V                                                                                     |                        |             | 40   |             | 40   | μA   |      |
|                   |                                                     |   | V <sub>CC±</sub> = MAX, V <sub>IH(S)</sub> = MAX V <sub>CC±</sub>                                                                      |                        |             | 1    |             | 1    |      |      |
| I <sub>IL</sub>   | Low-level input current into 1G or 2G               |   | V <sub>CC±</sub> = MAX, V <sub>IL(S)</sub> = 0.4 V                                                                                     |                        |             | -1.6 |             | -1.6 | mA   |      |
|                   |                                                     |   |                                                                                                                                        |                        |             |      |             |      |      |      |
| I <sub>IH</sub>   | High-level input current into S                     |   | V <sub>CC±</sub> = MAX, V <sub>IH(S)</sub> = 2.4 V                                                                                     |                        |             | 80   |             | 80   | μA   |      |
|                   |                                                     |   | V <sub>CC±</sub> = MAX, V <sub>IH(S)</sub> = MAX V <sub>CC±</sub>                                                                      |                        |             | 2    |             | 2    |      |      |
| I <sub>IL</sub>   | Low-level input current into S                      |   | V <sub>CC±</sub> = MAX, V <sub>IL(S)</sub> = 0.4 V                                                                                     |                        |             | -3.2 |             | -3.2 | mA   |      |
|                   |                                                     |   |                                                                                                                                        |                        |             |      |             |      |      |      |
| V <sub>OH</sub>   | High-level output voltage                           |   | V <sub>CC±</sub> = MIN, V <sub>IH(S)</sub> = 0.8 V, V <sub>IDH</sub> = 10 mV, I <sub>OH</sub> = -400 μA, V <sub>IC</sub> = -3 V to 3 V |                        | 2.4         |      |             |      | V    |      |
| V <sub>OL</sub>   | Low-level output voltage                            |   | V <sub>CC±</sub> = MIN, V <sub>IH(S)</sub> = 2 V, V <sub>IDL</sub> = -10 mV, I <sub>OL</sub> = 16 mA, V <sub>IC</sub> = -3 V to 3 V    |                        |             | 0.4  |             | 0.4  | V    |      |
| I <sub>OH</sub>   | High-level output current                           |   | V <sub>CC±</sub> = MIN, V <sub>OH</sub> = MAX V <sub>CC±</sub>                                                                         |                        |             |      |             | 250  | μA   |      |
| I <sub>OS</sub>   | Short-circuit output current¶                       |   | V <sub>CC±</sub> = MAX                                                                                                                 |                        | -18         | -70  |             |      | mA   |      |
| I <sub>CCH+</sub> | Supply current from V <sub>CC+</sub> , outputs high |   | V <sub>CC±</sub> = MAX, T <sub>A</sub> = 25°C                                                                                          |                        |             | 18   | 30          | 18   | 30   | mA   |
| I <sub>CCH-</sub> | Supply current from V <sub>CC-</sub> , outputs high |   | V <sub>CC±</sub> = MAX, T <sub>A</sub> = 25°C                                                                                          |                        |             | -8.4 | -15         | -8.4 | -15  | mA   |

<sup>‡</sup>For conditions shown as MIN or MAX, use the appropriate value specified under recommended operating conditions.

<sup>§</sup>All typical values are at V<sub>CC+</sub> = 5 V, V<sub>CC-</sub> = -5 V, T<sub>A</sub> = 25°C.

<sup>¶</sup>Not more than one output should be shorted at a time.

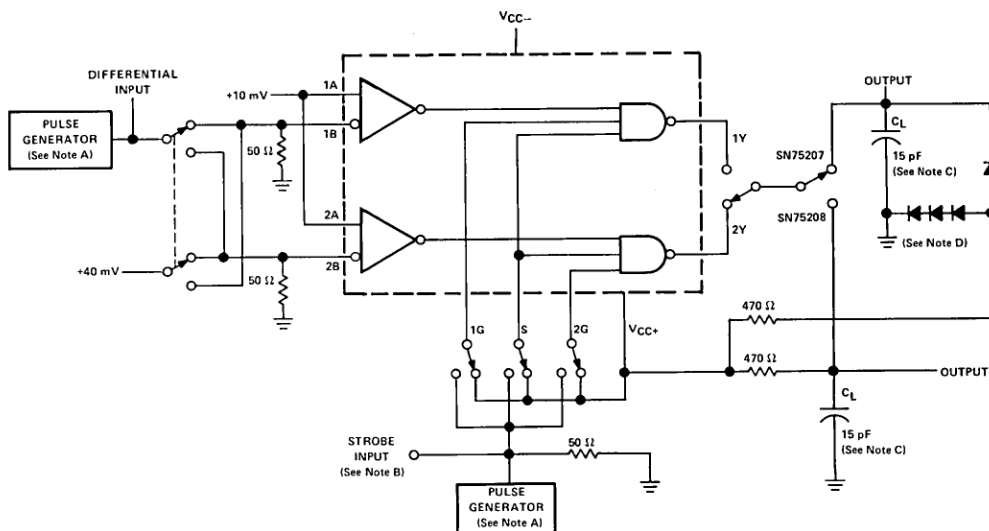
switching characteristics, V<sub>CC+</sub> = 5 V, V<sub>CC-</sub> = -5 V, T<sub>A</sub> = 25°C

| PARAMETER           |                                                                                    | TEST CONDITIONS                                              | '207, '207B |     |     | '208, '208B |     |     | UNIT |
|---------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------|-------------|-----|-----|-------------|-----|-----|------|
|                     |                                                                                    |                                                              | MIN         | TYP | MAX | MIN         | TYP | MAX |      |
| t <sub>PLH(D)</sub> | Propagation delay time, low-to-high-level output, from differential inputs A and B | R <sub>L</sub> = 470 Ω, C <sub>L</sub> = 15 pF, See Figure 1 |             | 35  |     |             | 35  |     | ns   |
| t <sub>PHL(D)</sub> | Propagation delay time, high-to-low-level output, from differential inputs A and B |                                                              |             | 20  |     |             | 20  |     | ns   |
| t <sub>PLH(S)</sub> | Propagation delay time, low-to-high-level output, from strobe input G or S         |                                                              |             | 17  |     |             | 17  |     | ns   |
| t <sub>PHL(S)</sub> | Propagation delay time, high-to-low-level output, from strobe input G or S         |                                                              |             | 17  |     |             | 17  |     | ns   |

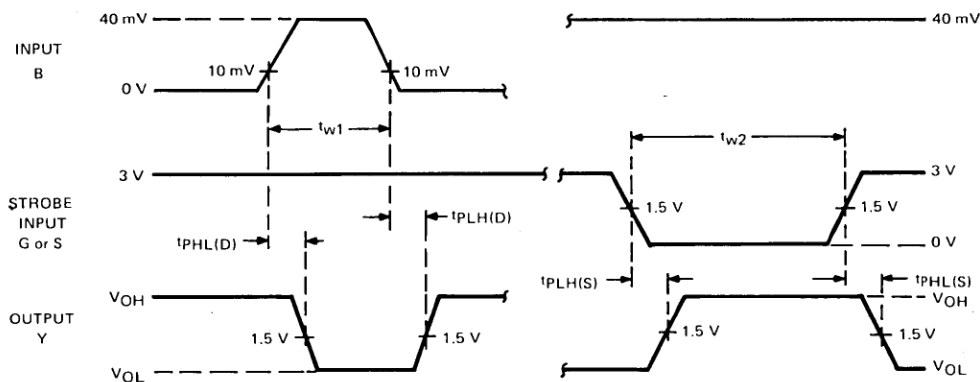


# **TYPES SN75207, SN75207B, SN75208, SN75208B** **DUAL SENSE AMPLIFIERS FOR MOS MEMORIES** **OR DUAL HIGH-SENSITIVITY LINE RECEIVERS**

## **PARAMETER MEASUREMENT INFORMATION**



## **TEST CIRCUIT**



- NOTES: A. The pulse generators have the following characteristics:  $Z_{out} = 50 \Omega$ ,  $t_r \leq 5$  ns,  $t_f \leq 5$  ns,  $t_{w1} = 500$  ns with PRR = 1 MHz,  $t_{w2} = 1$  ms with PRR = 500 kHz.
- B. Strobe input pulse is applied to Strobe 1G when inputs 1A-1B are being tested, to Strobe S when inputs 1A-1B or 2A-2B are being tested, and to Strobe 2G when inputs 2A-2B are being tested.
- C.  $C_L$  includes probe and jig capacitance.
- D. All diodes are 1N916.

**FIGURE 1—PROPAGATION DELAY TIMES**

# TYPES SN75207, SN75207B, SN75208, SN75208B DUAL SENSE AMPLIFIERS FOR MOS MEMORIES OR DUAL HIGH-SENSITIVITY LINE RECEIVERS

## TYPICAL APPLICATION DATA

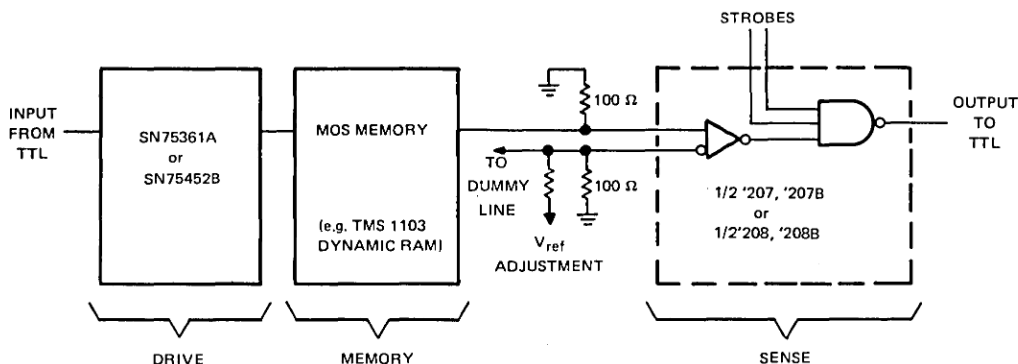
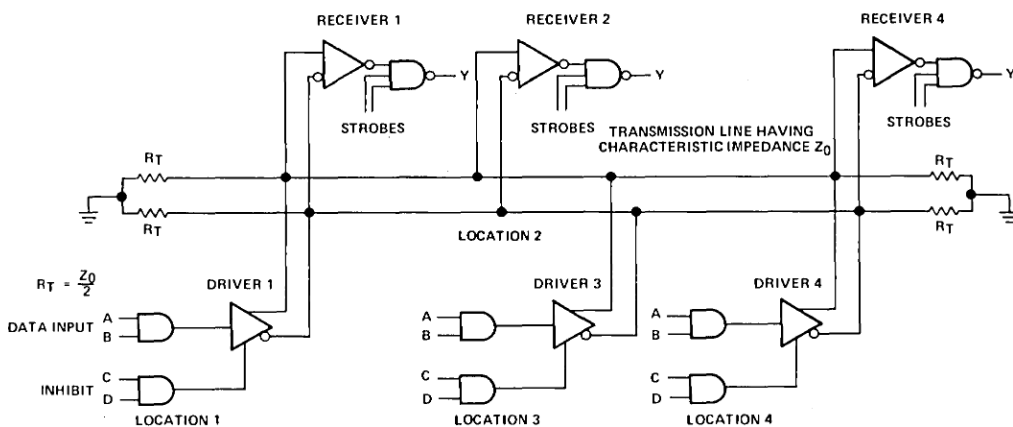


FIGURE 2—MOS MEMORY SENSE AMPLIFIER



Receivers are '207, '207B, '208, or '208B; drivers are SN55109A, SN75109A, SN55110A, SN75110A, or SN75112.

FIGURE 3—DATA-BUS OR PARTY-LINE SYSTEM

**PRECAUTIONS:** When only one receiver in a package is being used, at least one of the differential inputs of the unused receiver should be terminated at some voltage between -3 volts and +3 volts, preferably at ground. Failure to do so will cause improper operation of the unit being used because of common bias circuitry for the current sources of the two receivers. Strobe G of the unused channel should be grounded.

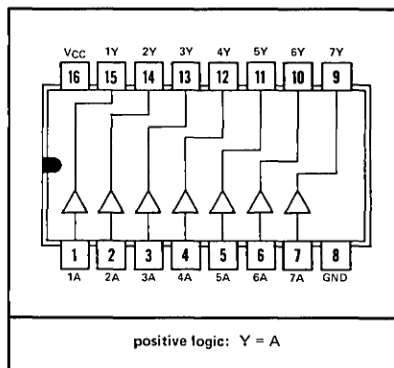
## INTERFACE CIRCUITS

## TYPE SN75270 7-UNIT MOS-TO-TTL CONVERTER AND THERMAL PRINthead DRIVER ARRAY

BULLETIN NO. DL-S 7712061, SEPTEMBER 1973—REVISED APRIL 1977

JOR N  
DUAL-IN-LINE PACKAGE (TOP VIEW)

- 7 Single-Ended Noninverting Drivers Per Package
- Inputs Compatible with MOS
- TTL-Compatible Outputs
- Single 5-V Supply

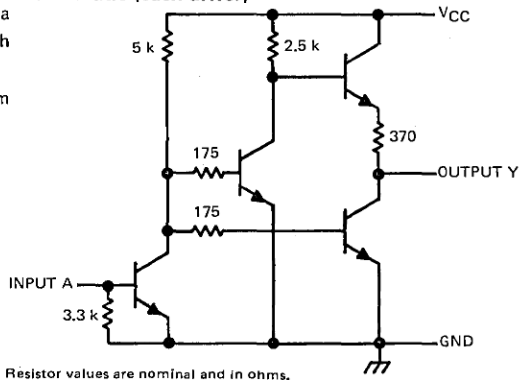


### description

The SN75270 is a monolithic integrated circuit designed for use as a sense amplifier or thermal printhead driver. As a sense amplifier, the device can be used to convert from MOS to TTL levels. As a thermal printhead driver, this device is used with EPN3600-type thermal printheads.

The SN75270 is characterized for operation from 0°C to 70°C.

### schematic (each driver)



### absolute maximum ratings over operating free-air temperature range (unless otherwise noted)

|                                                                |                |
|----------------------------------------------------------------|----------------|
| Supply voltage, $V_{CC}$ (see Note 1)                          | 7 V            |
| Input current                                                  | 4 mA           |
| Continuous total dissipation                                   | 800 mW         |
| Operating free-air temperature range                           | 0°C to 70°C    |
| Storage temperature range                                      | -65°C to 150°C |
| Lead temperature 1/16 inch from case for 60 seconds: J package | 300°C          |
| Lead temperature 1/16 inch from case for 10 seconds: N package | 260°C          |

### recommended operating conditions

|                                       | MIN  | NOM | MAX  | UNIT |
|---------------------------------------|------|-----|------|------|
| Supply voltage, $V_{CC}$              | 4.75 | 5   | 5.25 | V    |
| High-level input current, $I_{IH}$    | 0.5  |     | 2    | mA   |
| Low-level input current, $I_{IL}$     | 0    |     | 0.1  | mA   |
| Operating free-air temperature, $T_A$ | 0    |     | 70   | °C   |

NOTE 1: Voltage values are with respect to network ground terminal.

# TYPE SN75270

## 7-UNIT MOS-TO-TTL CONVERTER AND THERMAL PRINthead DRIVER ARRAY

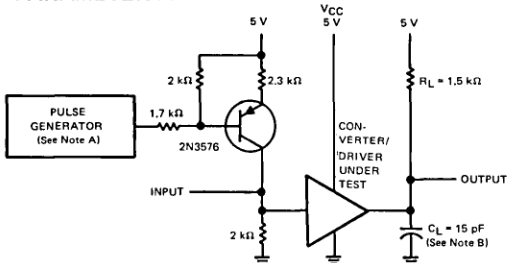
electrical characteristics over recommended operating free-air temperature range

| PARAMETER                                       | TEST CONDITIONS                                                                                     | MIN | TYP | MAX | UNIT |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------|-----|-----|-----|------|
| $V_{OH}$ High-level output voltage              | $V_{CC} = 4.75\text{ V}$ , $I_{IH} = 500\text{ }\mu\text{A}$ ,<br>$I_{OH} = -80\text{ }\mu\text{A}$ | 2.4 |     |     | V    |
| $V_{OL}$ Low-level output voltage               | $V_{CC} = 4.75\text{ V}$ , $I_{IL} = 100\text{ }\mu\text{A}$ ,<br>$I_{OL} = 3.2\text{ mA}$          |     |     | 0.4 | V    |
| $I_{OH}$ High-level output current              | $V_{CC} = 4.75\text{ V}$ , $I_{IH} = 500\text{ }\mu\text{A}$ ,<br>$V_O = 1\text{ V}$                | -5  |     |     | mA   |
|                                                 | $V_{CC} = 5.25\text{ V}$ , $I_{IH} = 500\text{ }\mu\text{A}$ ,<br>$V_O = 0.25\text{ V}$             |     | -15 |     |      |
|                                                 |                                                                                                     |     |     |     |      |
| $I_{CCL}$ Total supply current, all outputs low | $V_{CC} = 5\text{ V}$ , $I_{IL} = 100\text{ }\mu\text{A}$ ,<br>$I_O = 0$                            | 20  | 35  |     | mA   |

switching characteristics,  $T_A = 25^\circ\text{C}$

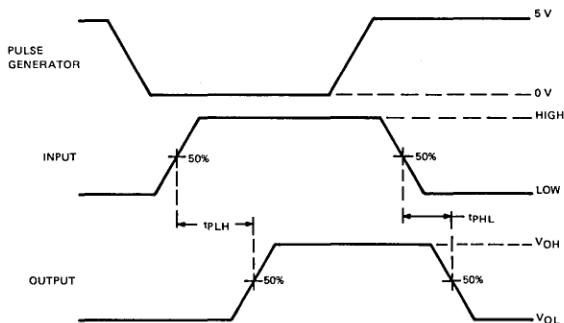
| PARAMETER                                                  | TEST CONDITIONS                                                                             | MIN | TYP | MAX | UNIT |
|------------------------------------------------------------|---------------------------------------------------------------------------------------------|-----|-----|-----|------|
| $t_{PLH}$ Propagation delay time, low-to-high-level output | $V_{CC} = 5\text{ V}$ , $C_L = 15\text{ pF}$ ,<br>$R_L = 1.5\text{ k}\Omega$ , See Figure 1 |     | 30  |     | ns   |
| $t_{PHL}$ Propagation delay time, high-to-low-level output |                                                                                             |     | 8   |     | ns   |

### PARAMETER MEASUREMENT INFORMATION



NOTES: A. The pulse generator has the following characteristics:  $Z_{out} = 50\text{ }\Omega$ ,  $t_r \leq 10\text{ ns}$ ,  $t_f \leq 10\text{ ns}$ ,  $PRR = 500\text{ kHz}$ ,  $t_w = 500\text{ ns}$ .  
B.  $C_L$  includes probe and jig capacitance.

### TEST CIRCUIT

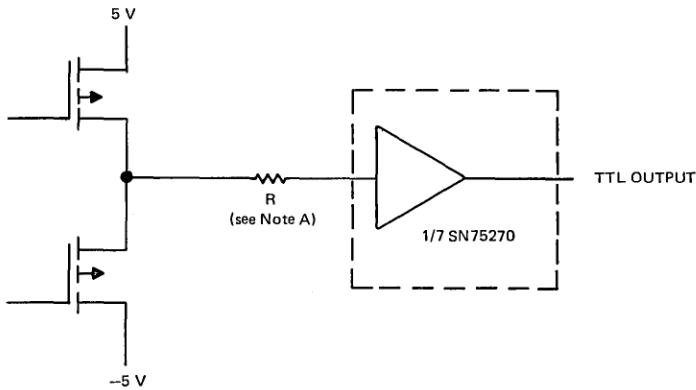


### VOLTAGE WAVEFORMS

FIGURE 1

# TYPE SN75270 **7-UNIT MOS-TO-TTL CONVERTER AND THERMAL PRINthead DRIVER ARRAY**

## TYPICAL APPLICATION DATA



Note A:

$$R = \frac{V_{OH} - V_{BE}}{I_{OH}}$$

$V_{OH}$  = High-level output voltage of MOS device

$V_{BE}$  = Base-Emitter voltage of input transistor of SN75270

$I_{OH}$  = High-level output current of MOS device

example: let  $V_{OH} = 4 \text{ V}$

$I_{OH} = 1 \text{ mA}$

$V_{BE} = 0.7 \text{ V}$

$$R = \frac{4 - 0.7}{1} = 3.3 \text{ k}\Omega$$

FIGURE 2—MOS TO SN75270 CONNECTION

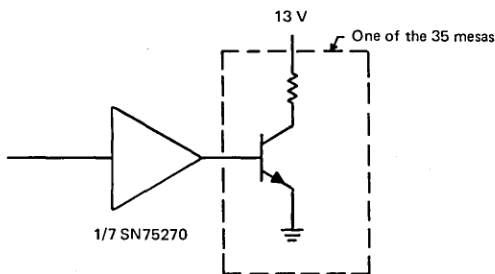


FIGURE 3—THERMAL PRINthead DRIVER FOR  
THE EPN3600 THERMAL PRINthead